



BASE II Clearing Services

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Visa Supplemental Requirements

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Chapter 1

BASE II Clearing Services Overview

BASE II Clearing System uses client-provided information and processing parameters to provide transaction clearing services for clients.

Clients provide information using:

- Client Processing Selections—Clients provide data, optional selections, and processing limits to BASE II Clearing to enable BASE II to provide client-selected services.
- Message Content—Many data elements of the BASE II record provide the client's preferences for the processing of that record. For example: merchant information in a BASE II record enables the processing of merchant services in addition to basic clearing services.

BASE II Clearing System Enrollment Process

BASE II Clearing System uses an enrollment process to provide services to clients.

Most services provided by BASE II Clearing System are available to all BASE II clients. Other services are optional.

To participate in services provided by BASE II Clearing System, clients should contact their Visa representatives, who coordinate service participation. Enrollment processes differ for each region. Visa representatives can explain the regional requirements and enrollment specifics.

BASE II Clearing System Testing

BASE II Clearing System enables clients to test their changes in the VisaNet Certification Management Service (VCMS) environment.

Testing is a process by which clients use specially designed test cases to verify the integrity of their VisaNet-connected systems. The test cases are specific for each Visa service. Clients perform testing using the Client Testing System environment that simulates (as closely as possible) the production environment.

Clients can use VCMS to verify the BASE II service requirements and test the processing of transactions through VisaNet.

VCMS:

- Improves client readiness for transaction processing.
- Facilitates implementation of new services for clients.
- Provides end-to-end testing and transaction lifecycle testing for Custom Payment Service (CPS)/ATM and Custom Payment Service (CPS)/POS.
- Reduces the risk of incorrect processing in authorization and clearing production systems both for Visa and for the client.

Clients should refer to the VisaNet Testing Guide for more information on VisaNet Testing processes and best practices involved with testing their systems with VCMS VIP and BASE II.

BASE II Clearing System Interactions

BASE II Clearing System is the international electronic clearing system that supports transaction-based payment exchange between Visa clients. BASE II Clearing supports all cards (Visa, Mastercard, American Express, etc.) and several non-card payment services.

Clearing Overview

Clearing is the process by which Visa collects data about a transaction from the source, validates the information and values, calculates fees and charges as appropriate, and then

delivers it to the destination. The receiving institution uses information to post accounts and reconcile activity and settlement.

Authorization is the process by which an issuer approves or declines a cardholder's transaction before a purchase is finalized or cash is disbursed.

Settlement is the process of calculating and determining the net financial position of each client for all transactions that are cleared.

This manual describes BASE II processing and clearing and settlement of client transactions.

BASE II Clearing System and VisaNet

The VisaNet Clearing System is referred to as BASE II. VisaNet is the processing platform that connects Visa and its worldwide client banks and financial institutions so that they can authorize and settle electronic payments quickly and securely. VisaNet is comprised of the VisaNet Integrated Payment (V.I.P.) System, BASE II Clearing, Visa communications network, VisaNet Settlement Service (VSS), and ancillary services, such as Fraud Reporting Service (FRS), merchant services, Visa Access, Visa Resolve Online (VROL), and VisaNet Certification Management Service (VCMS).

VisaNet supports a wide variety of transactions, including purchase, cash, push payments, and bill payment transactions made with any card; cardholder transfer transactions; recurring payment transactions; and ATM transactions for other networks.

Visa supports this variety of transactions with unique clearing rules based on brand, network, product, and jurisdiction.

VisaNet runs on a Central Processing Day (CPD) based upon GMT.

Clearing Benefits

BASE II Clearing is one of Visa's core processing functions—it provides guaranteed settlement between clients, timely information for cardholder posting, and merchant settlement.

Chapter 2

BASE II Clearing System Interactions

BASE II Clearing System Workflow

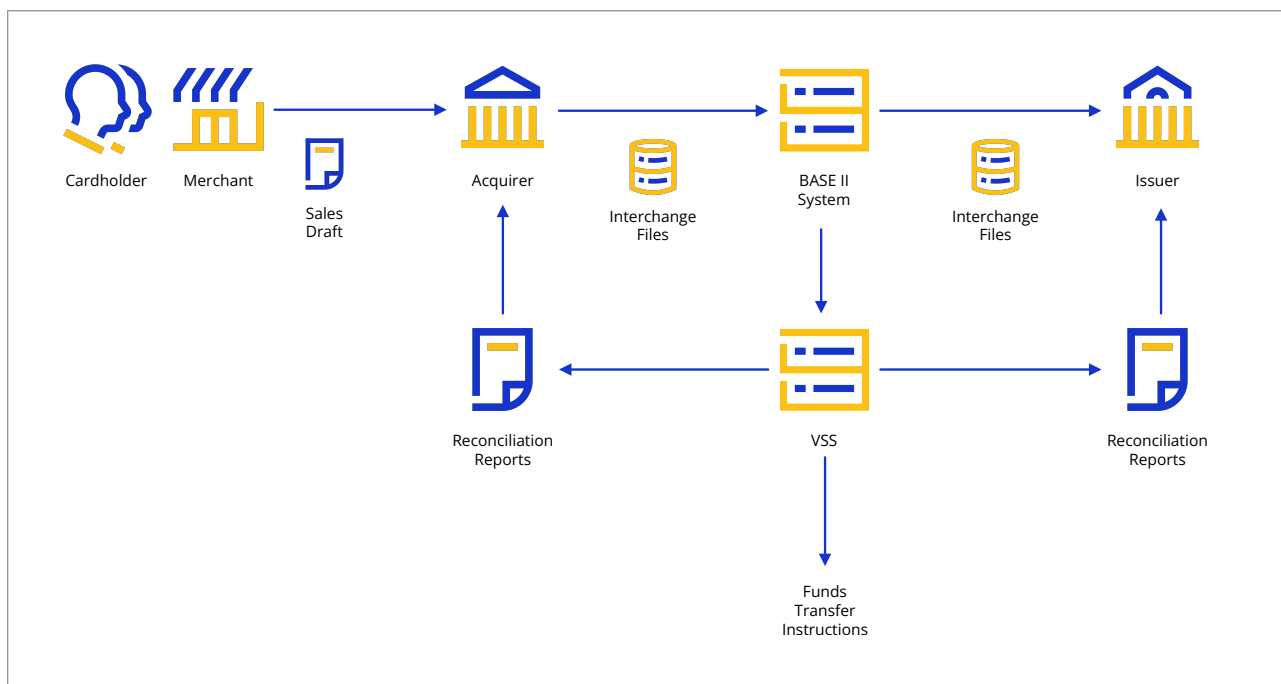
BASE II Clearing System receives and processes interchange files from clients at intervals specified by the clients and provides the settlement information at the appropriate time to receiving institutions, clients, and processors.

The workflow between merchants, clients, and Visa can be summarized as follows:

- **Merchant**—Transmits the electronic sales drafts to their acquiring institution or processor.
- **Acquiring Institution or Processor**—Edits, formats, and sends the merchant's transactions electronically as interchange files to a VisaNet Data Center.

- **VisaNet Interchange Center**—Clears the acquiring institution or processor's interchange files by:
 - Collecting, editing, and validating the interchange files, transaction by transaction.
 - Performing any necessary currency conversion.
 - Calculating interchange fees and Visa's daily transaction charges.
 - Creating settlement records for VisaNet Settlement Service, (VSS).
 - Creating and routing delivery interchange files to the appropriate issuing institutions and processors
- **Issuing Institution or Processor**—Reviews the clearing transactions and, if appropriate, removes the hold from the cardholder's funds and posts the amount of the transaction to the cardholder's account.

BASE II Clearing Workflow



BASE II Transaction Reconciliation

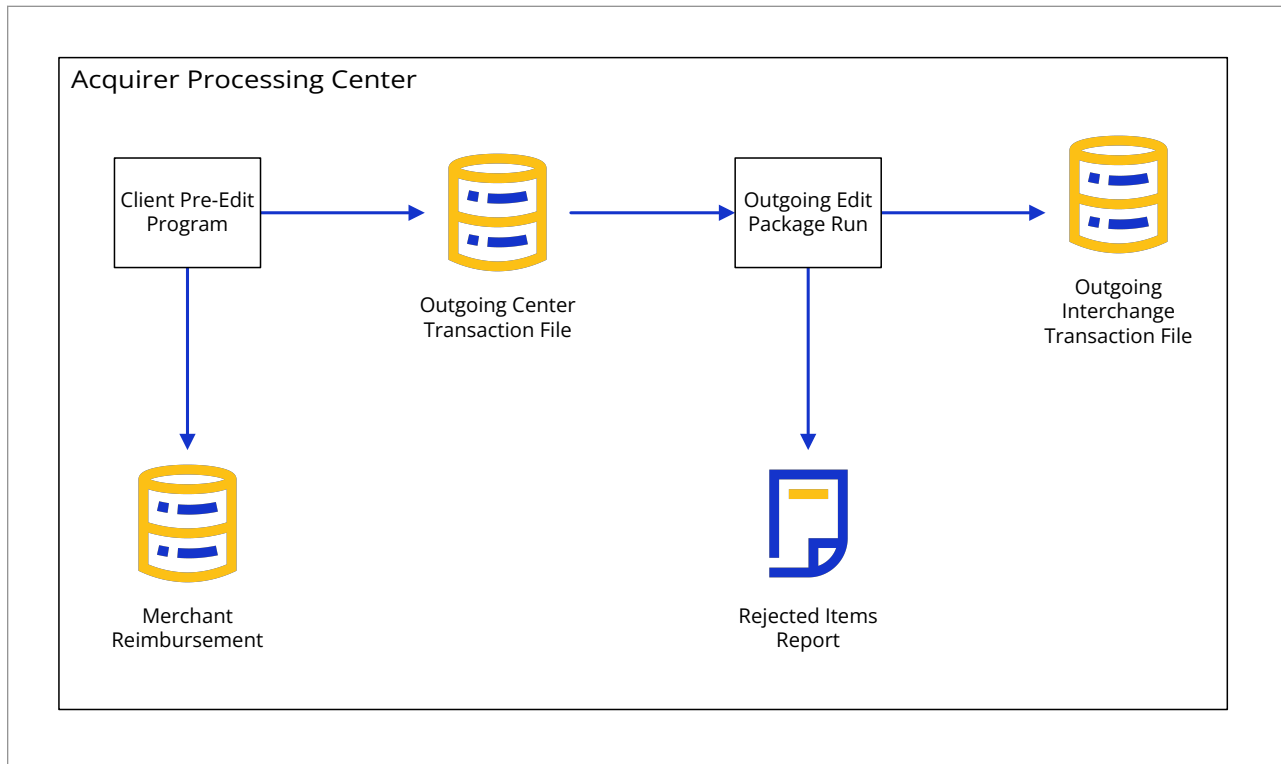
At a client option, Visa can provide transaction-level reconciliation through Clearing and Settlement Advices. These advices are created, one advice per transaction.

BASE II Interchange Transaction File

The acquirer or its processor converts the merchant's transaction data into Visa's format. Some clients and processors run the data through Visa's Edit Package application, while others have

created their own editing applications. The edit application rejects items not suitable for collecting into BASE II.

BASE II Interchange Flow



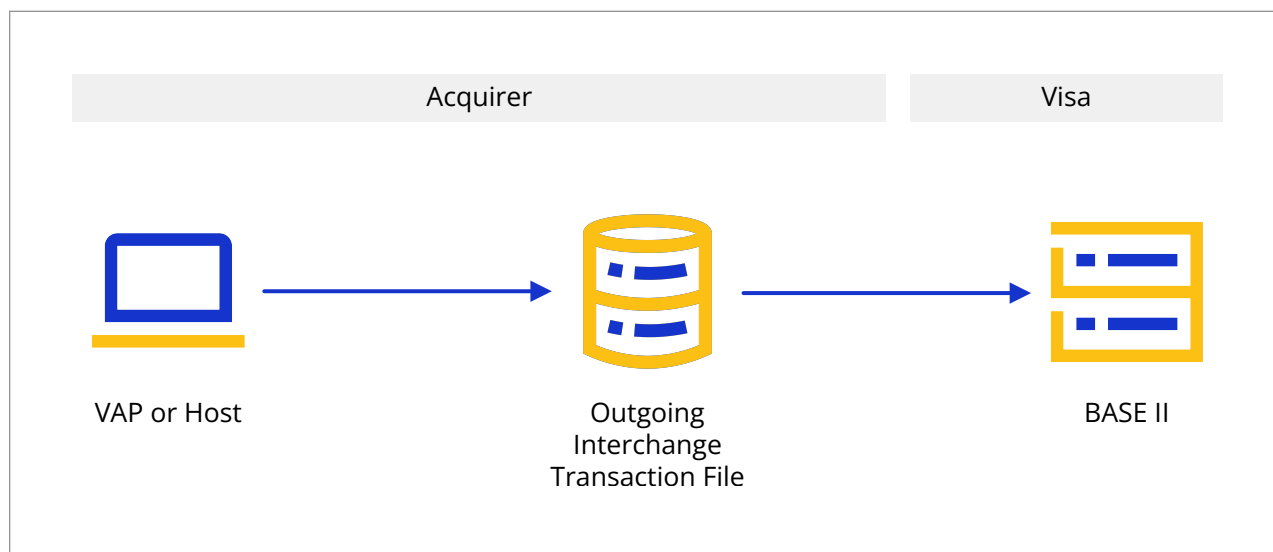
VisaNet Interchange Centers (VIC)

Throughout the day, VisaNet Interchange Centers (VIC) collect and receive interchange files containing transactions from a number of types of connections:

- **VisaNet Access Point (VAP)**—The Application Platform Facility (APF) dials out to the client/processor's access point and collects the files per daily schedule, as specified by the client.
- **Extended Access Server (EAS)**—Extended Access performs similarly to VAP, but includes more advanced security functionality.
- **Visa Direct Exchange (DEX) network**—The client/processor initiates the exchange, sending the files to Visa.
- **VisaNet Integrated Payment (V.I.P.) System**—Generates a record for full financial transactions to the clearing and settlement system.

Collection can be performed at any time specified by the client. BASE II polls the clients/processors to see if they have anything to collect at the specified time.

BASE II Outgoing Interchange from Acquirer



BASE II Interchange Transaction File Processing

BASE II Clearing System receives the Interchange Transaction File and processes the transactions.

When an interchange file reaches BASE II, the records are stripped out independently and processed through the system; by transaction, not by file or batch. Each transaction passes through the clearing system in a matter of milliseconds.

- **Validates Source**—BASE II ensures that it recognizes the parties to the transaction and determines that they have the right to submit interchange.
- **Determines Jurisdiction**—BASE II determines where the transaction took place (domestic, regional, interregional) to establish what rules the transaction falls under and what fees are appropriate.
- **Edits the Transaction**—The data is validated to ensure its integrity (for example, is the card number correct, is the country code valid, etc.)
- **Determines Settlement Service**—The clearing system determines the settlement service to use (international and national, area, and product settlement services), the currency used, and the timing of the settlement. Clients can request a particular settlement service, or allow the system to choose the most appropriate settlement service.

- **Values the Transaction, including Currency Conversion**—The system converts the transaction amount into the possible currencies involved in the transaction and settlement.
- **Calculates the Transaction's Fees and Charges**—Interchange fees are what one client pays another: for purchase transactions, the acquirer pays the issuer, and for ATM/Cash transactions, the issuer pays the acquirer. Interchange fees vary by jurisdiction based on regulations, product, settlement service, network, timeliness of clearing, etc. Daily transaction charges are calculated by Clearing.

VSS Transactions

BASE II Clearing System provides the information that VisaNet Settlement System (VSS) needs for settlement.

BASE II produces these records in the process of clearing a transaction:

- Settlement records for VSS.
- SMS reporting records for transactions that involve a full-service client.

VSS Processing

Visa Settlement System (VSS) uses the individual settlement records to calculate one net position for each funds transfer endpoint in the currency the endpoint has requested.

For example, Globalbank has two endpoints with different currency requirements: Globalbank Europe wants its position in euros, while Globalbank US wants its position in dollars.

VSS produces:

- Reconciliation reports for the issuer and acquirer, summarizing the activity underlying the net settlement amounts.

Note:

Reconciliation reports are also sent to “processing” institutions, which some issuers and acquirers designate to handle settlement for them.

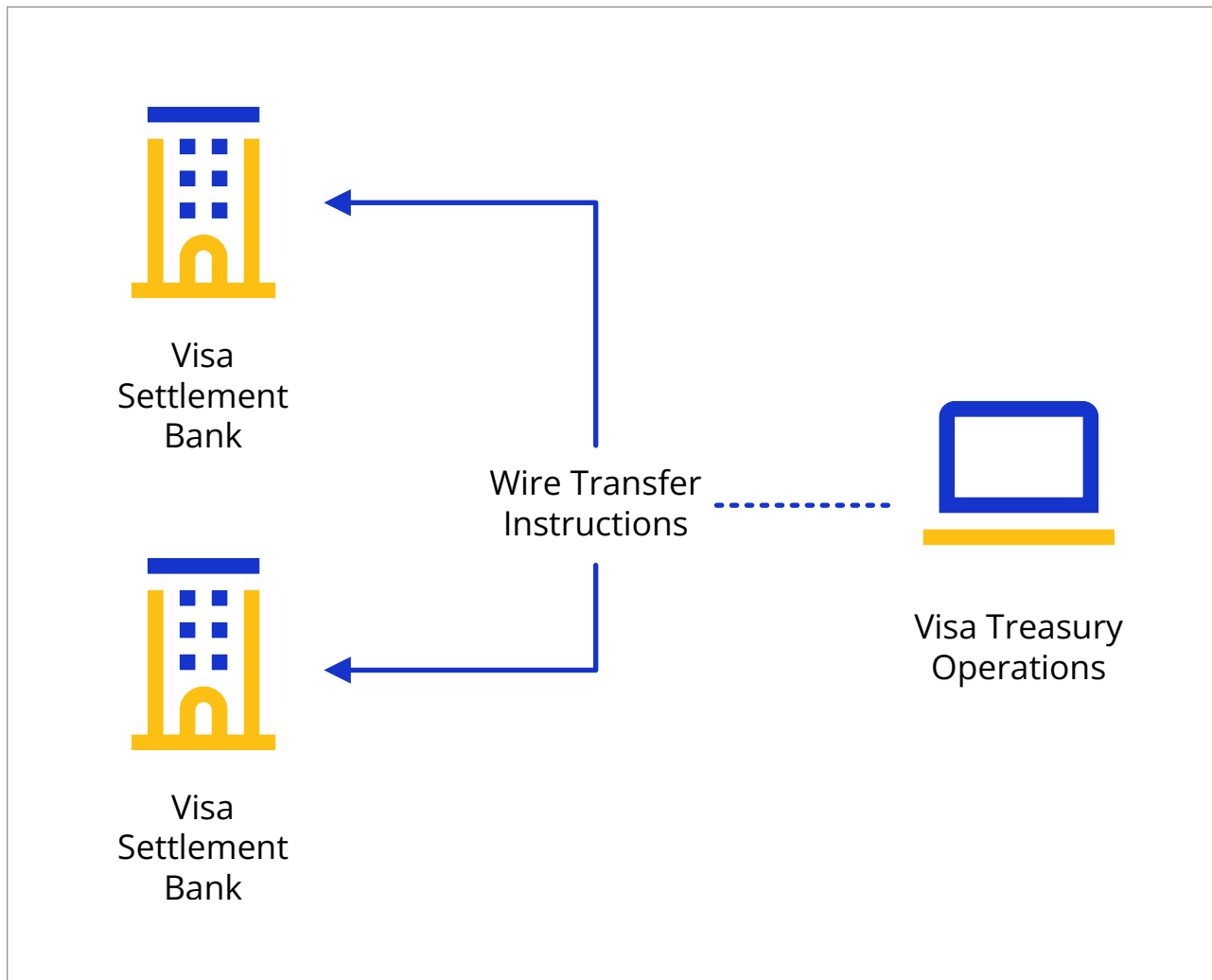
BASE II Visa Treasury Management Department

The Visa Treasury Management Department uses the net position amounts to settle funds for the international settlement service with clients.

Treasury sends funds wire instructions to:

- **Visa settlement banks**—Banks receive settlement wires with funds transfer instructions.
- **Clients that owe money for transactions**—Clients receive requests for funds.
- **Clients that are owed money**—Clients are informed how much they are owed.

Visa Treasury Wires Transfer Instruction to Settlement Banks



Note: Visa supports additional settlement banks depending on the settlement service and financial arrangements

The Visa settlement bank initiates and administers the actual collection of the owed funds:

1. The Visa settlement bank sends the owing client (typically the issuer) a wire, instructing them to pay the net settlement amount:
 - US clients receive their initial instructions via a Federal Reserve wire.
 - Clients outside the US receive a Global wire (aka Swift).
2. The client asks that their bank to wire a specified amount to Visa's settlement account.
3. Visa's Treasury Management Department instructs the settlement banks to pay out the settlement amounts to the clients that are owed money.

Throughout the day, Visa Treasury performs reconciliation and matching the funds requested to the payments. VisaNet Settlement Service (VSS) posts regional fees and charges to the appropriate General Ledger accounts. When wire instructions are sent out, the "value date" on

the wire is the date the client is supposed to pay (1 day for countries that settle using the US dollar, 2 days for countries that settle using other currencies). When a client does not pay on time, Visa will assess late payment fees.

Clearing Settlement Windows

Visa settles with clients per settlement service. Each settlement service has one or more specified times when settlement occurs, referred to as a settlement window. Visa also offers intra-day settlement services for specific conditions.

International Settlement is initiated daily at approximately 03:00 a.m. Pacific time (PT) (10:00 GMT April through October; 11:00 GMT October through April). All client files that have not completed transmission at the time the International Settlement window initiates at approximately 03:00 a.m. PT will be partially processed for inclusion in settlement that day, with the remainder of the file being processed for inclusion in settlement the following calendar day. Clients of other settlement services can designate the time at which these settlement services should close prior to international settlement.

BASE II National Net Settlement

Client banks may opt to participate in National Net Settlement Services (NNSS). VisaNet Settlement Service (VSS) determines their positions and transmits their settlement information at the time they have specified through their settlement agent. Each NNSS/Area Net Settlement Service (ANSS), therefore, has a settlement window time.

Related Information:

[Clearing Settlement Window Closures](#)

Clearing Settlement Window Closures

This table contains the settlement window in which each settlement service participates, listed by settlement service.

Settlement Window Closures

Settlement Service ID	Settlement Service Name	Time of Closure (GMT)
1	INTERNATIONAL SETTLEMENT	10:00/11:00 ¹
3	SPECIAL SERVICE 3	07:00/08:00 ²
4	SPECIAL SERVICE 4	06:00/07:00 ²
5	SPECIAL SERVICE 5	06:00/07:00 ²
10	AUSTRALIA NATIONAL NET	15:00
11	BARBADOS NATIONAL NET	10:00/11:00 ¹
12	BAHAMAS NATIONAL NET	10:00/11:00 ¹
13	BOLIVIA NATIONAL NET	10:00/11:00 ¹

Settlement Window Closures

Settlement Service ID	Settlement Service Name	Time of Closure (GMT)
14	BRAZIL NATIONAL NET ³	10:00/11:00 ¹
15	CANADA NATIONAL NET	10:00/11:00 ¹
16	COSTA RICA NATIONAL NET	10:00/11:00 ¹
17	CZECH REP NATIONAL NET	10:00/11:00 ¹
18	DOMINICAN R NATIONAL NET	10:00/11:00 ¹
19	ECUADOR NATIONAL NET	10:00/11:00 ¹
20	U.K. NATIONAL NET	06:00/07:00 ²
21	GREECE NATIONAL NET ³	10:00/11:00 ¹
22	GUATEMALA NATIONAL NET	10:00/11:00 ¹
23	HUNGARY NATIONAL NET	10:00/11:00 ¹
24	INDONESIA NATIONAL NET	10:00/11:00 ¹
25	INDIA AREA NET	03:30
26	JAMAICA NATIONAL NET	10:00/11:00 ¹
27	JAPAN NATIONAL NET	02:00
28	SOUTH KOREA NATIONAL NET ³	02:00
29	MEXICO NATIONAL NET	09:00 GMT
30	MALAYSIA NATIONAL NET	10:00/11:00 ¹
31	NEW ZEALAND NATIONAL NET	02:00
32	PERU NATIONAL NET	10:00/11:00 ¹
33	PARAGUAY NATIONAL NET	10:00/11:00 ¹
34	SAUDI AR NATIONAL NET	6:00/7:00 ¹
35	SWEDEN NATIONAL NET	07:00/08:00 ²
36	THAILAND NATIONAL NET	23:00
37	TRINIDAD NATIONAL NET	10:00/11:00 ¹
38	VENEZUELA NATIONAL NET	10:00/11:00 ¹
39	UAE NATIONAL NET	02:00
40	OECS AREA NET	10:00/11:00 ¹
41	KENYA NATIONAL NET	02:00
42	MAURITIUS NATIONAL NET	10:00/11:00 ¹
43	SEYCHELLES NATIONAL NET	10:00/11:00 ¹
44	HONG KONG NATIONAL NET	21:00
45	SINGAPORE NATIONAL NET	23:00

Settlement Window Closures

Settlement Service ID	Settlement Service Name	Time of Closure (GMT)
46	ROMANIA NATIONAL NET	10:00/11:00 ¹
49	RUSSIA NATIONAL NET	03:30
50	EGYPT NATIONAL NET	03:30
51	SRI LANKA NATIONAL NET	23:00 GMT ¹
52	POLAND NATIONAL NET	07:00 GMT/06:00 GMT
53	HONG KONG AND CHINA MAINLAND MNSS ³	10:00/11:00 ¹
54	COLOMBIA NNSS	06:00/07:00 ²
55	EL SALVADOR NATIONAL NET	10:00/11:00 ¹
56	HONDURAS NATIONAL NET	10:00/11:00 ¹
57	PAKISTAN NNSS	03:30
58	CHILE NATIONAL NET	10:00/11:00 ¹
59	ZIMBABWE NATIONAL NET ³	10:00/11:00 ¹
64	PHILIPPINES NATIONAL NET	10:00/11:00 ¹
65	BELIZE NATIONAL NET	10:00/11:00 ¹
67	KAZAKHSTAN NATIONAL NET	02:00
69	CROATIA NATIONAL NET	10:00/11:00 ¹
70	NICARAGUA NATIONAL NET	10:00/11:00 ¹
71	CAYMAN IS NATIONAL NET	10:00/11:00 ¹
72	MACEDONIA NATIONAL NET	23:00
73	BRAZIL CASH DISB NATL NET	06:00/08:00 ⁴
74	MACAU NATIONAL NET ³	10:00/11:00 ¹
76	SERBIA NNSS	06:00/07:00 ²
77	BOSNIA & HERZEGOVINA NNSS	06:00 GMT/07:00 GMT ¹
78	BULGARIA NNSS ³	10:00/11:00 ¹
79	BOTSWANA NNSS	03:30 GMT
80	EURO AREA NET	06:00/07:00 ²
81	UKRAINE NNSS	03:30
82	BELARUS NNSS	10:00/11:00 ¹
84	BANGLADESH NATIONAL NET	23:00 ¹
85	SOUTH AFRICA NNSS	06:00/07:00 ²
86	GEORGIA NNSS	06:00/07:00 ²
87	ALBANIA NNSS	06:00 GMT/07:00 GMT ¹

Settlement Window Closures

Settlement Service ID	Settlement Service Name	Time of Closure (GMT)
88	MOZAMBIQUE NNSS	06:00/07:00 ¹
89	GHANA NNSS	10:00/11:00 ¹
90	ZAMBIA NNSS	3:30 ¹
91	TANZANIA NNSS	10:00/11:00 ¹
92	NIGERIA NNSS	02:00
93	VIETNAM NATIONAL NET	02:00
94	UGANDA NNSS	10:00/11:00 ¹
95	UZBEKISTAN NNSS	03:30 GMT ¹
102	ARMENIA NNSS ³	10:00/11:00 ¹
103	LEBANON USD NNSS ³	06:00/07:00 ²
104	LEBANON LBP NNSS	06:00/07:00 ²
106	MONTENEGRO NNSS ³	10:00/11:00 ¹
107	AZERBAIJAN NNSS	03:30
109	MOLDOVA NNSS	03:30
110	MALAWI NNSS	10:00/11:00 ¹
112	QATAR NNSS	02:00
116	KYRGYZSTAN NNSS	02:00
119	TAJIKISTAN NNSS	10:00/11:00 ¹
120	ANGOLA NNSS ³	03:30
122	TURKEY NNSS ³	06:00/07:00 ²
124	BRAZIL REDECARD NNSS	10:00/11:00 ¹
125	BRAZIL CIP NNSS	06:00/08:00 ⁴
126	RWANDA NNSS	10:00/11:00 ¹
127	BRAZIL CDP NNSS ³	10:00/11:00 ¹
128	GUYANA NNSS	10:00/11:00 ¹
129	MOROCCO NNSS ³	06:00/07:00 ²
131	ICELANDIC NATIONAL NET	07:00
132	MYANMAR NNSS	23:00
133	HAITI NNSS	10:00/11:00 ¹
134	COLOMBIA VCSS NNSS	06:00/07:00 ²
135	DUTCH CARIBBEAN ANSS	10:00/11:00 ¹
136	NEPAL NNSS	03:30

Settlement Window Closures

Settlement Service ID	Settlement Service Name	Time of Closure (GMT)
137	ARGENTINA NNSS	10:00/11:00 ¹
138	CENTRAL AFRICAN AREANET ³	06:00/07:00 ²
139	CAMBODIA NNSS	02:00
140	DENMARK NNSS	07:00/08:00 ²
141	MALDIVES NNSS	02:00
142	ARUBA NNSS ³	10:00/11:00 ¹
143	NORWAY NNSS	07:00/08:00 ²
144	URUGUAY UYU NNSS	10:00/11:00 ¹
145	URUGUAY USD NNSS	10:00/11:00 ¹
146	CAMBODIA (GOV) NNSS	02:00
147	PAPUA NEW GUINEA NNSS	19:00
148	JORDAN NNSS	03:30
149	FIJI NNSS	19:00
150	IRAQ NNSS ³	03:30
151	ARGENTINA NNSS USD	10:00/11:00
152	BOLIVIA NNSS	10:00/11:00

1 Adjusts with Pacific daylight savings time.

2 Adjusts with Central Europe daylight savings time.

3 Inactive as of this publication's effective date.

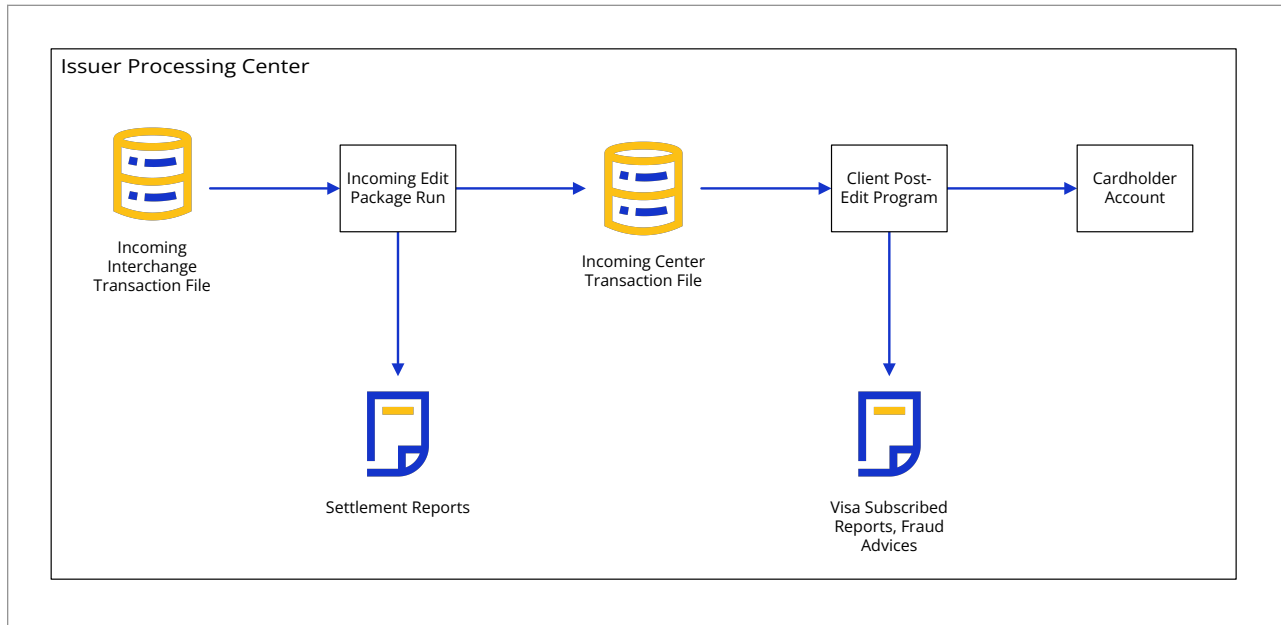
4 Adjusts with Brazil daylight savings time.

Client files containing transactions for a given window not received by Visa at the time of Settlement Window closure will not be processed until the next Central Processing Date (CPD).

BASE II Issuer Transaction Cycle

The issuer's incoming Interchange Transaction File (ITF) is fed through the Incoming Edit Package Run from which clients can extract their own settlement reports. The incoming ITF, after editing, becomes the incoming Center Transaction File (CTF), which is fed into the issuer's post-edit program, for the posting of financial transactions to the cardholder account, as well as other non-financial information, such as fraud advices or Visa subscribed reports.

Issuer Processing Flow



The issuer or its processor reviews the clearing batches, removes the hold from the cardholder's funds, and posts the amount of the transaction to the cardholder's account.

At the end of each monthly billing cycle, the issuer prepares a statement for the cardholder, listing all activity on the account during the billing period. When the cardholder pays his or her bill to the issuer, the transaction cycle is complete.

Chapter 3

BASE II Clearing System Edit Package Service

BASE II Clearing System Edit Package Overview

Clients use the Edit Package service to prepare interchange data files for the BASE II Clearing System. Edit Package also receives and reports on incoming interchange from Visa, preparing it for the processing center's Host system.

BASE II Clearing System Edit Package Description

Edit Package is software supplied by Visa to endpoints worldwide. It is operated by processing centers to prepare their interchange data files and to verify the integrity of the files sent to Visa.

Note: An endpoint is a client, merchant, or third-party processor connected to VisaNet.

The Edit Package is the interface between the endpoints' internal payment processing systems and the file transfer process used to interchange files with BASE II. The Edit Package software runs on mainframe and Windows-based PC. Edit Package also receives and reports on incoming interchange from Visa.

BASE II Clearing System Edit Package Benefits

The Edit Package Service offers several benefits.

- **Improved system reliability**—Increases the reliability of the interface between the processing center and the BASE II Clearing System.
- **Greater data integrity**—Ensures the integrity of transactions and information submitted to Visa.
- **Improved processing accuracy**—Pre-edits transactions before being sent to Visa. Invalid or erroneous data is detected before sending the transactions to VisaNet.
- **Enhanced operations**—Provides reports and reconciliation of interchange activity, as well as extracts of reports and tables.

Edit Package Features

Edit Package has several key features which provide maximum flexibility for both installation and daily processing to all supported platforms.

Features include:

- Data extracts
- Run control options
- Error, audit, reconciliation and summary reports
- Utilities

PC Edit Package for Windows has additional features and components:

- Data Entry
- File Transfer
- Application Configuration
- Batch Facility

Edit Package Data Extracts

Edit Package data extracts provide access and extracts of Visa reference data such as: Routing Table, Currency Rate, ARDEF, VID, etc.

Edit Package Run Control Options

Access to most functions in the Edit Package is available through run control options. The use of run control options allows processing centers to customize the Edit Package in a way best suited to their specific processing needs. Run control options can be modified at installation time or before each Edit Package processing run.

Run control options offer a variety of functions, including:

- Setting tables' effective date to test future business enhancements.
- Establishing duplicate batch detection settings.
- Requesting automatic table and software updates to keep the Edit Package system current.
- Splitting incoming Interchange Transaction Files (ITFs) into multiple Center Transaction Files (CTFs.)

Edit Package Reports

Edit Package reports information about various Edit Package processes.

- **Outgoing Edit Package Reports**—Provides the reconciliation and summary information obtained from outgoing Edit Package processing. A report on system activity during outgoing runs is also included.

Examples of outgoing reports are EP-100A through EP-199.

- **Incoming Edit Package Reports**—Provides the reconciliation and summary information obtained from incoming Edit Package processing. A report on system activity during incoming runs is also included.

Examples of incoming reports are EP-200A through EP-299.

- **On-Request Reports**—Clients can request the information stored in the Edit Package files:

- ACQ/ISS Identifier Table (can be referred to as the Visa Identifier (VID) Table)
- Account Range Table
- Value Tables
- Processing Log

Examples of on-request reports are EP-300 through EP-399.

- **Optional Transaction-Based Reports**—Contains displays of record data for specific transactions.

Examples of transaction-based reports are EP-701 through EP-758.

See *BASE II Edit Package (Release 4.0) Reports*.

Edit Package Utilities

BASE II Clearing System uses several support utilities to assist clients during installation, application configuration, and system administration.

Utilities are used to initialize key files, generate default run control options, apply table updates, load, recover and synchronize table-related files. They can also be used to extract information from Edit Package value tables, such as the Country, Currency, MCC, and ARDEF information, as well as generate on-request reports.

PC Edit Package for Windows Features

PC Edit Package for Windows includes a number of features that make information transfer reliable and convenient.

Edit Package Data Entry

Clients can use Edit Package data entry to cut and paste transactions between batches, employ default field values, and employ user-specific and BASE II CIB-specific data entry options.

- Allows correction of Edit Package rejects, BASE II returns, and entries of new transactions.
- Specification File Updates, which are usually enhanced twice a year during each business release, allow easy maintenance of Data Entry screens.
- Allows multiple users to perform data entry for each BASE II CIB. Locking mechanisms ensure data integrity and a LAN monitor window tracks user activity.
- Data Entry Date override feature allows users to test future Edit Package changes without modifying the computer system date.

Edit Package File Transfer

Clients can use the Edit Package File Transfer Interface to exchange outgoing and incoming ITFs with VisaNet.

PC Edit Package Application Configuration

PC Edit Package combines the administrative functions into a user-friendly interface that offers a number of options to customize the Edit Package.

The system includes user interface to define:

- Processors (CIBs)
- Operators
- Interfaces (VAP, EAS, or DEX)

- Batch Facility
- Data Entry

PC Edit Package Batch Facility

PC Edit Package includes a wizard to aid the set up of key processes, such as Outgoing, Incoming, and File transfer in batch mode.

The system also provides a *.bat command file that can be initiated through the Windows scheduler or through client-customized procedures. It may also be submitted within the Batch Facility wizard.

Edit Package Process Flow

Outgoing and incoming processing is performed by Edit Package for all supported platforms. These key processes perform specific functions for each run.

Edit Package provides software used by processing centers to process daily Visa transaction files on a mainframe or PC. These Visa transaction files are communicated from a processing center's internal payment processing systems to the BASE II System via the VAP, EAS, or DEX.

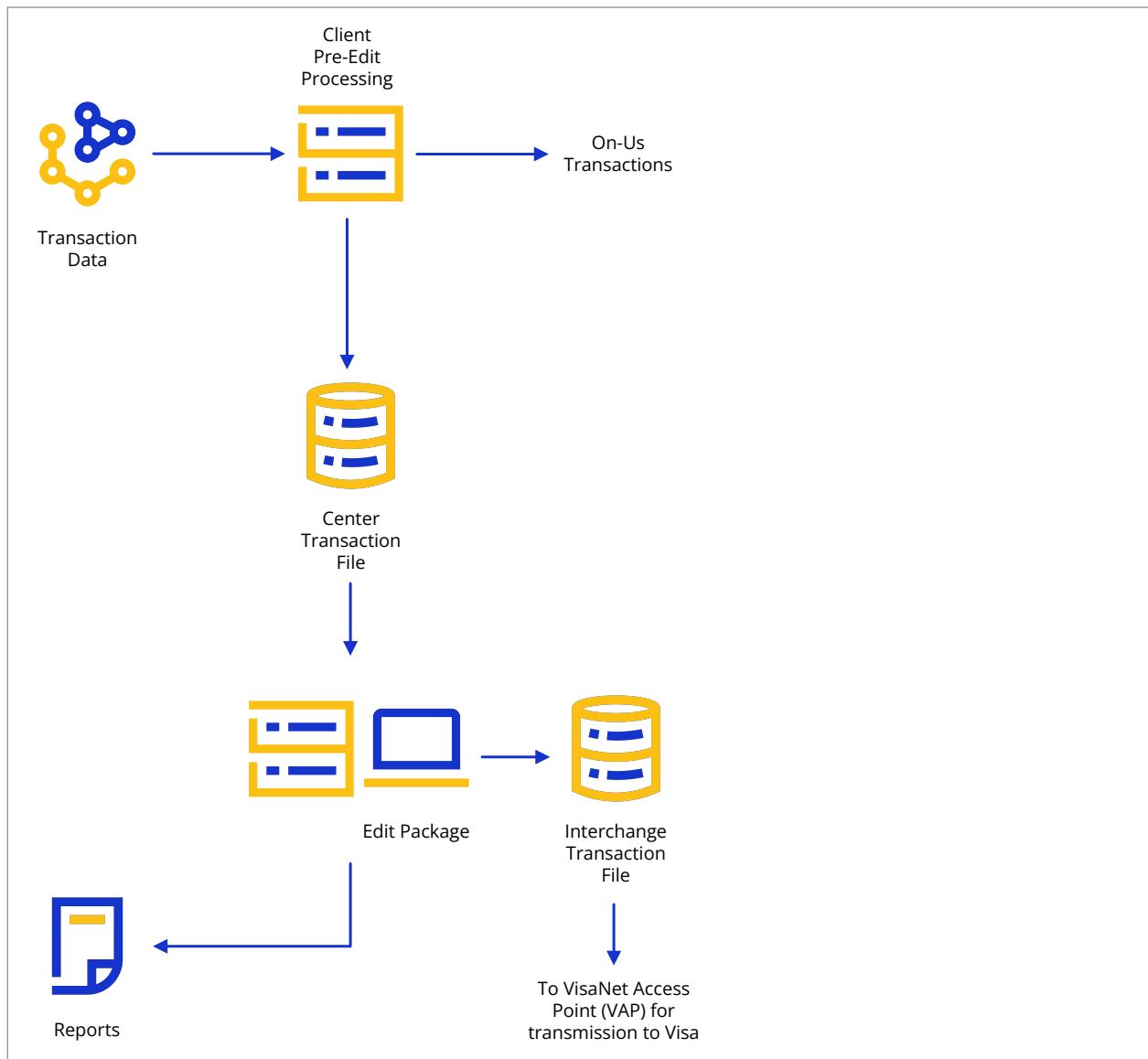
Edit Package Outgoing Process

The outgoing Edit Package is a single-step process controlled by an overall program that calls many sub-modules.

The E3OUT outgoing control program performs these functions:

- Calls initialization functions, as needed, at the beginning of each run, which includes searching and validating run control options, loading of internal tables, and assigning run and starting batch numbers.
- Processes each transaction found in the CTF by checking the optional file header, building logical transactions, validating logical transactions using the rules and value tables, and verifying all batch and file trailers balance.
- Performs any optional processing, such as calculating hash totals if duplicate batch checking via hashing is selected and saving rejected transactions if center has selected SAVEREJECT run control option.
- At the end of processing, if there are no severe errors, the control program produces the necessary reports and makes the ITF available for staging on the VAP, EAS, or DEX.

Edit Package Outgoing Interchange



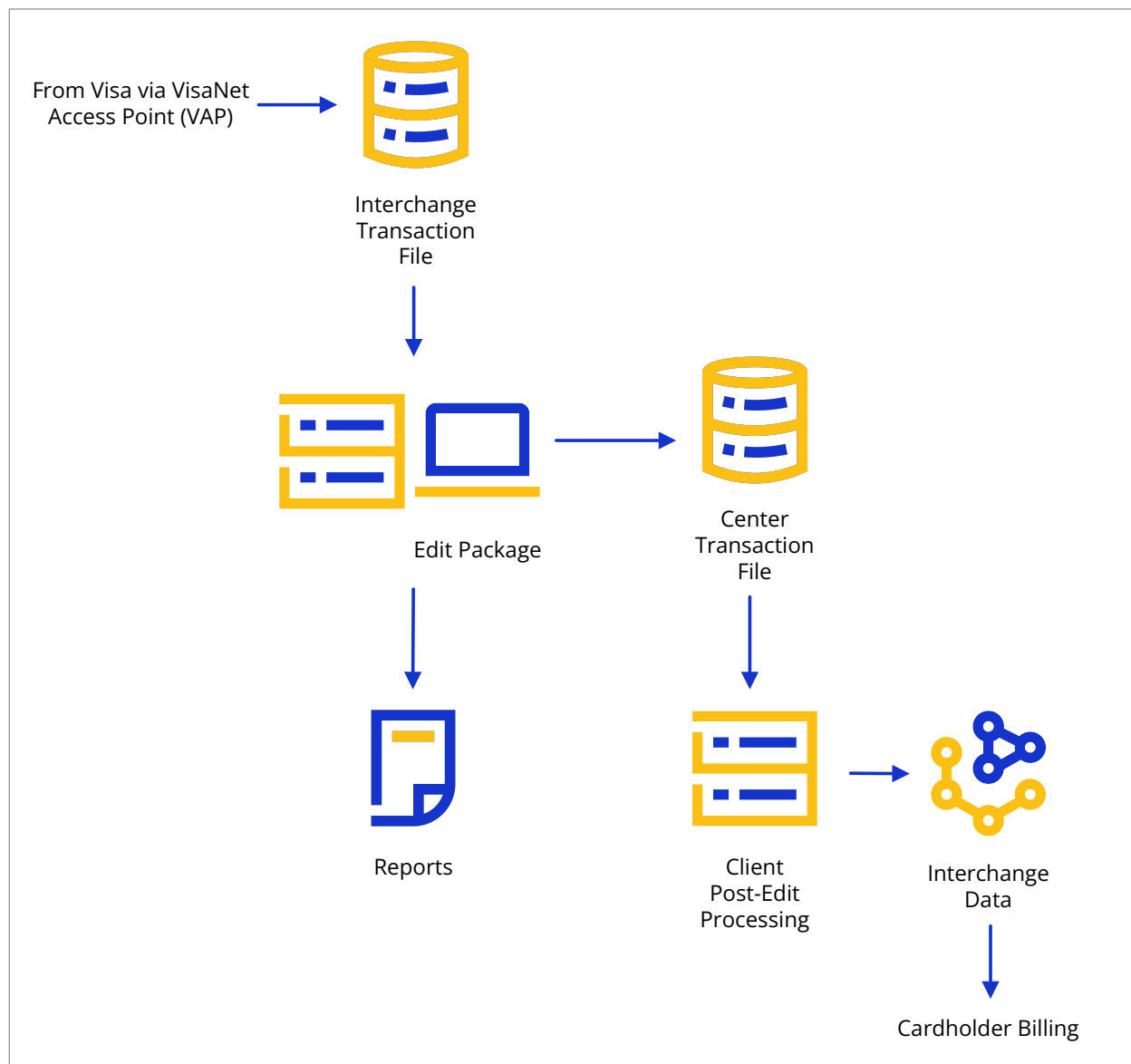
Edit Package Incoming Process

The incoming Edit Package is a single-step process controlled by an overall program that calls many submodules.

The E3IN incoming control program performs these functions:

- Calls initialization functions, as needed, at the beginning of each run, which includes searching and validating run control options, loading of internal tables and assigning a run number:
- Processes each transaction found in the ITF by checking the file header, building logical transactions, validating the logical transactions using the rule and value tables, formatting and extracting the transaction for reporting, validating the incoming file trailer and verifying that batch and file trailers balance.
- Performs any optional processing based on run control options specified. Examples are creating multiple CTFs and saving returned items to a separate file.
- During processing, Edit Package will update/create these files:
 - Optional TC33
 - Currency Conversion Rates File
 - Control Tables
 - History files
 - Environmental Run Information
 - Engine Table Usage Metric File
- At the end of successful processing, the control program produces the necessary reports and makes the CTF available for use by the processing center.

Edit Package Incoming Interchange

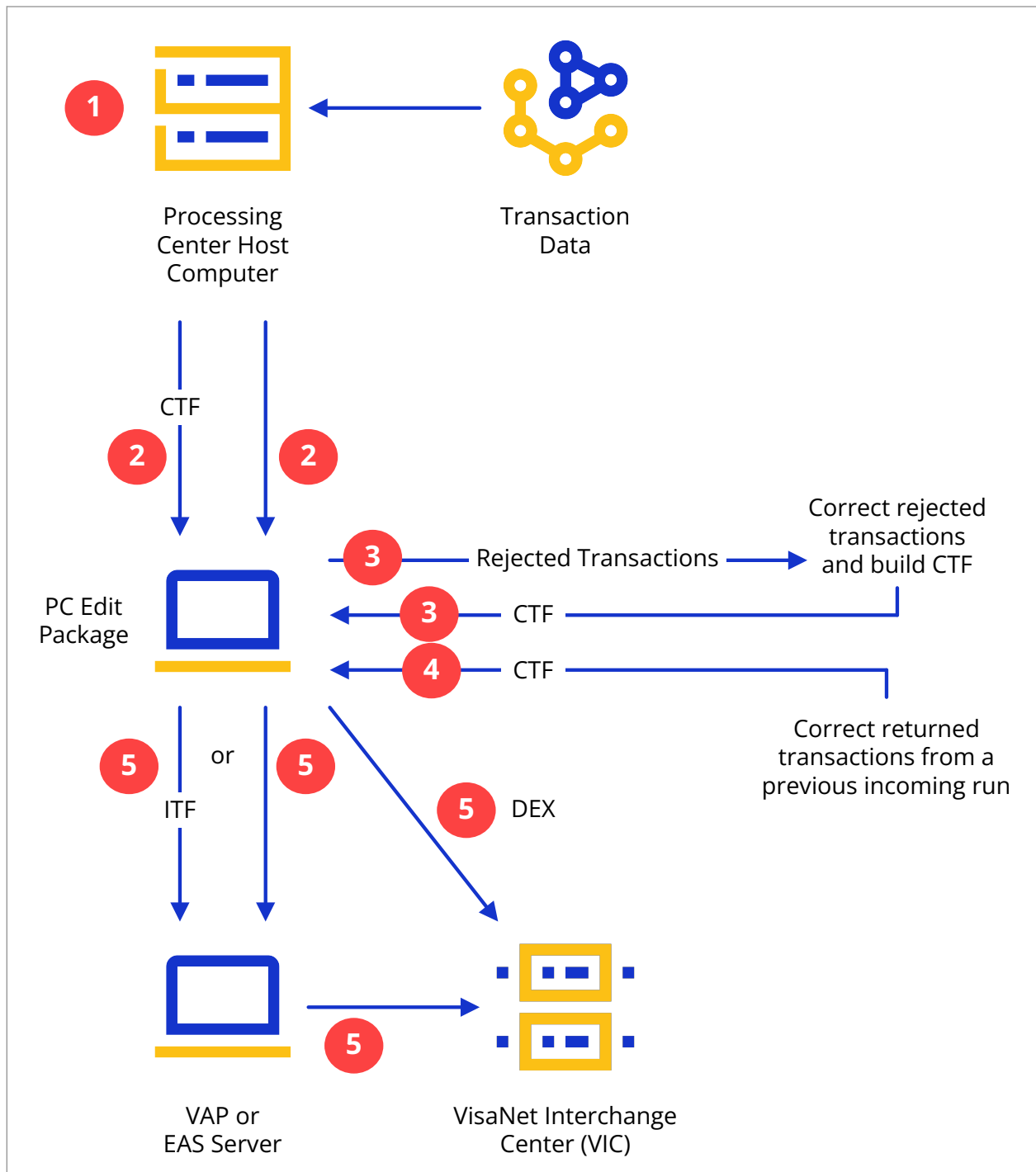


VIC Outgoing Process

The term *outgoing* refers to interchange sent from the client's processing center to the VisaNet Interchange Center (VIC).

This graphic shows the flow of electronic data from the client's processing center to the VIC.

Outgoing Interchange Process



- The processing center's pre-edit software formats the transaction data into a CTF.
- The client's computer transfers the CTF to Edit Package, which validates the information and ensures that it is free of common errors.
- If a transaction has data errors, it fails Edit Package validation and is placed in a reject batch on disk to be corrected.
 - PC Edit Package for Windows

After correcting each transaction, build a CTF and re-send it through PC Edit Package for Windows's validation procedure.
 - Mainframe Edit Package

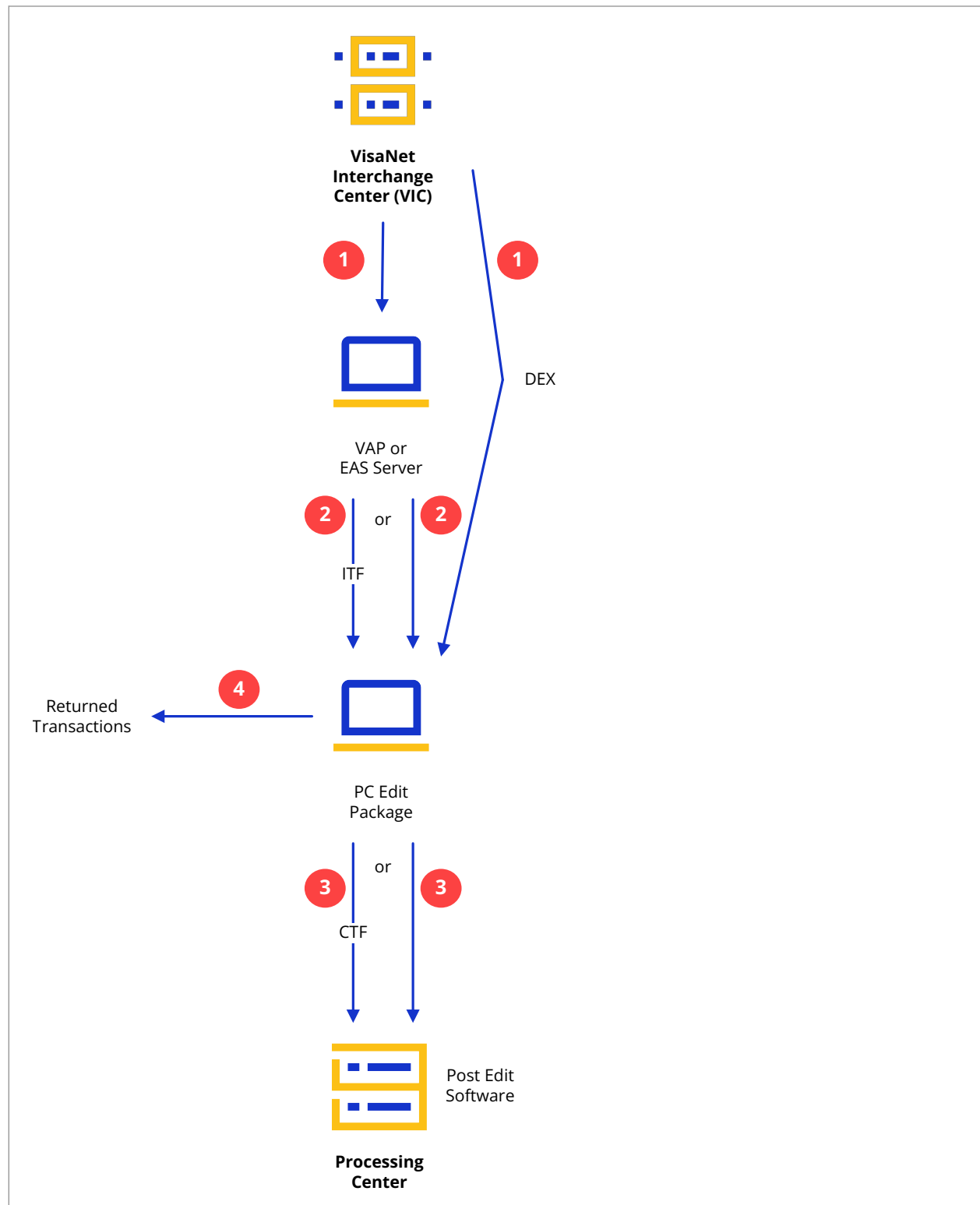
Rejected item files created from the mainframe Edit Package processing can be optionally downloaded into the PC Edit Package for Windows Data Entry for correction. After correcting each transaction, build CTF, upload CTF to the mainframe and send it through the mainframe Edit Package outgoing processing.
- Similarly, If there were any returned transactions from a previous incoming run, correct them and include them in a CTF to go through the validation procedure.
- Once the transactions have been validated, Edit Package builds an ITF. PC Edit Package for Windows or Visa File transfer or client file transfer software sends the ITF electronically to a VAP, EAS, or DEX. The VAP, EAS, or DEX transmits the ITF to the VIC.

VIC Incoming Process

The term *incoming* refers to interchange coming into the client's processing center from the VisaNet Interchange Center (VIC).

This graphic illustrates the steps in an incoming interchange process.

Incoming Interchange Process



- The VIC sends clearing transactions in ITF format to the VAP, EAS, or DEX.
 - PC Edit Package for Windows or Visa File transfer or the processing center's software electronically transfers the ITF from VAP or EAS. For the DEX interface, PC Edit Package retrieves the ITF from the DEX server for processing.
- Edit Package validates the transactions and converts them to CTF format.
- The CTF is processed by the processing center's post-edit software.
 - Returned items are corrected using the PC Edit Package for Windows and validated during Outgoing processing.

Edit Package Requirements

This includes system requirements for Edit Package and PC Edit Package for Windows.

PC Edit Package Supported Windows Operating Systems

For information about supported Windows operating systems for Edit Package, contact your Visa representative.

See *BASE II Clearing Edit Package Operations Guide Release 4* and *BASE II Clearing PC Edit Package for Windows User Guide Release 4* for more detail.

Processing Center Certification

All processing centers, regardless of platform, are required to follow BASE II Clearing System certification procedures.

- New processing centers must be certified when installing Edit Package software for the first time.
- Centers must be certified when installing a new Edit Package software version.

Certification Process

Certification is arranged through the processing center's Visa representative. All necessary information for certification and implementation is provided by Visa.

The certification process must include tests for:

- Processing center pre-edit software compatibility with the Edit Package
- Edit Package data exchange with BASE II
- Processing center post-edit software compatibility with Edit Package

Test data is created by the processing center, Visa, or both.

Once certification procedures are executed, the processing center must verify test results. This summarizes the Edit Package certification steps.

1. Determine if certification is required.
2. Certify the Edit Package.
 - a. Verify pre-edit software
 - b. Exchange data with BASE II
 - c. Verify post-edit software
 - d. Reinitialize the Edit Package

Edit Package Service Documentation

These documents are available.

- *BASE II Clearing Edit Package Operations Guide Release 4*
- *BASE II Clearing Edit Package Messages Reference Guide Release 4*
- *BASE II Clearing Edit Package Reports Reference Guide Release 4*
- *BASE II Clearing PC Edit Package for Windows User Guide Release 4*
- *BASE II Clearing Edit Package Run Control Options Quick Reference Release 4*

Chapter 4

BASE II Endpoint Alternative Transaction File Processing Option

Visa offers this alternate file transaction processing option to endpoints that do not wish to use the Visa Edit Package to build BASE II outgoing files and process BASE II incoming files.

Please refer to your Visa Representative for full details.

Chapter 5

BASE II File Collection and Delivery Services

BASE II Clearing System File Collection and Delivery Service uses client preferences to collect outgoing interchange files and delivers the incoming interchange and reports to the destination.

BASE II File Collection and Delivery Services Overview

BASE II Clearing System uses collection times to connect with a client's VisaNet Access Point (VAP) or Visa Extended Access Server (EA Server). Other connection options for interchange (DEX and internal Visa processors) do not require collection times.

BASE II automatically delivers files post-settlement; however, clients can choose among several delivery options that allow for the scheduling of extra deliveries in addition to the end-of-cycle files. These options provide faster access to files in a sequence that best meets a client's needs.

BASE II allows clients to choose how information is received, package critical data separately, designate certain files for early or customized delivery, and route specific transactions to alternate locations for faster processing.

File collection and delivery options are available to all clients and their processors in all Visa regions.

BASE II File Collection and Delivery Services Description

File Collection and Delivery Service collects and receives interchange files and delivers transactions and information to processors.

BASE II supports file collection 24 by 7, 365 days a year. BASE II file delivery has many client delivery options by volume, by time, or by event. Clients can specify the type of data they require in a specific file for customized, split routing, and expedited file delivery. Split Routing determines who will process the data, while Custom Delivery is how the data will be received. A collection option supports multiple files to be staged in a given day.

BASE II File Collection

Throughout the day, Visa collects and receives interchange files containing transactions from these sources:

- VisaNet Access Point (VAP)—The Application Platform Facility (APF) dials out to the client's or processor's VAP and collects the files per a daily schedule.
- Extended Access Server (EA Server)—The EA Server performs similarly to the VAP, but includes more advanced security functionality.
- Direct Exchange (also known as Visa Gateway)—Processors initiate the exchange, sending the files to Visa.

Processors who have VAPs or EA Server connections can establish times when they want the system to poll their files for processing. Processors who use DEX can send their files to Visa at any time.

There are two collection options: standard and split collect.

BASE II Standard File Collection

Visa collects files from clients' VAPs or EA Servers based on the day and hour set in the processor's configuration.

Processors have the option of sending outgoing interchange multiple times during the cycle. Splitting transaction data into multiple files that are collected at different times enables processors to better balance their workload, facilitate global processes, and speed settlement.

Processors that use Direct Exchange internally control the day and times the files are sent to Visa.

BASE II Split Collect File Collection

Split Collect (SC) allows a client to have a file collected from a source other than its usual processor. For example, a client can have an alternate processor process its back-office transactions.

Split Collect is a file collection service.

Related Information:

[BASE II File Delivery Options](#)

BASE II File Delivery

Processors can choose to have their transaction data delivered before settlement is completed or to have specific types of data delivered in advance of settlement.

Normally, transaction data and settlement reports are delivered simultaneously. However, a client can choose to have data in advance, for example, in order to balance the workload, determine whether any transactions were rejected in a file, or package data critical to a client's processing.

Clients can define how they want their files delivered based on time, day, volume, and event. In addition, delivery can be initiated by request to Visa. Clients can also customize the delivery of BASE II files based on a list of predefined file types. Some of those files can be expedited for immediate delivery.

At the end of the processing cycle, Visa delivers any pending data at the close of the International Settlement window, which is 03:00 PT (10 or 11 GMT, depending on U.S. daylight savings time).

BASE II File Delivery Options

BASE II provides clients with multiple customized options to receive their files.

These options enable clients to schedule file delivery at a preselected time or when the client's interchange has reached a certain volume.

Clients can select any combination of these options. Normal processing occurs when the settlement window for that service closes. Clients have options to receive files in a variety of ways.

- Time

Processors can receive their files according to a predetermined schedule that best meets their needs and processing schedule. For example, a client can request that all data processed so far be sent at 15:00, 17:00, etc.

- Volume

Clients can request the delivery of data based on the volume processed. For example, deliver a file as soon as 1,000,000 transactions are completed.

- Event

- Operations command

Visa allows processors to call with a request to deliver a file immediately. The VisaNet operator will initiate delivery of available data. This does not override the delivery option for that file—any delivery options established by the client will remain in effect.

Clients have a number of options to choose how the data in their files are presented and delivered from BASE II. This allows clients to customize data to meet their own processing needs.

BASE II Customized File Delivery Description

Clients can facilitate their internal processing by requesting Visa to categorize data in custom files.

For example, a client receives all of their POS sales drafts in one file and has all ATM credit card transactions separated from ATM debit card transactions. In addition to this type of customization, clients can add customization of dispute transactions and have it routed to an alternate destination.

This streamlines client processes customized to client needs, improves delivery speed, and allows clients to prioritize their processing.

BASE II Customized File Delivery Options

Customized File Delivery provides a number of delivery options.

This includes expedited delivery. This option provides immediate delivery of selected customized delivery files, regardless of the normal delivery schedule. Expedited delivery allows processors to receive selected files immediately after a file is completed.

BASE II Split Routing Description

Split Routing enables BASE II processors to select alternate BASE II file delivery locations for certain transaction types. Clients can use these routing options individually or in combination.

A VSS relationship is created based on the routing option for the processor to which they are routing.

BASE II Split Routing Process Flow

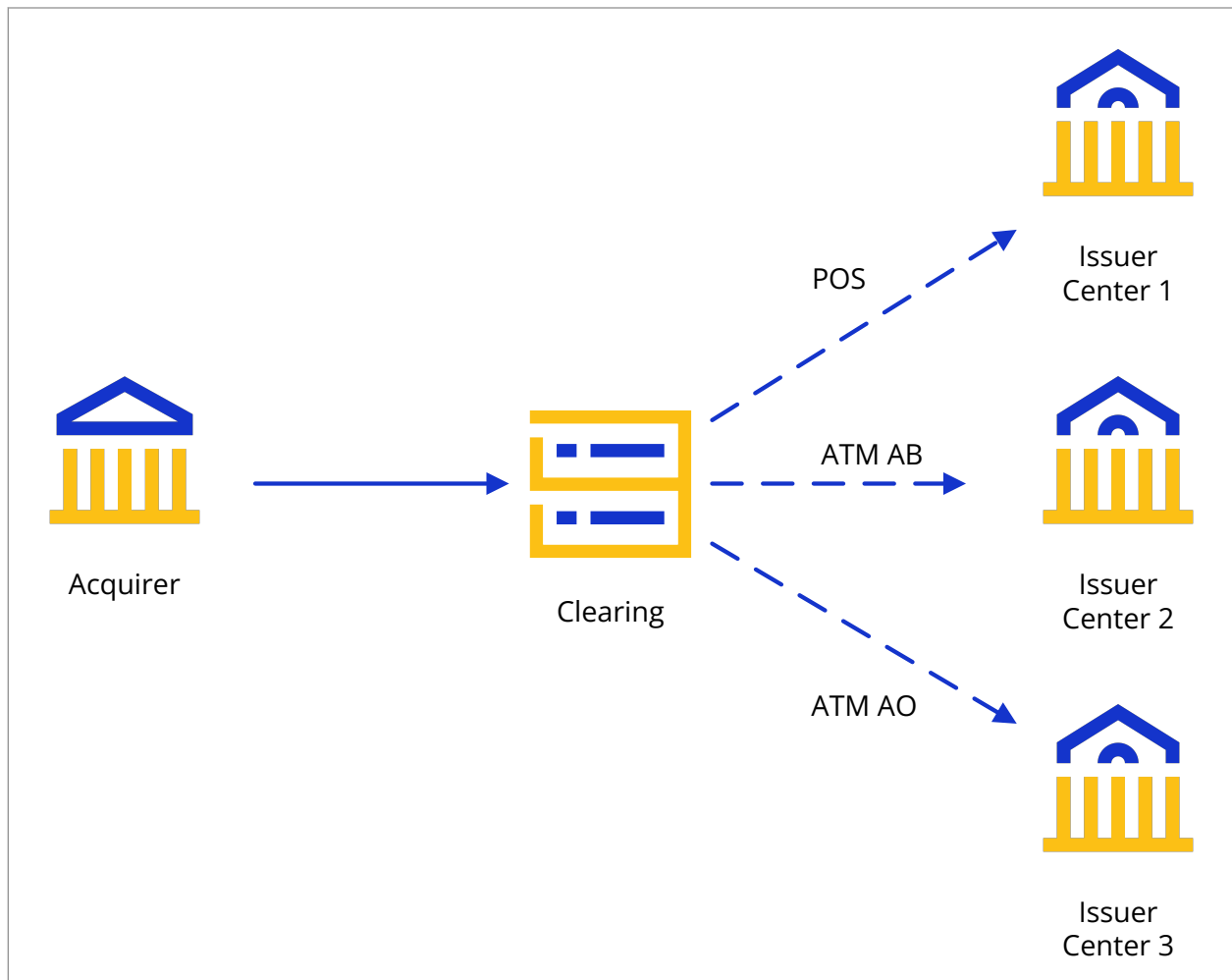
Split Routing codes can be set up from the Clearing Business or Account Range.

Where two split routing options have overlapping selection criteria, the routing option with the higher priority is used to route the transactions qualifying for both options.

Example: Both options set at BASE II CIB level

The ATM Originated (AO) file type selects all ATM transactions—both issuer and acquirer, original and back-office—and it has a selection sequence number. The ATM Back-Office (AB) split routing code selects only back-office ATM transactions, but has a higher selection sequence number than the AO file types. Issuer Center 1 has been set with AO routing to Issuer Center 3 and AB routing to Issuer Center 2.

Example Illustration



BASE II Custom Delivery and Split Routing File Types

BASE II Clearing processors may select customized delivery file types as needed to meet their requirements (e.g., financial, fraud, and back-office). This table contains the file types used for customized file delivery.

Note: BASE II uses consolidated file delivery to place all of the data that is undifferentiated (data left over after customization) from the VisaNet Interchange Centers (VICs) in a single logical file with a file type code of IC.

Custom Delivery and Split Routing File Types

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
ACKNOWLEDGEMENT		MA AK				Retired.
Acquirer Services Reporting Advices	ASRA	AS	Yes	Yes	No	Acquirer services reporting
Assessment — Generic		AG				Retired.
ATM Back-Office	ATMBO	AB	Yes	No	Yes	Back-office ATM transactions; dispute financials, dispute response financials, and their reversals.
ATM Back-Office Credit	ATMBC	BC	Yes	No	Yes	Back-office ATM transactions where the credit account was selected.
ATM Back-Office Debit	ATMBD	BD	Yes	No	Yes	Back-office ATM transactions where the savings or checking account was selected.
ATM Format Conversion: ALL	ATMAF	AF	No	No	Yes	Used for ATM Format Conversion Service.
ATM Format Conversion: Credit	ATMFC	FC	No	No	Yes	Used for ATM Format Conversion Service where credit account is selected.
ATM Format Conversion: Debit	ATMFD	FD	No	No	Yes	Used for ATM Format Conversion Service where debit account is selected.
ATM Originated	ATMOR	AO	Yes	No	Yes	Original and back-office ATM transactions, regardless of account type.
ATM Originated Credit	ATMOC	AC	Yes	No	Yes	Original and back-office ATM transactions where the credit account was selected.

Custom Delivery and Split Routing File Types

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
ATM Originated Debit	ATMOD	AD	Yes	No	Yes	Original and back-office ATM transactions where the savings or checking account was selected.
Back-Office	BAKOF	BK	Yes	No	Yes	All back-office transactions, both POS and ATM.
BASE I Advices	B1ADV	B1	Yes	No	No	Bulk file delivery of authorization Stand-in Advices
BASE II Returns	RTRN	RT	Yes	Yes	No	Submitted transactions that failed clearing edits.
BASE I TC33 Report	B1T33	BA	Yes	Yes	No	Specialized authorization reporting, such as special airline data.
CAPTURE		MC				Retired.
CAST33 NN/AN ADV	T33NN	D1	Yes	Yes	Yes	Clearing and Settlement TC33 Advice – Allows Clearing and Settlement (CAS) TC33 advices for NNSS and ANSS transactions only to be sent in a separate delivery file. This will include transactions settled in a national or area net. Transactions will be combined if the same settlement window applies to both the national and area net.

Custom Delivery and Split Routing File Types

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
CAS TC33 ADV ISS	T33IS	D2	Yes	Yes	Yes	Clearing and Settlement TC33 Advice – Allows Clearing and Settlement (CAS) TC33 advices for international transactions only to be sent in a separate delivery file. This will include transactions settled in the international settlement service. This can apply to domestic transactions settled in international settlement service.
CDM ACQUIRER	CDMAQ	CA	No	No	Yes	VDAS TC38 Flexible Routing options for Acquirers.
CDM ISSUER	CDMIS	CI	No	No	Yes	VDAS TC38 Flexible Routing options for Issuers.
Chargeback Reversals	CBREV	CB	Yes	Yes	Yes	Used when an issuer wants to send dispute financials and their reversals and receive dispute response financials and their reversals and receive dispute financials.
Clearing Data Capture	CDC	CD	Yes	Yes	No	Data capture advices.

Custom Delivery and Split Routing File Types

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
Clearing Settlement Advice	CAS33	CS	Yes	Yes	Yes	TC33 Clearing and Settlement Advice – Allows Clearing and Settlement (CAS) TC33 Advices for ISS, NNSS, and ANSS transactions to be sent in a separate delivery file. This file will include transactions settled in an international and national or area net settlement service.
Combo Card	COMBO	CM	No	No	Yes	Reserved for Brazil.
Currency Rates	CURR	CU	Yes	Yes	No	Daily (M-F) currency trading rates.
Debit Raw Data	DBRAW	RW	Yes	Yes	No	Full Service detailed transaction data in a machine-readable format.
Debit Reports	DBRPT	DR	Yes	Yes	No	SMS detail reports.
Delta Balance File		DB				UK Delta Service—Retired
Detokenization	TKEXA	TA	Yes	Yes	No	Support new TC33 Token Exchange Activity File to be used to populate on-us Clearing system data store.
DISP CM SEND	DCMSO	MS	No	No	Yes	Reserved for Brazil.
DISP CM SENDREC	DCMSR	MB	No	No	Yes	Reserved for Brazil.
Edit Package ADP		EA				Retired.
Edit Package ADT		ET				Retired.

Custom Delivery and Split Routing File Types

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
Edit Package MF		EM				Retired.
Edit Package PC		EC				Retired.
Edit Package Updates	EPUPD	EP	Yes	No	No	Daily Edit Package VID and ARDEF table changes (TC54).
Edit Package Updates (VCX)	VCX54	CX	Yes	Yes	No	VCX Daily Edit Package VID and ARDEF table changes (TC54).
EP40 Code Update	EP4CB	ES	Yes	Yes	No	Edit Package version 4 software updates (PC or Mainframe). Separates EP 4.0 Code update (software updates) from endpoint's regular daily incoming (UNDIF) files.
Fee/Funds	FEFUN	FF	Yes	Yes	Yes	Used for issuers and acquirers to send and receive client-to-client Fee Collections and Funds Disbursements.
Financial	FINAN	FI	Yes	No	No	Any financial activity, segregated from nonfinancial transactions.
Fraud	FRAUD	FR	Yes	Yes	No	Fraud service data.
ICS Expedited	ICSX	IX	Yes	Yes	No	Expedited issuer clearing house.
ICS Standard	ICS	IS	Yes	No	No	Issuer clearing house.
IRF Detail Intl	IRFDI	II	Yes	Yes	No	Reserved for Japan.
IRF Detail NNSS	IRFDN	IN	Yes	Yes	No	Reserved for Japan.

Custom Delivery and Split Routing File Types

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
ISO Formatted		SO				Retired.
Issuer Fee Service Report		FS				Retired.
Japan National Net reports	JNS	JN	Yes	Yes	Yes	Japan NNSS reports.
NARS Expedited	NARS	NX	Yes	Yes	No	National Application Review Service expedited.
National Net Settlement	NNSS	NN	Yes	No	Yes	Allows clients within a country to settle domestic transactions through a local settlement agent bank. The service can be used for transactions where the merchant, acquirer, and issuer are in the same country. The custom file type puts all national net transactions into a separate file.
National Settlement Reports	NSRPT	NS	Yes	Yes	No	National Net Settlement reports.
NCRF Data	NCRF	NC	Yes	Yes	Yes	Europe CRB.
Nonfinancial	NONFI	NF	Yes	No	No	All nonfinancial data segregated from financial transactions.
Non-interchange	NONIC	NI	Yes	No	No	Non-interchange Report File.
Original Drafts	ORIG	OD	Yes	No	No	Originals segregated from exception items and nonfinancial.
PFX CAS Advice	PFX33	PF	Yes	Yes	Yes	Alternate routing of the TC33 – PFX CAS Advice.

Custom Delivery and Split Routing File Types

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
POS Originals	POSOR	PO	No	No	Yes	POS original transactions; that is, original presentments of clearing drafts and reversals.
RC Adv Amt Total	REHV2	R2	Yes	Yes	No	Reclassification advices reason code HV2.
Reclassify Advices	READV	R1	Yes	Yes	No	Reclassification advices.
Settlement report	SETLR	SR	Yes	No	No	Settlement reports – machine-readable and print-ready (TC46 and TC47 combined)
Settlement Reports (Machine-Readable)	SETLM	MR	Yes	No	No	Settlement reports – machine-readable (TC46)
Settlement Reports (Print)	SETLP	LR	Yes	No	No	Print ready settlement reports (TC47).
SWEEPSTAKE	SSWSTK	SW	Yes	No	Yes	Sweepstakes – TC10 and TC20.
Sweepstakes VML Format		SG				Retired.
TRID TC33 Report	TRRPT	TR	Yes	Yes	No	Token Requestor ID (TRID) TC33 Report.
UK Electronic HCF	UKNEG	UN	Yes	Yes	No	UK electronic hot card file.
Universal Biller File		UB				Retired.
VCR Claim CO	VROLC	VL	No	No	Yes	Split collection-only VCR disputes.
VCR Claim Msg	DFT33	DF	Yes	Yes	Yes	Determine TC33 advice delivery for Dispute Financial to VROL.
Visa Capture File	VCAPF	VC	Yes	Yes	No	CyberSource Enhanced TC33 Capture File

Custom Delivery and Split Routing File Types

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
Visa Rewards Assessment File	ASSMT	AT	Yes	No	Yes	Visa Rewards fee collections/funds disbursements with TCQ 1.
Car Rental ¹	CARNT	CC	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = CORPCA
Car Rental ¹	CRADE	CE	No	No	Yes	Level 3 commercial card data. Car Rental (ADE). TC50s with TCR0 Service Identifier = CORPCD
Car Rental ¹	CRAD3	CT	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = COMMCL
Car Rental – Line Item Detail ¹	CRLID	CL	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = CORPCL
Car Rental – Line Item Detail Third-Party ¹	CR3RD	CN	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = COMMCL
Electronic Bill Pres ¹	EBFIL	EB	No	No	Yes	TC50
Fleet Service ¹	FLEET	FT	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = COMFLT
General Use – Line Item Detail ¹	GNLID	GL	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = COMMXL

Custom Delivery and Split Routing File Types

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
Generic Data ¹	GENER	GN	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = COMMGN
Lodging ¹	LODGE	LG	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = CORPLG
Lodging (ADE) ¹	LGADE	LA	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = CORPLA
Lodging (ADE) Third-Party ¹	LGAD3	L3	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = COMMLA
Lodging – Line Item Detail ¹	LGLID	LL	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = CORPLL
Lodging – Line Item Detail (Third-Party) ¹	LG3RD	LT	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = COMMLL
Open Format	OPNFM	OF	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = OPNFMT
Passenger Itinerary Data ¹	PASID	PI	No	No	Yes	Level 3 commercial card data. TC50s with TC0 Service Identifier = CORPAI
Passenger Itinerary Data Leg Pecific (Third-Party) ¹	PASL3	P3	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = COMMAS

Custom Delivery and Split Routing File Types

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
Passenger Itinerary Data Leg Specific ¹	PASLS	PL	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = CORPAS
Passenger Itinerary General ¹	PASIG	PG	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = COMMAG
Purchase Txn ADE (TC50) ¹	PRCHA	PA	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = PURCHA
Purchasing Transaction (ADE) ¹	GNADE	GA	No	No	Yes	Level 3 commercial card data. TC50s with a TCR0 Service Identifier = COMMXA
Purchasing Transaction – Line Item Detail ¹	PRCHL	PD	No	No	Yes	Level 3 commercial card data. TC50s with TCR0 Service Identifier = PURCHL

¹ Enables clients and processors to specify different delivery endpoints for Commercial Card transactions so that they can be delivered from the issuer to the acquirer or to a third party. The enhanced data is delivered directly to the selected issuing endpoint instead of sending the data to the application processor (CIB) and then to the recipient

Custom and Split Route File Types—Visa Use Only

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
CRIS Reports	CRIS	CR	Yes	Yes	No	Visa use only.
RFC	RFC	RC	Yes	Yes	Yes	Visa use only.
RFC ACQUIRER	RFCAQ	RA	No	No	Yes	Visa use only.
RFC ISSUER	RFCIS	RI	No	No	Yes	Visa use only.
VCMS Responder	VCMSR	VR	Yes	No	No	Visa use only.

Other Platform Files

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
Commercial Choice Select Reference Data	COMCH	CH	Yes	Yes	No	The file type COMCH is used by the Visa Commercial Choice Select participating acquirers that subscribes to Commercial Choice Select reference data in RDMS. The file type of COMCH in the BASE II customized delivery file type identifies files containing the TC33 Commercial Choice Select reference data. The file is delivered weekly.
CYB Capture File		CY				Retired.
DUP TRANS RPTS	DUPCA	DC	Yes	Yes	No	Duplicate Transaction Identification Detection and Reporting Service (GDTID) program reporting.
Generic GC File	GMRGC	GC	Yes	Yes	No	TC50 BC file (EAS to APF).
Generic GD File	GMRGD	GD	Yes	Yes	No	Generic GD file (APF to Global Merchant Repository AMMF Server).
GMR TC50 BC File	GMRBC	GM	Yes	Yes	No	TC50 BC file (EAS to APF).
GMR TC50 BD File	GMRBD	RB	Yes	Yes	No	TC50 BD file (APF to Global Merchant Repository AMMF Server).
International Fee Advice	IFADV	IF	Yes	Yes	No	Delivery of International Transaction Fee changes reports (ITFR).
Monthly Enrollment Detail		NE				Retired.
Monthly Enrollment TC33		ME				Retired.

Other Platform Files

File Type	BASE II Customized Delivery File Type (ITF TC90)	File Type Code— Visa Use Only	Customized	Expedite	Split Route	Business Description
Rewards Enrollment		RE				Retired.
Rewards Points TC33		RP				Retired.
RDMS	RDMS	RD	Yes	No	No	Client reports from the Report Distribution Management System.
Risk Adj Fee Rpt	FEE40	RF	Yes	Yes	No	FEE 400 and FEE 500 series reports for Risk Adjusted fees.
Token Report	TKPRV	TV	Yes	Yes	No	TC33 Token Provisioning Results (ODF to EAS)

BASE II Consolidated File Delivery

BASE II uses consolidated file delivery to place all of the data that is undifferentiated (data left over after customization) from all the VisaNet Interchange Centers (VICs) in a single logical file.

Clients can choose to consolidate data (collapse files into a single file). This option causes all data that is undifferentiated (data left over after customization) from all the VisaNet Interchange Centers (VICs) to be placed in a single logical file. Without this option, a client or processor may receive at least one data file from each VIC and one settlement report file. Because the consolidation process runs after the settlement event occurs, there is a delay for these clients—while the settlement event is the trigger, the delivery is not immediate.

Clients using a VAP or EA Server have the option of enabling consolidation from this service, reducing delivery delays from VisaNet.

Chapter 6

BASE II Collection-Only Service

All BASE II clients and processors are required to send all intraprocessor transactions to Visa for reporting. Intraprocessor transactions are those transactions that involve two different clients serviced by the same processing center.

BASE II Collection-Only Overview

In most cases, intraprocessor transactions are submitted to the BASE II System for clearing and settlement; however, the processor may clear and settle the transaction and send a collection-only transaction to Visa.

Note:

All U.S. domestic transactions must be cleared and settled through Visa.

Collection-only transactions are identified by an entry of "C" in the Collection-Only Flag of the transaction message. These transactions have limited BASE II editing:

- Entries for Source and Destination identifiers and for account number fields must be the same as for non-collection-only transactions.
- The Source Currency Code must be spaces or a valid currency. If spaces are entered, the Merchant Country Code must be valid.
- The Destination Amount must be zero.

- The Destination Currency code must be spaces.
- These fields must be numeric:
 - Acquirer Reference Number
 - Acquirer's Business ID
 - Purchase Date
 - Merchant Category Code
 - Reason Code
 - Central Processing Date
 - Settlement Flag
 - Source Amount

Collection-only transactions can be combined with other interchange or can be submitted as a separate file. BASE II does not calculate or assess BASE II processing charges or interchange reimbursement fees for collection-only transactions. The transactions are shown as nonfinancial transactions on Edit Package reports EP-110B, EP-110C, EP-110D, EP-221B, and EP-221D.

BASE II Collection-Only Eligibility

BASE II Collection-Only Service is available to all BASE II clients and processors.

Chapter 7

BASE II Clearing-Only Service

BASE II Clearing-Only is a client-to-client transaction exchange using VisaNet clearing services for data quality and reporting; the transactions are settled outside Visa. The Multicurrency Conversion Service is not supported for Clearing-Only transactions.

BASE II Clearing-Only Service Overview

Transactions settled outside of Visa are identified by an entry of 3 (Clearing-Only International Settlement) or 4 (Clearing-Only National Net Settlement) in the Settlement Flag (TCR 0 position 150) of the transaction message. These appear on the VisaNet Settlement Service (VSS) reports as transactions that are not settled and appear as a unique line on the VSS-900 report.

BASE II acquirers must send an authorization source code value of "8" (authorization not performed by VisaNet) in the TCR 1 Additional Data, position 128 of the draft data Clearing-Only transactions when the original authorization is obtained outside of the VisaNet system.

BASE II Clearing-Only Service Eligibility

This service is available only if an issuer card range is registered for the BASE II Clearing-Only Service. Issuer card ranges participating in Clearing-Only Service are identified in the ARDEF. Transactions sent with one of the Clearing-Only settlement flags of 3 or 4 for an issuer card range not participating in this service will be returned with a BASE II return reason code 55 (Account range/transaction not allowed for Clearing-Only Service).

Sales drafts, merchandise credit and their reversals, and original credits transactions (OCT) are valid message types for Clearing-Only. Dispute processing is not supported.

BASE II originated Clearing-Only transactions that fail the NNSS qualification criteria will be returned with a BASE II return reason code 48. (The source identifier, destination identifier, and merchant in a national net or bilateral interchange transaction are not all in the same country or the source currency is not the local currency).

Chapter 8

BASE II Currency Conversion Service

The Currency Conversion Service provides clients the ability to clear transactions in any currency recognized by the International Organization for Standardization (ISO). Visa settlement currencies are a subset of ISO 4217 and can be found in *BASE II Data Codes*.

BASE II Currency Conversion Service Overview

Currency Conversion Service includes these transaction currency processing functions.

- **Clearing Conversion**—This feature automatically converts the transaction currency to the issuer destination currency for the cardholder. The transaction currency is generally the currency in which a transaction takes place. The issuer destination currency for the cardholder as provided to Visa. VisaNet supports virtually every currency recognized by the International Organization for Standardization (ISO).
- **Settlement Conversion**—This feature automatically converts the transaction currency to the acquirer's settlement currency (if the two are different), and to the issuer's settlement currency (if the two are different). Twenty-four settlement currencies are supported.
- **Currency Rate Delivery**—This feature provides clients (five days a week, Monday through Friday) with the currency conversion rates Visa uses to process transactions.

Currency Conversion Service is mandatory for all issuers, acquirers, processors, and merchants in all Visa Regions.

BASE II Currency Conversion Service Functions Overview

Visa uses buy and sell rates for currency conversion.

These rates are paired into:

- **USD-based rates** that are used when converting non-USD currencies against the U.S. dollar.
- **Cross rates (non-USD-based rates)** that are used for selected currencies for which the rates quoted are against a currency other than the U.S. dollar

BASE II Currency Conversion Service Description

Visa uses buy and sell rates to process currency conversion.

These are the basic formulas that VisaNet uses for all transactions when converting currencies:

- When converting from the counter currency to the base currency, the formula is:
 $\text{Counter currency} \times \text{rate} = \text{Base currency}$
- When converting base currency to counter currency, the formula is: $\text{Base currency} \div \text{rate} = \text{Counter currency}$

From these basic formulas, Visa has established sets of processing rules for each type of transaction and has categorized them into exchange direction groupings:

- Buy—Sell
- Sell—Buy

VisaNet uses exchange direction processing rules for the rate pair type that is appropriate for the transaction and for the type of conversion required when performing the currency conversion.

Currency Conversion Service Monetary Units

Currencies are defined as having zero, two, or three minor units of currency. For instance, the U.S. dollar has two minor units of currency (the two positions to the right of the decimal point); the Japanese yen has no minor units.

BASE II always assumes two minor units:

- If the transaction amount is 20492, it must be entered in the request as 2049200.
- If the transaction amount is 67.89, it must be entered in the request as 6789.
- If the transaction amount is 3.129, it must be entered in the request as 313.

Currency Conversion Service Settlement Currencies

Currency Conversion Service automatically converts client or processing center net financial positions into the client's settlement currency supported by the International Settlement Service.

- If the local currency is one of the settlement currencies, settlement will be in the local currency.
- If the local currency is not one of the settlement currencies, the client can choose any supported settlement currency.

BASE II Currency Conversion - Visa Issuer FX Calculator (formerly known as OIF)

Issuers can choose to charge a Visa Issuer FX Calculator amount to the cardholder for transactions that require currency conversion. The issuer can specify an intraregional or interregional Visa Issuer FX Calculator amount, or both.

The Visa FX Calculator issuer-defined parameters are maintained in VisaNet databases according to issuing identifiers. The Visa Issuer FX Calculator amount is calculated at the time of currency conversion using the percentage rate established by the issuer and is included in the transaction destination amount. To implement the Visa Issuer FX Calculator, the client must contact Visa.

All VisaNet systems that support Multicurrency Service processing use common conversion rates.

In cross-border money transfer OCTs, the Visa Issuer FX Calculator amount is not deducted from the amount that the recipient receives.

BASE II Currency Rate Delivery

Monday through Friday, BASE II Clearing System obtains and verifies international currency rates from various sources around the world. Visa determines the wholesale rate Monday through Friday based on the cost to Visa for buying and selling currencies on the foreign exchange markets. Visa uses these currency rates for that day's financial transactions.

Visa offers these rates to subscribing clients as part of the VisaNet BASE II Clearing currency rate delivery service. The service allows processors to choose to receive the rates with their interchange files or immediately after the rates are available in BASE II Clearing System.

BASE II Currency Rate Calculation Process

Visa uses the calculations performed by BASE II to provide currency conversions.

BASE II uses two components for the currency conversion calculation:

- A buy and/or sell rate to obtain the Transaction Amount in Destination Currency (TADC).
- For currency conversion transactions, BASE II calculates the Visa Issuer FX Calculator amount (if any) for sales drafts, credit vouchers, cash advances, and their reversals.

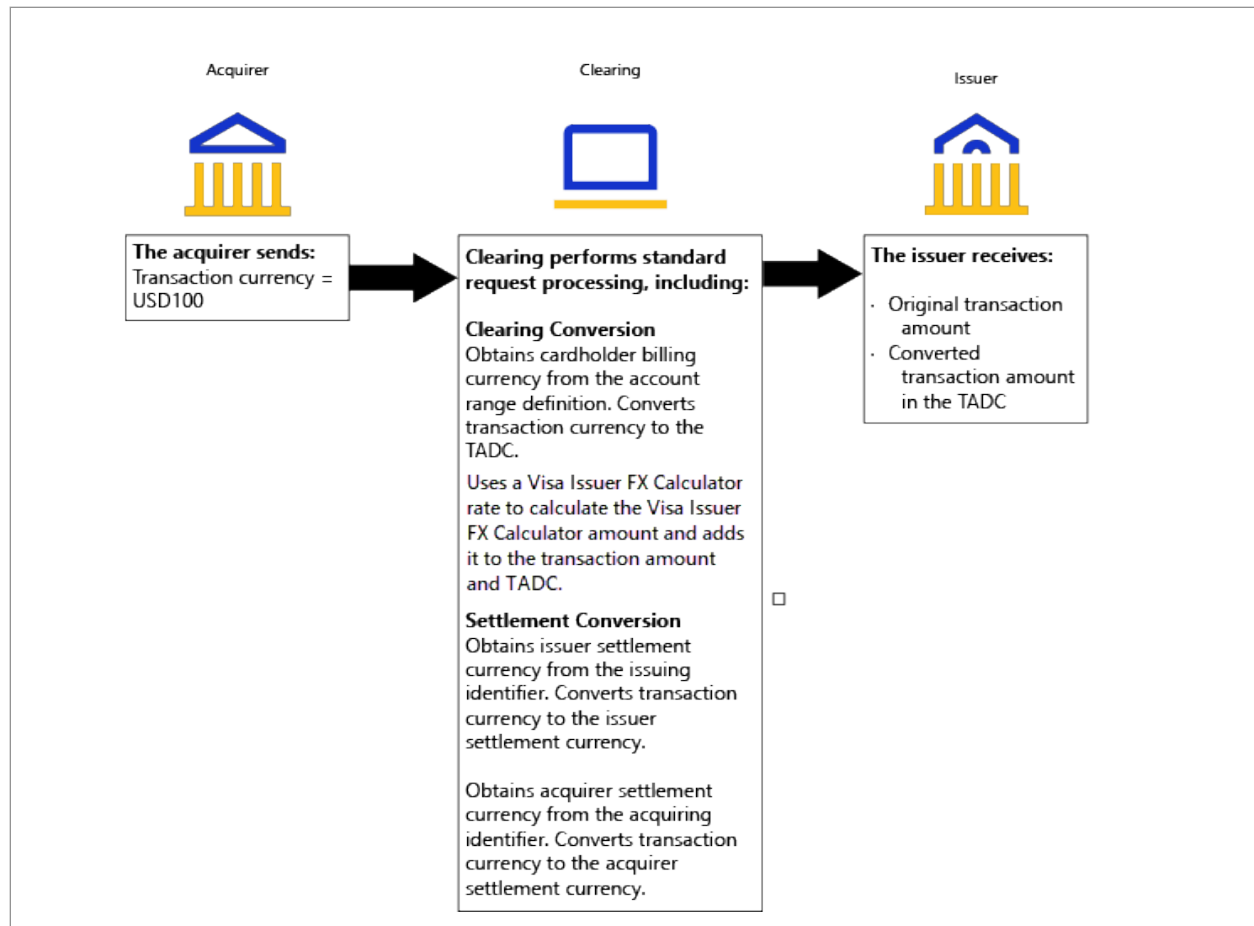
The sum of TADC and Visa Issuer FX Calculator amount is carried in the destination amount field.

BASE II Currency Conversion Process Flow

BASE II Clearing System applies the currency conversion calculation to the transaction flow to manage the sending and receiving currency conversion information.

This figure illustrates a Currency Conversion Service process flow for a transaction acquired in U.S. dollars for an Australian cardholder.

Currency Conversion Service Process Flow



Within the currency conversion transaction process flow:

- The acquirer sends transaction that includes the transaction currency used at the point of sale or point of service (POS).
- Clearing performs standard clearing processing and currency conversion, populating the destination amount field with the transaction amount in the destination currency.
- Clearing populates the destination currency in the transaction to the issuer.

Currency Rates

Standard currency rates are published by Visa Treasury on a daily basis. Clearing will use the rates for the current central processing date (CPD).

Persistent FX is a feature that applies the exchange rate from the time of authorization throughout the entire clearing and settlement process for POS and ATM original transactions. This feature provides issuers and cardholders with greater transparency regarding exchange rates and currency-converted amounts.

Persistent FX is applicable to VisaNet transactions where cardholders make purchases in currencies different from their billing currency. It does not apply to dispute-related transactions such as dispute financials and dispute response financials, nor to advice messages.

For additional transaction visibility, clients have the option to receive BASE II TC33 advices containing detailed Persistent FX and exchange rate information.

BASE II Buy and Sell Currency Rates

VisaNet applies buy and sell rates to transactions during clearing and settlement.

The Visa currency conversion rate system involves specific terminology as defined in this table.

Term	Description
Base	This term is associated with a currency. Within VisaNet, variable units of a base currency are equivalent to one unit of a counter currency.
Counter	This term is associated with a currency. Within VisaNet, one unit of a counter currency is equivalent to variable units of a base currency.
Buy Rate	The buy rate in VisaNet is the number of units of base currency required to buy one unit of the counter currency.
Sell Rate	The sell rate in VisaNet is the number of units of base currency received from selling one unit of the counter currency.
USD-Based Rate Pair	In this type of pair, the U.S. dollar is always the base currency and fluctuates against the counter currency. The counter currency will be a currency other than USD.
Exchange Direction	This term is used to designate the set of currency conversion processing rules that will be applied to a transaction based on the transaction type.
Transaction Amount in Destination Currency (TADC)	This term is used to identify the submitted transaction amount in the currency that is appropriate to the destination endpoint. The TADC is included in the BASE II destination amount field. Besides the TADC, the fields may contain the Visa Issuer FX Calculator amount. The BASE II destination amount field is provided in clearing transactions to the destination client.

This table provides a mapping of the terminology used to the corresponding fields in BASE II.

Article Terminology	BASE II TCR/Position
Source amount	Draft Data TCR 0, Positions 77–88
TADC	Draft Data TCR 0, Positions 62–73 ¹

¹ In this section, only the TADC component is described and displayed in these fields. The Visa FX Calculator and currency conversion charge are not addressed.

BASE II Exchange Direction Classifications

The exchange direction classification, either buy-sell or sell-buy, is used to group types of transactions that use the same set of currency conversion processing rules.

All original transactions, and their dispute financials and reversals are assigned the same exchange direction classification.

—> Indicates that the flow is from the acquirer to the issuer.

<— Indicates that the flow is from the issuer to the acquirer.

<-> Indicates that the flow can be from either acquirer to issuer or issuer to acquirer.

This table shows the buy-sell exchange direction type.

Transaction Type	Transaction Flow	Money Flow
TC 05—Sales Drafts	—>	<—
TC 25—Sales Draft Reversals	—>	—>
TC 15—Sales Draft Dispute Financials	<—	—>
TC 35—Sales Draft Dispute Financial Reversals	<—	<—
TC 07—Cash Disbursements	—>	<—
TC 27—Cash Disbursement Reversals	—>	—>
TC 17—Cash Disbursement Dispute Financials	<—	—>
TC 37—Cash Disbursement Dispute Financial Reversals	<—	<—
TC 10—Fee Collections	<->	<->

This table shows the sell-buy exchange direction type.

Transaction Type	Transaction Flow	Money Flow
TC 06—Credit Vouchers	—>	—>
TC 26—Credit Voucher Reversals	—>	<—
TC 16—Credit Voucher Dispute Financials	<—	<—
TC 07—Cash Disbursements Note: ATM Credit Adjustments only that originate from V.I.P. Full Service ATM acquirers.	—>	<—
TC 27—Cash Disbursement Reversals Note: ATM Credit Adjustments only that originate from V.I.P. Full Service ATM acquirers.	—>	—>
TC 36—Credit Voucher Dispute Financial Reversals	<—	—>
TC 20—Funds Disbursements	<->	<->

BASE II USD-Based Rates

VisaNet uses USD-based rates to determine amounts in transactions, regardless of transaction type, when either or both the source amount or the TADC is specified in a non-USD currency and a non-USD-based rate pair that matches the currencies in the transaction is not on the rate file.

VisaNet uses USD-based rates when these conditions are met:

- Either or both the source amount or the TADC is specified in a non-USD currency.
- A non-USD-based rate pair that matches the currencies in the transaction is not on the rate file.

VisaNet uses the source amount submitted in the transaction to calculate the TADC.

BASE II USD-Based Rate Processing Rules

The Processing Rules for USD-Based Rate Calculations table shows the processing rules that are applied to determine the TADC using USD-based rates. The information in the Exchange Direction column indicates a grouping of transaction types to which the currency conversion processing rules will apply.

The information in the Processing Rule column is provided to indicate which rule was applied to a transaction in the scenarios that follow in this section. The information in the Currency Combination column indicates the type of conversion required.

Exchange Direction	From Field	To Field	Rule ¹	Currency Combination	Processing Rule
Buy-Sell	Source Amount	TADC	DB-1	Non-USD to USD	Source amount x buy rate of source currency.
			DB-2	USD to non-USD	Source amount ÷ sell rate of destination currency.
			DB-3	Non-USD to same non-USD	No conversion required.
			DB-4	Non-USD to other non-USD	This type of transaction requires this conversion process: - Source currency to USD: Source amount x buy rate of source currency. - USD to destination: USD amount ÷ sell rate of destination currency.
Sell-Buy	Source Amount	TADC	DB-5	Non-USD to USD	Source amount x sell rate of source currency.
			DB-6	USD to non-USD	Source amount ÷ buy rate of destination currency.
			DB-7	Non-USD to same non-USD	No conversion required.
			DB-8	Non-USD to other non-USD	This type of transaction requires this conversion process: - Source currency to USD: Source amount x sell rate of source currency. - USD to destination: USD amount ÷ buy rate of destination currency.

1 DB = U.S. dollar-based

BASE II USD-Based Currency Conversion Scenarios

These scenarios illustrate how VisaNet applies USD-based rates to original purchase transactions to calculate the TADC. The calculated TADC only reflects the conversion from the amount submitted in the source amount field and does not include any assessments or charges.

BASE II USD-Based Currency Conversion Scenario #1

In this scenario, an original purchase transaction is submitted to VisaNet for a purchase made by a USD cardholder at an ABC merchant. The purchase amount of the transaction is ABC 1,000.00. The relationship between the originator of the transaction and the type of currency in the transaction is as follows:

- Source = ABC = Counter currency
- Destination = USD = Base currency

This table lists the USD-based rate pairings and rates that are used to calculate the TADC in these scenarios.

Scenario	Counter Currency	Base Currency	Buy Rate	Sell Rate
1 & 2	ABC	USD	Buy-A	Sell-A
2 & 3	XYZ	USD	Buy-B	Sell-B

Because the currency of the TADC is USD, only the buy rate is applied to this transaction. VisaNet converts the source amount to the TADC using the:

- USD-based rate pairing of ABC/USD
- DB-1 rule: Non-USD to USD formula: source amount x buy rate of source currency

As applied to this transaction, the calculation is:

ABC		1,000.00	
	x	Buy-A	(Buy rate)
USD	=	TADC	

This table shows the content of the source amount and TADC fields and corresponding currency code fields in BASE II sales draft transactions. All positions referenced are carried in the Draft Data TCR 0 record.

Populated by Acquirer		Populated by Visa		Visa Performed Calculation
Positions 77-88	000000100000	Positions 62-73	Amount in U.S. dollars	DB-1 rule: Non-USD to USD: Source amount x buy rate of source currency. As applied to the transaction: ABC 1,000.00 x Buy-A = USD TADC.
Positions 89-91	Numeric value for currency code ABC	Positions 74-76	840	

BASE II USD-Based Currency Conversion Scenario #2

In this scenario, an original purchase transaction is submitted to VisaNet for a purchase made by an XYZ cardholder at a USD merchant. The purchase amount of the transaction is USD\$400.00. The relationship between the originator of the transaction and the type of currency in the transaction is as follows:

- Source = USD = Base currency
- Destination = XYZ = Counter currency

This table lists the USD-based rate pairings and rates that are used to calculate the TADC in these scenarios.

Scenario	Counter Currency	Base Currency	Buy Rate	Sell Rate
1 & 2	ABC	USD	Buy-A	Sell-A
2 & 3	XYZ	USD	Buy-B	Sell-B

Because the currency of the source amount is USD, only the sell rate is applied to the transaction. VisaNet converts the source amount to the TADC using the:

- USD-based rate pairing of XYZ/USD
- DB-2 rule: USD to non-USD formula: source amount ÷ sell rate of destination currency

As applied to this transaction, the calculation is:

USD		1,000.00	
	x	Sell-B	(Sell rate)
XYZ	=	TADC	

This table shows the content of the source amount and TADC fields and corresponding currency code fields in BASE II sales draft transactions. All positions referenced are carried in the Draft Data TCR 0 record.

Populated by Acquirer		Populated by Visa		Visa Performed Calculation
Positions 77-88	000000400000	Positions 62-73	Amount in XYZ currency	DB-2 rule: USD to non-USD currency: Source amount ÷ sell rate of destination currency. As applied to the transaction: USD 400.00 ÷ Sell-B = XYZ TADC.
Positions 89-91	840	Positions 74-76	Numeric value for currency code XYZ	

BASE II USD-Based Currency Conversion Scenario #3

In this scenario, an original purchase transaction is submitted to VisaNet for a purchase made by an XYZ cardholder at an ABC merchant. The purchase amount of the transaction is ABC 1,000.00. The relationship between the originator of the transaction and the type of currency in the transaction is as follows:

- Source = ABC = Counter currency
- Destination = XYZ = Counter currency

This table lists the USD-based rate pairings and rates that are used to calculate the TADC in these scenarios.

Scenario	Counter Currency	Base Currency	Buy Rate	Sell Rate
1 & 2	ABC	USD	Buy-A	Sell-A
2 & 3	XYZ	USD	Buy-B	Sell-B

In this scenario, the source currency and the TADC are both counter currencies to the U.S. dollar. However, the rate file does not contain a non-USD-based rate pair that matches the currencies in the transaction, so USD-based rate pairs are used for the conversion. Because the source currency and the TADC are in currencies other than USD, both buy and sell rates are applied to this transaction. To convert the source amount to the TADC, VisaNet performs two calculations converting the currencies through the U.S. dollar using the:

- USD-based rate pairing of ABC/USD
- USD-based rate pairing of XYZ/USD
- DB-4 rule: Non-USD to other non-USD formula:
 - Source amount x buy rate of source currency
 - USD amount ÷ sell rate of destination currency

As applied to this transaction, the calculations are:

ABC		1,000.00	
	x	Buy-A	(Buy rate)
USD	=	VisaNet interim amount	
	÷	Sell-B	(Sell rate)
XYZ	=	TADC	

This table shows the content of the source amount and TADC fields and corresponding currency code fields in BASE II sales draft transactions. All positions referenced are carried in the Draft Data TCR 0 record.

Populated by Acquirer		Populated by Visa		Visa Performed Calculation
Positions 77-88	000000100000	Positions 62-73	Amount in XYZ currency	DB-4 rule: Non-USD to other non-USD currency:
Positions 89-91	Numeric value for currency code ABC	Positions 74-76	Numeric value for currency code XYZ	• Source amount x buy rate of source currency. • USD amount ÷ sell rate of destination currency. As applied to the transaction: • ABC 1,0000 x Buy-A = USD amount. • USD amount ÷ Sell-B = XYZ TADC.

BASE II Non-USD-Based Currency Conversion Scenarios

VisaNet uses non-USD-based rates to determine the TADC in transactions, regardless of transaction type, when the source amount and the TADC are in different non-USD currencies and a non-USD-based rate pair exists. The base currency will be in a currency other than U.S. dollars.

Non-USD-based rate pairs are also used to calculate settlement amounts under these conditions:

- A non-USD-based rate pair was used to calculate the TADC.
- A non-USD-based rate exists that matches the currencies of the source amount and the specified settlement amount, either source settlement amount or destination settlement amount.

USD-based rates are applied to all other amounts in the transaction.

VisaNet uses the source amount submitted in the transaction to calculate the TADC. The Processing Rules for Non-USD-Based Rate Calculations table shows the processing rules that are applied to determine the TADC using non-USD-based rate pairs. The information in the Exchange Direction column indicates a grouping of transaction types to which the currency conversion processing rules apply.

The information in the Processing Rule column is provided to indicate which rule was applied to a transaction in the scenarios that follow. The information in the Currency Combination column indicates the type of conversion required.

Exchange Direction	From Field	To Field	Rule ¹	Currency Combination	Processing Rule
Buy-Sell	Source Amount	TADC	NDB-1	Counter currency to base currency	Source amount x buy rate of source currency.
			NDB-2	Base currency to counter currency	Source amount ÷ sell rate of destination currency.
Sell-Buy	Source Amount	TADC	NDB-3	Counter currency to base currency	Source amount x sell rate of source currency.
			NDB-4	Base currency to counter currency	Source amount ÷ buy rate of destination currency.

¹ NDB = non-U.S. dollar-based

These scenarios are provided to illustrate how VisaNet applies the non-USD-based rate pairs in original purchase transactions to calculate the TADC. The calculated TADC only reflects the conversion from the amount submitted in the source amount field and does not include any service assessments or charges.

BASE II Non-USD-Based Currency Conversion Scenario #1

In this scenario, an original purchase transaction is submitted to VisaNet for a purchase made by an MNO cardholder at an XYZ merchant. The purchase amount of the transaction is XYZ 700.00. The relationship between the originator of the transaction and the type of currency in the transaction is:

- Source = XYZ = Counter currency
- Destination = MNO = Base currency

This table lists the non-USD-based rate pairings and rates used to calculate the TADC in the scenarios.

Scenario	Counter Currency	Base Currency	Buy Rate	Sell Rate
1	XYZ	MNO	Buy-C	Sell-C
2	MNO	ABC	Buy-D	Sell-D

Because the currency of the TADC is the same as the base currency, only the buy rate is applied to this transaction. VisaNet converts the source amount to the TADC using the:

- Non-USD-based rate pairing of XYZ/MNO
- NDB-1 rule: Counter currency to base currency formula: source amount x buy rate of source currency

As applied to this transaction, the calculation is:

XYZ		700.00	
	x	Buy-C	(Buy rate)
MNO	=	TADC	

This table shows the content of the source amount, TADC fields, and corresponding currency code fields in BASE II sales draft transactions. All positions referenced are carried in the Draft Data TCR 0 record.

Populated by Acquirer		Populated by Visa		Visa Performed Calculation
Positions 77-88	000000070000	Positions 62-73	Amount in MNO – source amount x buy rate of source currency	NDB-1 rule: Counter currency to base currency: Source amount x buy rate of source currency.
Positions 89-91	Numeric value for currency code XYZ	Positions 74-76	Numeric value for currency code MNO	As applied to the transaction: XYZ 700.00 x Buy-C = MNO TADC.

BASE II Non-USD-Based Currency Conversion Scenario #2

In this scenario, an original purchase transaction is submitted to VisaNet for a purchase made by an MNO cardholder at an ABC merchant. The purchase amount of the transaction is ABC 500.00. The relationship between the originator of the transaction and the type of currency in the transaction is:

- Source = ABC = Base currency
- Destination = MNO = Counter currency

This table lists the non-USD-based rate pairings and rates that are used to calculate the TADC in the scenarios.

Scenario	Counter Currency	Base Currency	Buy Rate	Sell Rate
1	XYZ	MNO	Buy-C	Sell-C
2	MNO	ABC	Buy-D	Sell-D

Because the currency of the source amount is the same as the base currency, only the sell rate is applied to the transaction. VisaNet converts the source amount to the TADC using the:

- Non-USD-based rate pairing of MNO/ABC
- NDB-2 rule: Base currency to counter currency formula: source amount ÷ sell rate of destination currency

As applied to this transaction, the calculation is:

ABC		500.00	
	x	Sell-D	(Sell rate)
MNO	=	TADC	

This table shows the content of the source amount and TADC fields and corresponding currency code fields in BASE II sales draft transactions. All positions referenced are carried in the Draft Data TCR 0 record.

Populated by Acquirer		Populated by Visa		Visa Performed Calculation
Positions 77-88	000000050000	Positions 62-73	Amount in MNO currency	NDB-2 rule: Base currency to counter currency: Source amount ÷ sell rate of destination currency. As applied to the transaction: ABC 500.00 ÷ Sell-D = MNO TADC.
Positions 89-91	Numeric value for currency code ABC	Positions 74-76	Numeric value for currency code MNO	

BASE II Currency Conversion Rate Update Records

The Currency Conversion Rate Update Records (TC 56) are sent by VisaNet to transmit daily updates to the client. VisaNet provides subscribing clients with the same Currency Conversion Rate information that it uses, so that clients can have the actual conversion rate used. Clients must subscribe to this service to receive the conversion rates.

Each currency conversion rate update transaction contains eleven different entries; the last record in the file may contain fewer than eleven. Each entry contains the ISO numeric Currency Code of the counter currency, the ISO numeric Currency Code of the base currency, VIC Processing Date, the buy conversion and sell conversion rate applied to the currency that day, and the currency scale factor identifier for each of the conversion rates.

See *BASE II Clearing Interchange Formats, TC 50 to TC 92*.

Chapter 9

VisaNet Settlement Service

VisaNet Settlement Service (VSS) performs settlement for BASE II clearing transactions and V.I.P. full financial transactions in a single, centralized service, enabling clients to manage settlement processes more effectively.

VSS Overview

Issuers, acquirers, and processors in all regions are eligible to use the VisaNet Settlement Service. VSS is for V.I.P. full financial transactions and BASE II clearing records.

Visa offers these settlement services:

- International Settlement Service
- National Net Settlement Service (NNSS)
- Area Net Settlement Service (ANSS)

VSS Description

Once clearing has been performed (on all BASE II clearing records and V.I.P. full financial transactions), the cleared transactions are sent to VSS for settlement cut off.

The International Settlement Service is used to settle all international transactions and domestic transactions for clients that do not participate in National Net Settlement Service (NNSS) or Area Net Settlement Service (ANSS).

The NNSS or ANSS allows clients within a country or defined area to settle qualifying domestic transactions through a central settlement agent bank. Qualifying transactions are those for which the merchant, acquirer, and issuer are in the same country, or defined area, and the transaction currency is the local currency for that country or defined area.

VisaNet Settlement Service Settlement Relationships

VisaNet Settlement Service (VSS) provides flexibility when defining settlement relationships. Clients can define from two to eight levels of settlement relationships. The different levels allow clients to build and maintain the most appropriate settlement relationships for their business needs. For example, the settlement relationship levels can be used to reflect the products in a client's organization. With this flexibility, clients can easily and efficiently manage settlement functions.

VisaNet Settlement Service Reporting Entities

VisaNet Settlement Service (VSS) provides flexibility to clients through the use of settlement reporting entities (SREs) and settlement hierarchies. The SRE represents a business entity for which Visa performs settlement. SREs are linked together to form a structure called a settlement hierarchy. A settlement hierarchy is designed by a client to support the client's business needs.

VisaNet Settlement Service Report Layouts and Formats

VisaNet Settlement Service (VSS) reports provide a common layout for BASE II- and V.I.P. Full Service clients. This common layout allows all clients to streamline their internal procedures.

All VSS reports are available in print-ready and machine-readable formats. VSS reports use business-oriented terminology to better reflect the business needs of clients.

VisaNet Settlement Service Report Delivery

Clients can have their VisaNet Settlement Service (VSS) reports sent to multiple locations of their choice, including locations other than their processing centers. Interchange routing does not determine the routing of settlement information.

Funds Transfer Point

For VisaNet Settlement Service (VSS), the funds transfer point can be defined at any level in the settlement structure. This flexibility allows clients using third-party processors to be responsible for their own funds transfer.

Alternately Routed and Split-Routed Transactions

Clients can specify how they want to settle their alternately-routed or split-routed transactions. This option is called **alternate routing** for V.I.P. Full Service and **split routing** for BASE II. They can elect to settle these transactions with their normally routed transactions or separately.

National Settlement Cutoff Time

Clients that participate in NNSS or ANSS have the option to settle domestic activity at a time that is appropriate to their business day. The specific settlement time can be different from the International Settlement Service cutoff time. To use this feature, all clients participating in a particular NNSS or ANSS must agree on a common settlement cutoff time.

VSS Processing

The settlement records generated by BASE II are processed by VisaNet Settlement Service (VSS), which accumulates them and produces reconciliation reports for the issuer and acquirer detailing the net settlement amount.

VSS Settlement Timing

Visa settles with its clients on a daily basis through the VisaNet Settlement System (VSS). Settlement reports are run and transmitted after the close of the clearing cycle so that the settlement information is available to clients at the beginning of their business day.

VSS Funds Transfer

The final step in the VisaNet Settlement Service (VSS) settlement process is **funds transfer**, during which funds are collected from settlement entities with a net debit position, and paid to settlement entities with a net credit position.

Funds transfer refers to the movement of funds between the client's settlement bank and the settlement agent's settlement bank for the purpose of settlement. Funds transfers represent the net position of the client's credits and debits.

For international settlement, the settlement account can be either a Demand Deposit Account (DDA) or a general ledger account depending on the client's business requirements. Whichever is used, the account must be open and funded before settlement begins. The account type must be set up to receive and process drawdown requests. For funds transfer to take place, the authority for Visa to draw down against the client's settlement account must be granted through the completion of Settlement Funds Transfer (SFT) forms and the exchange of authorization letters when required.

The currency of the net funds transfer amount must be associated to a settlement account of like currency. Several funds transfers can be associated to the same bank account.

For international settlement, funds can be settled with a client-selected settlement bank in the client's settlement currency, but the currency must be one that has been established by Visa as a settlement currency.

VSS Benefits

VisaNet Settlement Service (VSS) provides clients with significant flexibility and control in organizing and reporting settlement data and funds transfers. Multiple reporting levels that can be mapped to the clients' business structure gives better financial control over funds transfers. Clearly designed reports make reconciliation easier and more efficient. Because all reports are available in machine-readable format, clients can develop an automated reconciliation system that can save substantial time and resources in back-office processing.

VSS Features

VisaNet Settlement Service (VSS) offers clients these features:

- Flexibility in establishing settlement relationships.
- Standardized report layouts in print-ready and machine-readable formats.
- Several report delivery options.
- Choice of settlement options for alternately routed and split-routed transactions.
- Ability to specify end-of-day cutoff times for all clients participating in National Net Settlement Service (NNSS) or Area Net Settlement Service (ANSS).
- Alternative report formats.

VSS Requirements

In order to participate in the VisaNet Settlement Service (VSS), clients must fulfill the testing and implementation requirements.

VSS Testing

VisaNet Settlement Service (VSS) testing is available for clients preparing for VSS implementation or for modifying their existing VSS hierarchies or report subscriptions. Testing enables clients to receive files and reports that can be used to test their internal systems and procedures.

During VSS testing with VisaNet, the client is responsible for validating test results and ensuring the VSS reports and files can be processed correctly.

VSS Implementations

Clients can use either a standard or custom method for implementing VisaNet Settlement Service (VSS). It is up to the client to decide which strategy to use.

- **Standard**—This implementation type creates a two-level hierarchy with one top settlement reporting entity (SRE) that includes parameters that define the entire hierarchy, such as the settlement currency code. The top SRE also represents a summary of all SREs below the top SRE. In a standard hierarchy, the top SRE represents all activity cleared (processed) by the processor. The lowest-level SREs are either clearing identifiers (acquirers and issuers) or account ranges (issuers). Funds transfer can occur at the top level or at the lowest level (client-level settlement). Reporting options for a standard hierarchy can be defined by groups of acquiring or issuing identifiers or can apply to all SREs within the hierarchy. Multiple report delivery destinations must be requested as a custom implementation.
- **Custom**—Customizing a standard hierarchy may be needed for many reasons, including:
 - Using mid-level summary reporting
 - Establishing additional funds transfer points
 - Sending reports to additional endpoints
 - Creating a VSS hierarchy that involves acquiring or issuing identifiers clearing through multiple processors.

VSS Documentation

This documentation is available:

- *VisaNet Settlement Service (VSS) User's Guide, Volume 1, Specifications*
- *VisaNet Settlement Service (VSS) User's Guide, Volume 2, Reports*
- *VisaNet Settlement Service (VSS) Quick Look Cards*

Chapter 10

VisaNet Copy Request & Fulfillment Service

VisaNet Copy Request and Fulfillment Service (VCRFS) allows clients to request and deliver sales drafts via electronic exchange of Request for Copy (RFC) messages among issuers and acquirers.

Note: RFC messages are no longer valid and will be returned as of October 2021.

VCRFS Overview

VisaNet Copy Request and Fulfillment Service (VCRFS) is an automated process that supports the electronic exchange of Request for Copy (RFC) message-related information among issuers and acquirers. VCRFS increases the efficiency of a client's back-office operations, reduces operating costs, and improves customer service by facilitating the process of requesting and delivering sales drafts between clients.

Participation in VCRFS is mandatory for all issuers and acquirers in all Visa Regions.

VCRFS-Issuers

VisaNet Copy Request and Fulfillment Service (VCRFS) provides issuers with a complete set of electronic functions that can make their back-office processing simpler and more efficient.

Issuers can:

- Send a Request for Copy (RFC) requesting automated fulfillment.
- Receive RFC advices.
- Receive automated fulfillments.
- Receive nonfulfillment messages.
- Route fulfillment and nonfulfillment details automatically to an imaging system.

VCRFS-Acquirers

VisaNet Copy Request and Fulfillment Service (VCRFS) offers acquirers a full complement of electronic functions to simplify and enhance back-office processing.

Acquirers can:

- Receive Request for Copy (RFC) advices.
- Send automated fulfillments.
- Transfer fulfillments directly from imaging systems to client's host system.
- Send nonfulfillment messages.

VCRFS Description

VisaNet Copy Request and Fulfillment Service (VCRFS) provides both issuers and acquirers with a fast and easy way to send and receive Request for Copy (RFC) transactions.

VCRFS enables clients to request and deliver copies of sales drafts among issuers and acquirers. The issuer sends an RFC and the acquirer responds by sending a fulfillment or nonfulfillment message through VROL.

VCRFS Issuer Perspective

VisaNet Copy Request and Fulfillment Service (VCRFS) provides issuers with faster responses to copy requests, which improves productivity and enhances customer service.

A Request for Copy (RFC) message can be submitted with normal outgoing BASE II interchange or through VROL when a copy of a transaction receipt or other sales document is needed because of a cardholder dispute or fraud inquiry. The message must contain a code asking for an electronic response to the request.

VCRFS enhances the message by adding a unique identification number for tracking purposes. It also verifies that the acquirer can fulfill the request electronically. If the request is for a domestic transaction in a Custom Payment Service (CPS) country (Brazil and the U.S.), it is validated by the Enhanced Chargeback Reduction Service. The enhanced request is then delivered to the acquirer. VCRFS also sends the issuer and acquirer an electronic advice to indicate a request has been initiated. These advices are delivered through VisaNet interchange to VROL, VIEW, or VIEW OnLine.

Upon receipt, the acquirer fulfills the Request for Copy.

VCRFS Acquirer Perspective

Acquirers that participate in VisaNet Copy Request and Fulfillment Service (VCRFS) receive electronic copy requests through their existing VisaNet connection.

Once the acquirer determines by service reports that the issuer is a VCRFS participant, the first step after receiving a copy request is to locate the requested document. The acquirer then scans the document (or imports it from the acquirer's host imaging system) into VROL. If the request cannot be fulfilled, an electronic nonfulfillment message is sent to the issuer to explain the reason. Upon receipt of the response, the issuer can forward a printed copy of the image to the cardholder or follow up in the case of a nonfulfillment message.

VCRFS Benefits

VisaNet Copy Request and Fulfillment Service (VCRFS) not only reduces operating costs, but also significantly improves a client's customer service by expediting the flow and timeliness of information.

This service:

- Delivers copy requests and sales draft copies using BASE II Clearing and guarantees delivery of copies.
- Gives clients the choice of hardware used to connect to the service.
- Scans the sales draft and sends the image, along with identifying data, through VROL (either from merchant to requesting acquirer or from acquirer to issuer).
- Provides reports with VCRFS activity.

VCRFS Features

VisaNet Copy Request and Fulfillment Service (VCRFS) features include guaranteed delivery, fast response time, efficient processing, comprehensive reports, and full customer support. The implementation of VCRFS increases the efficiency of back-office operations, reduces operating costs, and improves customer service.

- **Guaranteed delivery**—VCRFS eliminates the possibility that requests or fulfillments could be lost in the mail because copy requests and fulfillments are delivered electronically. Each copy request is assigned a unique identification number through VCRFS that follows it throughout the request and fulfillment lifecycle and provides an audit trail that saves time and resources in researching requests.
- **Fast response time**—The electronic delivery of fulfillments significantly improves response time as compared to mail delivery. Acquirers receive requests faster and the time it takes to deliver copies or nonfulfillment messages to issuers is quicker, which enables a faster response to cardholder inquiries, resulting in improved cardholder service.
- **Efficient processing**—As an automated delivery mechanism, VCRFS positively affects back-office operations through reduced float and lower processing costs. By shortening the time it takes for issuers to receive fulfillments and resolve disputes, provisional credits remain in cardholder accounts for less time, so float costs are lower. Processing costs are also reduced because fewer manual resources are needed to process, track, and research requests.
- **Full customer support**—VCRFS participants have access to ongoing support from Visa. VCRFS support is available to provide any information or assistance needed.

VCRFS Delivery Options

VisaNet Copy Request and Fulfillment Service (VCRFS) can be accessed and utilized in multiple ways.

VCRFS-VROL

Visa Resolve Online (VROL) is a web-based service that enables issuers to create Request for Copy (RFC) messages. When an RFC transaction is sent either from VROL or from a client's host-based VisaNet connection, RFC advices (one for the issuer and another for the acquirer) are created and delivered to VROL.

Issuers and acquirers see RFC advices in the work queues. Acquirers can use VROL to respond to RFC advices by either fulfilling with a copy of the sales draft or sending a nonfulfillment message.

The Bulk SI Upload Request file contains dispute information sent to the opposing clients. The issuer's file contains outgoing RFC transactions. The acquirer's file contains outgoing RFC fulfillments and nonfulfillments.

The Bulk SI Download Response is generated automatically; however, it can be rerequested by a Bulk SI Download Request. It contains a client's incoming Dispute Resolution forms and supporting documentation.

- For an issuer, this contains incoming documents related to RFC fulfillments and nonfulfillments, RFC advices, RFC rejects.
- For an acquirer, this contains RFC advices.

VCRFS-VisaNet

VisaNet is comprised of networks and systems, through which Visa delivers authorization clearing and settlement services to clients.

For VisaNet Copy Request and Fulfillment Service (VCRFS), VisaNet provides:

- Delivery and validation of Request for Copy (RFC) messages between issuers and acquirers.
- Delivery of RFC advices to issuer and acquirer workstations.
- Support for the various routing options that allow VCRFS to facilitate client workflows.

VCRFS Requirements

Clients must define their business requirements for implementing VisaNet Copy Request and Fulfillment Service (VCRFS). Visa works with clients to develop project plans and complete required tasks for successful implementation.

BASE II issuers must change their host processing environments to support automated Request for Copy (RFC) messages by setting the TC 52 (TCR 1, Position 39) with a value of **1**.

Acquirers automatically have TCR 1, Position 40 populated with the value of "1". Therefore, acquirers do not have to do coding changes to their host environments.

VCRFS Fees and Charges

VisaNet Copy Request and Fulfillment Service (VCRFS) sets fees and charges for using the service. Fees are the amounts clients pay each other. Charges are amounts clients pay to Visa for the use of the service.

VCRFS Fees

VisaNet Copy Request and Fulfillment Service (VCRFS) charges retrieval request and incentive request fees to clients that use the service.

Retrieval Request Fees

Issuers pay Retrieval Request fees to the acquirer for each request made. VisaNet collects and disburses the fees to both the issuer and the acquirer. The fees are determined and settled during the central processing date (CPD) cycle in which the request is processed.

Incentive Fees

Issuers pay incentive fees to the acquirer. VisaNet generates fees according to the timeliness of an acquirer's response to an RFC item that is non-U.S. domestic (where only one side is in the U.S. or both parties are outside the U.S.). For U.S. Domestic, incentive fees are included in the Interchange Reimbursement Fee of the actual TC 52 based on the request reason code.

Each fee is based on the elapsed time between the central processing date (CPD) of the retrieval request and the CPD of the response. The shorter the response elapsed time for the response, the higher the incentive fee.

VCRFS Charges

Visa assesses VisaNet Copy Request and Fulfillment Service (VCRFS) participants per-transaction charges for the messages routed through VisaNet from their VCRFS endpoints.

Chapter 11

Visa Resolve Online Service

Visa Resolve Online (VROL) is a web-based service that automates and simplifies the dispute resolution process.

Information about VROL, including VROL service overview, features, enrollment, contact information, and available client documentation, can be found via **Visa Access > Operations > Dispute Resolution > Visa Resolve Online**.

VROL Overview

Visa Resolve Online (VROL) enables clients to retrieve transaction information online, exchange information electronically, send and receive financials and other types of dispute transactions, report fraudulent transactions, manage Account Screen Authorization File (ASAF) listings, submit arbitration and compliance cases, and receive rulings online.

VROL helps clients manage Visa transaction disputes and prevent disputes from occurring. VROL is designed to expedite information exchange and guide users through all stages of a dispute, from the initial inquiry through final resolution.

Clients may choose to have dispute financials generated through VROL. VROL routes these as BASE II financial transactions or as V.I.P. full financial transactions based on client configurations set up in VROL. Clients may also choose to send dispute transactions from their host system directly to BASE II or V.I.P. as full financial messages.

Acquirers, issuers, and their processors in all Visa regions are required to use VROL to exchange dispute-supporting documentation.

VROL Description

Visa Resolve Online (VROL) offers a set of user interface screens to manage the dispute process and a set of systems interfaces (SI) to allow integration with the client's existing applications on their host systems.

VROL is accessible through these interface alternatives:

- **User Interface:** Provides a set of user interface screens that are accessible from a common web browser.
- **Real-Time Systems Interface (RTSI):** Allows custom system integration between a client's system and VROL through a set of real-time web services.
- **Bulk Systems Interface (Bulk SI):** Uses one of Visa's secure file delivery services.

The user interface is the quickest and simplest interface to implement because it accesses VROL over an existing internet connection maintained by the client, which means there is no in-house system customization required.

VROL is accessible through a link on the main page of each regional Visa Access website. Each user must have a valid Visa Access user ID that entitles them to access VROL. Clients must log in and be authenticated by Visa Access before accessing VROL.

The Visa Administrator or Client Site Administrator grants VROL users access to specific features based on their assigned roles, as determined by the client's region. Details about each user and their roles are maintained in the user profiles.

VROL Features

Visa Resolve Online (VROL) provides a number of key features that automate and simplify the dispute resolution process. The use of these features is limited to the interface alternative selected by the client.

Retrieve and View Transaction Information

VROL enables clients to create a Transaction Inquiry (TI) to search Visa's databases for original authorization and settlement transaction information. Clients can obtain a history of activity for an account through the user interface for fraud reporting and Account Screen Authorization File (ASAF) listing transactions. Client staff can make inquiries about cardholder or merchant transactions and receive enough detailed information to help determine whether or not there is a dispute.

For merchants subscribed to the Order Insight service, digital receipts are available for transaction inquiries on both transactions processed by Visa and not processed by Visa.

Exchange Information Electronically

VROL automates the exchange of documentation and information, to improve and speed up the dispute process. Dispute resolution and imaging can be completed online by the issuer with information from the cardholder and then submitted to the acquirer electronically.

Imaging is the function that adds scanned documents to case folders or attaches them to questionnaires as supporting documentation, eliminating the mailing and matching of documentation.

Manage Work Flows

VROL provides work queues and reporting to manage and track cases. This process helps clients reconcile the delivery of the questionnaire and any attached documentation with the actual clearing of a dispute financial or dispute response financial.

VROL provides basic user, billing, processing, reconciliation, and statistical reports available for online display, print, or as data files for export to other applications.

Report Fraud Transactions

VROL enables clients to report fraudulent transactions to Visa. Fraud disputes deal with issues related to the compromise of account data; for example, stolen card numbers.

Manage Account Screen Authorization File (ASAF) Listings

VROL enables issuers to manage Account Screen Authorization File (ASAF) listings, including querying the Account Screen Authorization File (ASAF), creating new Account Screen Authorization File (ASAF) listings, updating or deleting existing listings, and repairing rejects.

Facilitating Rapid Dispute Resolution

Enables issuers to address cardholder disputes or facilitate an expedited resolution to resolve the dispute before entering into a dispute process.

Processing Disputes

The dispute process allows issuers to submit one or more transactions for the correction of liability as per the *Visa Core Rules and Visa Product and Service Rules*.

Dispute Pre-Filings

Dispute pre-filing is used to refer to dispute-related Pre-Arbitration or Pre-Compliance questionnaires found in VROL. Clients can initiate Pre-Filing questionnaires in VROL by continuing a dispute from within an existing case or by initiating a Pre-Compliance from TI. The user must give a reason for initiating the Pre-Filing. The opposing party will send a Pre-Filing response, either accepting full or partial liability or continuing the process by declining the Pre-Filing.

Dispute Case Filings

A client files dispute Arbitration and Compliance cases with Visa via VROL. This is used when clients have not been able to resolve a dispute without Visa's involvement.

Process Financial Transactions

VROL enables clients to create dispute financial, dispute response financial transactions, and corresponding reversal transactions to BASE II. This capability eliminates the need for clients to enter the financial transaction and original questionnaire in separate applications.

Financial transactions created in VROL that are rejected by BASE II can be corrected in VROL.

File Arbitration and Compliance Cases

VROL provides an online facility for the filing arbitration and compliance cases with Visa that includes notifying the opposing client as part of the process. Online notification of final decision to both parties helps complete the process in a timely manner.

VROL Requirements

Visa Resolve Online (VROL) clients must select the interface alternative and network connectivity that best meets their requirements. To implement VROL, there are several network, hardware, and software requirements clients must fulfill depending on the interface alternatives and network connectivity option they have selected. There are other requirements, including User ID and Routing ID.

VROL distinguishes between issuers, acquirers, and dual organizations. Clients who have separate issuer and acquirer organizations must have two Visa Access user IDs to access VROL, one as an issuer, the other as an acquirer. Clients in a dual organization need only one Visa Access user ID.

The VROL Routing ID (RID) is used by VROL to uniquely identify a client or processor organization and by BASE II to route transactions.

VROL Client Testing

Testing is available to clients implementing one or both of the systems interfaces available in Visa Resolve Online (VROL). Testing is not generally available to clients using only the VROL user interface.

Chapter 12

Custom Payment Service/ATM

Custom Payment Service (CPS)/ATM is an incentive program that provides accurate settlement and improved management of cardholders' accounts through better matching of messages, from authorization through clearing, using a unique transaction identifier (TID).

CPS Overview

Custom Payment Service (CPS) ensures more timely delivery of clearing records by acquirers and reduces exception item processing by improving transaction integrity and lifecycle control.

Custom Payment Service (CPS) is available to clients in all Visa regions.

CPS Description

Custom Payment Service (CPS) is a set of special transaction processing requirements that help clients reduce fraud and exception item processing costs by providing features, such as improved transaction integrity and lifecycle control. CPS supports domestic POS and international ATM transactions with custom rate incentive programs for different markets.

CPS/ATM-Qualified

International ATM transactions that meet Custom Payment Service (CPS)/ATM requirements in BASE II are considered CPS/ATM-compliant and the acquirer receives the Tier II cash disbursement fee.

- Authorization Characteristics Indicator (ACI) equals **E**.
- Requested Payment Service (RPS) equals **9**.
- Validation code in authorization message and clearing record match.
- Clearing timeliness equals 3 calendar days or less.
- Amount tolerance factor is met. the amount submitted in BASE II must be equal to or less than the amount authorized.

Note:

Full Service ATM acquirers' technical processing of international ATM transactions automatically qualifies for CPS/ATM and Tier II requirements; however, Full Service acquirers will need to qualify and meet the CPS/ATM business requirements in order to comply with *Visa Core Rules and Visa Product and Service Rules*.

Non-Participation in CPS Liability

Participation for international ATM clients is mandatory, ATM clients participate in either the:

- CPS ATM service, containing a qualified CPS ATM authorization message, followed by a CPS ATM BASE II clearing record.
- Or
- As a certified Full Service ATM endpoint (V.I.P. full financial ATM message) and be in compliance with the Tier II requirements.

Acquirers that meet both the business and technical qualifications receive Tier II fees.

Full Service processing of ATM transactions is mandated for all ATM acquirers in the U.S. Region and for all new ATM acquirers in all regions.

All existing ATM acquirers must be certified to use Full Service ATM processing or participate in the CPS ATM service.

International ATM transactions that do not meet these criteria lose all international cash disbursement fees. International acquirers that do not qualify for Full Service ATM or for the CPS/ATM service are assessed a minimum special ATM handling fee of USD.\$5.00.

CPS Benefits

Custom Payment Service (CPS) is a set of special transaction processing requirements that help clients reduce fraud and exception item processing costs by providing features, such as improved transaction integrity and lifecycle control.

CPS supports domestic POS transactions and international ATM transactions with custom rate incentive programs for different markets.

CPS Process Flow

Custom Payment Service (CPS) uses dual-message processing in which the acquirer submits separate authorization and clearing transactions. CPS/ATM acquirers must submit ATM authorization messages through the V.I.P. system and the clearing transaction through the BASE II system.

Note:

All international ATM transactions must qualify for CPS/ATM processing. Issuers not participating in CPS/ATM are not required to make changes either to their authorization systems or to their clearing systems. They will not receive any new fields relating to CPS/ATM.

The CPS/ATM process flow steps:

1. The acquirer submits an authorization request with an authorization characteristics indicator (ACI) to indicate that this is a CPS transaction.
2. V.I.P. ensures that the transaction meets the criteria and evaluates the characteristics of the request, then replaces the ACI value to reflect the authorization characteristics of the transaction.

V.I.P. generates a unique TID and passes it with the ACI to the issuer (if the issuer is participating in CPS/ATM and chooses to receive it).
3. The issuer approves the transaction, sends the transaction back to V.I.P. with the authorization response code, and optionally returns the ACI and the TID.
4. V.I.P. generates a validation code based on key authorization fields. V.I.P. forwards the validation code with the ACI, the TID, and the authorization response code to the acquirer.

Note:

The validation code is not present in advices to issuers.

5. The acquirer submits a clearing message to BASE II with a Requested Payment Service (RPS), which specifies CPS/ATM. The clearing transaction contains the ACI, TID, and validation code received from V.I.P. with the authorization response.
6. By regenerating the validation code, BASE II validates that the key authorization fields contained in the clearing transaction match those sent in the authorization request and in the CPS fields received in the outgoing response. BASE II applies edits to ensure that the transaction qualifies for CPS/ATM.

VisaNet then forwards the transaction to the issuer with the ACI, TID, and RPS (if the issuer participates in CPS/ATM).

The issuer uses the TID to match the clearing transaction to the authorization.

CPS Clearing

Custom Payment Service (CPS) uses an authorization message, followed by a BASE II clearing record. All Visa and Plus international ATM transactions using the CPS ATM Service must be authorized through the V.I.P. system and cleared through BASE II.

Each CPS/ATM clearing message must include the transaction identifier (TID), an authorization characteristics indicator (ACI) with the value **E**, the validation code from the authorization response, and key CPS/ATM fields used to develop the validation code. All authorization requests must have been approved and the clearing message submitted within three calendar days from the transaction date.

Additionally, the clearing transaction must include the acquirer's request for the CPS/ATM fee through the use of the Requested Payment Service (RPS). The RPS value **9** indicates that the transaction complies with the CPS/ATM authorization and clearing qualification rules.

To qualify a transaction for CPS/ATM, V.I.P. performs a validation code check to ensure that key authorization and clearing data elements are identical. A successful validation confirms that the transaction has been properly authorized.

Authorization vs. Clearing Amounts

Authorization amounts must be equal to (but can be greater than) their clearing amounts. If the source amount in the TCR 0 – BASE II Draft Data, is equal to or less than the authorized amount in the TCR 5 – Payment Service Data, BASE II allows the transaction to be eligible for the CPS/ATM Tier II rate. This processing supports misdisbursement transactions.

Reclassified Transactions

BASE II reclassifies transactions that do not meet the authorization-related edits for CPS/ATM as non-CPS transactions. BASE II updates the ACI to **X**, indicating that the transaction does not qualify for CPS/ATM. If a transaction fails to qualify for CPS/ATM, VisaNet applies non-CPS cash disbursement fees.

Nonqualified CPS ATM transaction under the 3-calendar-day limit may be returned to the acquirer for fixing and resubmission. The acquirer does not receive the international cash disbursement fee for returned CPS ATM nonqualified transactions.

Delivering the Transaction to the Issuer

If the issuer participates in CPS/ATM, a TCR 5 record is included as part of the draft data. This record includes the TID.

Special ATM Handling Fee

Visa charges a special USD\$5 handling fee for international ATM transactions not submitted for CPS/ATM consideration or that fail to meet CPS/ATM fee edit criteria. Visa does not charge the fee on a dispute response financial and does not return it to the acquirer on a dispute financial. The fee appears on the monthly Integrated Billing statement in addition to any other special ATM handling fees, such as that for submitting a transaction for clearing after the 3-calendar-day limit.

Data Requirements for Transaction Processing

This table identifies key Custom Payment Service (CPS) fields and explains how the authorization fields are carried forward by the acquirer to populate the related fields in the BASE II records. It also shows which key CPS/ATM fields are used to develop the validation code.

BASE II Transaction Fields		Used to Determine Validation Code
BASE II Field	Field Name	
TCR 0, positions 5–20	Account Number	Yes
TCR 0, positions 1–2	Transaction Code	
TCR 1, position 130	ATM Account Selection	Yes
TCR 5, positions 20–31	Authorized Amount	Yes
TCR 0, positions 133–136	Merchant Category Code	
TCR 0, positions 162–163	POS Entry Mode	
TCR 0, positions 28–33	Acquiring Identifier (in the Acquirer Reference Number)	Yes
n/a	n/a	
TCR 0, positions 152–157	Authorization Code	
TCR 5, positions 35–36	Authorization Response Code	
TCR 1, positions 96–103	Terminal ID	
TCR 1, positions 81–95	Card Acceptor ID	
TCR 0, positions 92–116	Merchant Name	Yes
TCR 0, positions 117–129	Merchant City	
TCR 0, positions 130–132	Merchant Country Code	
TCR 5, positions 32–34	Authorization Currency Code	Yes
n/a	n/a	

BASE II Transaction Fields		Used to Determine Validation Code
BASE II Field	Field Name	
TCR 0, positions 142–144	Merchant State/Province Code (U.S. region only)	
TCR 0, position 151	Authorization Characteristics Indicator (ACI)	Yes
TCR 5, positions 5–19	Transaction Identifier	Yes
TCR 5, positions 37–40	Validation Code	
TCR 0, position 168	Reimbursement Attribute	
TCR 4, positions 51–58	Surcharge Amount	Yes

Transaction Identifier

All transactions contain a TID in the TCR 5. V.I.P. assigns a TID to all transactions at the time of authorization. BASE II assigns a TID to all transactions.

Issuer processing centers receive the TCR 5 in incoming original ATM cash disbursement transactions. An issuer receiving the TID is able to better manage cardholders' open-to-buy balances by matching authorization and clearing records. If the issuer receives the TID in the TCR 5, it must retain it and return it in subsequent dispute financial.

CPS Requirements

In order to use Custom Payment Service (CPS), issuers and acquirers must meet the business and technical qualifications.

Issuers pay a fixed disbursement fee to the acquirer for each approved international Visa or Plus ATM cash disbursement. The acquirer receives an ATM cash disbursement fee if the transaction meets all the requirements for Tier II qualification.

Acquirers must have completed the necessary service certifications, demonstrating business and technical qualifications. Following successful certification, qualification flags are set in the VisaNet system.

This table displays the business and technical qualifications for Tier II fees.

Qualification Type	Qualification Description
Business	Support for these: • Provide cash withdrawal • Use Visa Routing Table and Plus Routing Table (if all transactions are not routed to Visa) • Accept all valid cards • Comply with the <i>Visa Core Rules and Visa Product and Service Rules</i>, including, but not limited to: – Reversal processing – No expiration date editing • Comply with acquirer quality service levels • Comply with customer account selection option standards
Technical	Acquirer must participate in Full Service ATM processing or meet CPS/ATM standards on each V.I.P. authorization message and BASE II clearing record: • V.I.P. – Meet CPS/ATM edit requirements – Participate in the CVV Service – Use the Multicurrency Service for authorization requests • BASE II – Match authorization and clearing message data (account number, account selection processing code, transaction amount, acquirer institution ID, merchant country code, and currency code) – Provide payment service fields, transaction identifier, validation code, authorization characteristics indicator, and required payment service fields – Supply ATM name and location fields – Process authorization through VisaNet if transaction cleared through VisaNet – Clear transaction within three days

If an ATM acquirer does not meet the Tier II standards, Visa may, at its discretion, remove the ATM acquirer from the Tier II level.

Custom Payment Service (CPS) acquirers and issuers must be certified to send and to receive CPS-specific fields. For international and intraregional CPS/ATM transactions, participation is mandatory. For domestic transactions, see your Visa Representative for domestic requirements by region.

Tier I requirements for acquirers include authorization and clearing processing through the same network.

Tier II requirements for acquirers include:

- Meet Tier I requirements
- Participate in Full Service ATM processing or, for dual-message acquirers, participate in the CPS/ATM Service.
- Comply with the Visa international acquirer quality service standards.
- Participate in the Card Verification Value (CVV) Service. Plus authorization requests must include code **05** or **90** in Field 22—Point-of-Service Entry Mode Code to be eligible for CVV checking.
- Comply with customer account selection standards.

CPS/ATM Documentation

This documentation is available:

- *V.I.P. Authorization-Only Online Messages Processing Specifications*
- *VisaNet Authorization-Only Online Messages Technical Specifications*
- *Full Service ATM Online Messages Processing Specifications (International)*
- *Full Service ATM Online Messages Technical Specifications*
- *V.I.P. System SMS Processing Specifications (U.S.)*
- *Visa Global ATM Member Guide*

Chapter 13

Custom Payment Service/POS

Custom Payment Service (CPS)/POS is a series of payment services customized to serve the needs of distinct merchant segments for point-of-sale (POS) transactions.

CPS/POS Overview

Custom Payment Service (CPS)/POS accommodates different merchant procedures and decreases fraud losses and operating expenses associated with each transaction. CPS/POS increases Visa client profitability by reducing client operating costs, improving risk management techniques, and increasing client revenues through increased card usage.

CPS/POS transactions can be processed as dual messages through V.I.P. and BASE II.

CPS/POS is available to clients in the U.S. region and Brazil.

Note:

United States (U.S.) participants should refer to the *U.S. Interchange Reimbursement Fee (IRF) Rate Qualification Guide*.

CPS/POS Description

Custom Payment Service (CPS) is a set of special transaction processing requirements that helps clients reduce fraud and exception item processing costs by providing features, such as improved transaction integrity and lifecycle control. Custom Payment Service (CPS) acquirers must submit authorization messages through the V.I.P. system and the clearing transaction through the BASE II system.

This service description describes the CPS/POS incentive program processing requirements for card-present (CP) and card-not-present (CNP) transactions. This service description applies only to non-U.S. participants, and only for domestic national transactions. The national market covered is Brazil.

CPS/POS National Market Program: Brazil

To qualify for the CPS/POS program and for its associated rate, a transaction must have these characteristics.

- Card present, cardholder signature obtained.
- One authorization per clearing transaction is allowed.
- Complete and unaltered contents of Track 1 or Track 2 of the card's magnetic stripe must be read and be transmitted.
- The POS entry mode code in field 22 must be **90** or **05** and must be included in the authorization request.
- Purchase date (transaction date) must be the authorization date.
- Transaction must clear within two days.
- For retail transactions including petrol, the clearing amount must equal the authorized amount.
- For restaurant transactions, the clearing amount must be equal to or not more than 20% above the authorized amount.
- The merchant category code must be valid:
 - Petrol must be **5541**.
 - Restaurant must be **5812** or **5814**.
- The validation code must be present and valid. For CPS originals and their reversals, this field must contain the validation Code received in the authorization response, (V.I.P. Field 62.3). If multiple authorizations were supplied, this field must contain the validation code from the *first* authorization response.

This table identifies the Custom Payment Service (CPS)/POS programs for the national markets of Brazil. This national market processes CPS/POS transactions with an authorization message followed by a BASE II clearing record:

CPS/POS Program	Description	Card Present	Card Not Present
Brazil			
CPS/Retail	For retail transactions, including those initiated at restaurants at which the magnetic stripe is read.	Yes	

CPS/POS Qualifications

A transaction must qualify for a Custom Payment Service (CPS)/POS program in its national market to receive authorization-related chargeback protection. It must then meet additional requirements to qualify for an applicable CPS rate.

All CPS/POS clearing transactions must contain the TID, the ACI, and the validation code from the authorization response. For transactions with multiple authorizations, with an authorization reversal, or with both, these data elements must be identical to those contained in the first authorization response.

The clearing transaction must include the acquirer's request for a specific CPS/POS program using the RPS field. The RPS value indicates that the transaction complies with the set of authorization and clearing qualification rules for the CPS program requested.

To qualify a transaction for any CPS/POS program, a validation code check ensures that certain authorization and clearing data elements are identical. A successful validation confirms that the transaction has been properly authorized.

CPS/POS to Non-CPS/POS Reclassification

Transactions that fail to meet Custom Payment Service (CPS)/POS program edit requirements are reclassified as non-CPS transactions.

Acquirers may choose to have these transactions returned. If acquirers do not request this option, these transactions are reclassified and are delivered to the issuer. BASE II updates the ACI to **X**, which indicates that the transaction does not qualify as a CPS transaction. An advice is sent to the acquirer as notification that the transaction was reclassified.

Reclassified transactions do not receive authorization-related chargeback protection and are not eligible for the requested CPS rate, but do continue to receive lifecycle control.

Transactions that are reclassified as non-CPS receive the next best rate that can be applied. The reimbursement attribute (RA) is updated to reflect the appropriate rate applied. The requested payment service (RPS) field is updated to a space.

Some national market participants participate in the National Net Settlement Service (NNSS). In those cases, the NNSS rate structure applies. Refer to the appropriate regional Operating Regulations for more information.

CPS/POS Fee Changes

Transactions that meet the requirements for the requested Custom Payment Service (CPS)/POS program, but fail to qualify for the CPS/POS rate, receive a fee change.

- The transaction fails to meet the clearing timeliness requirement for the CPS/POS rate associated with the program.
- The transaction fails to meet industry-specific clearing data requirements.

Transactions that meet the requirements for the requested CPS/POS program, but fail to qualify for the CPS/POS rate, still receive authorization-related chargeback protection. The RA is updated to reflect the appropriate rate applied. The RPS value and the ACI are not changed when a transaction qualifies for a particular CPS/POS program, but does not qualify for a CPS/POS rate.

All transactions that qualify for the CPS/POS program requested, but that fail either the clearing timeliness or the industry-specific data requirements for the interchange reimbursement fee (IRF) associated with the requested CPS program receive a fee change to the next best rate that can be applied. The RA is updated to reflect the rate at which the transaction was settled. The requested payment service field is not updated.

For specific national-market CPS information about fee changing circumstances and procedures, clients can contact their Visa representatives.

CPS/POS Ineligible Transactions

Ineligible Brazil national-market Custom Payment Service (CPS)/POS transactions are those not specifically identified as CPS/Retail purchases, including those for petrol and restaurant purchases. Ineligible transactions include cash disbursements and quasi-cash transactions. Credit vouchers are considered to be eligible transactions.

CPS/POS Amount Tolerance

Authorizations initiated by certain merchants, such as hotels and car rental companies, contain amounts that may vary significantly from the final transaction amounts. A transaction that fails

to meet the amount tolerance edits is reclassified or is returned. This table identifies the Custom Payment Service (CPS)/POS amount tolerances for Brazil.

CPS/POS Program	Amount Tolerance
Brazil (through October 2007)	
CPS/Retail, including Petrol	Clearing amount must equal the authorization amount.
CPS/Retail, Restaurant	Clearing amount must be equal to or no greater than 20% of the authorization amount.

Note:

After October 2007, Brazil eliminated CPS fees, but retained CPS lifecycle processing.

CPS/POS Program Clearing Times

Each Custom Payment Service (CPS)/POS program has clearing times between the transaction date and the BASE II central processing date (CPD).

This table lists the clearing times for specific CPS programs.

Custom Payment Service	National Market Clearing Times
	Brazil (through 10/07)
CPS/Retail Credit Card	2 days (retail, petrol, restaurant)
CPS/Card Not Present ¹	Not Applicable

¹ Clearing timeframe measured from the date of goods shipment, checkout, or return date, as opposed to the authorization date.

In calculating clearing times, the transaction date, BASE II CPD, Sundays, and specific holidays are not included. Holidays depend on the national market.

CPS Qualification Schedules

Refer to the *Visa Core Rules and Visa Product and Services Rules* for processing timeframes between transaction dates and clearing central processing dates for the Custom Payment Service (CPS). Also refer to the *Visa Core Rules and Visa Product and Services Rules* for holidays and for other days that may be excluded from the clearing time calculations.

CPS Chargeback Reduction Service

A chargeback is a transaction an issuer returns to an acquirer. An acquirer can accept financial liability for the transaction, or can re-present the transaction. The Chargeback Reduction Service (CRS) is designed to reduce client costs associated with dispute processing.

CPS Authorization-Related Chargeback Protection

Qualified Custom Payment Service (CPS) transactions are stored in a history database and are systematically protected by the Chargeback Reduction Service (CRS) from invalid authorization-related chargebacks.

These chargebacks fall into eight categories based on the authorization characteristics and the merchant category of the original purchase transactions (travel and entertainment [T&E] or non-T&E). CRS determines which category of protection a CPS transaction receives by using the ACI, the RPS, and the merchant category code (MCC).

CRS Life-Cycle Control

BASE II provides Chargeback Reduction Service (CRS) lifecycle control for all Brazil domestic purchase transactions. All purchase exception transactions (and the associated RFCs) are suspended for one settlement cycle. During this suspension, the exception transaction is checked against the history of the transaction to ensure that processing rules for the sequence of the transaction have been followed, and to determine whether the exception transaction complies with the *Visa Core Rules and Visa Product and Services Rules* for the different regions.

CPS/POS Benefits

Custom Payment Service (CPS)/POS helps clients reduce fraud and exception item processing costs.

For acquirers, CPS protects against authorization-related chargebacks by requiring the authorization request to contain certain key information that might not otherwise be present to provide a more complete picture of the transaction conditions and to help validate cardholder authenticity.

For issuers, CPS increases risk control and improves account balance management. Issuers can accurately match a transaction's clearing and authorization messages using a unique transaction identifier (TID) that is assigned by the Visa Integrated Payment (V.I.P.) System.

CPS/POS Process Flow

Custom Payment Service (CPS) POS acquirers must submit authorization messages through the V.I.P. system and the clearing transaction through the BASE II System.

The Custom Payment Service (CPS)/POS process flow:

1. The acquirer submits an authorization request with an authorization characteristics indicator (ACI) to indicate that this is a CPS transaction.
2. The V.I.P. System ensures that the transaction meets the CPS/POS criteria and evaluates the characteristics of the request, then replaces the value of the ACI to reflect the authorization characteristics of the transaction.

V.I.P. generates a unique transaction identifier (TID) and passes it to the issuer if the issuer is certified and has chosen to receive it. VisaNet may also pass the ACI.
3. The issuer approves the transaction, sends the transaction back to the V.I.P. System with the authorization response code, and optionally returns the ACI and TID.
4. The V.I.P. System generates a validation code based on key authorization fields. VisaNet forwards the validation code with the ACI, TID, and the authorization response code to the acquirer.
5. The acquirer submits a clearing message to BASE II with a value in the Requested Payment Service (RPS) field, which specifies the CPS program desired. The clearing transaction also contains the ACI, TID, and validation code received from V.I.P. with the authorization response.
6. BASE II validates that the key authorization fields in the clearing transaction match those used in the authorization message by recalculating the validation code. BASE II applies edits to ensure that the transaction qualifies for the requested CPS program.

The issuer uses the TID to match the clearing transaction to the authorization.

CPS/POS Requirements

Custom Payment Service (CPS)/POS testing requirements are generally the same for the different national markets. Clients can contact their Visa representatives to make arrangements for testing.

CPS sets clearing data requirements common to all CPS/POS transactions. Additional requirements specific to each CPS/POS program are listed in the section describing that particular program.

All clients in a CPS country must participate in CPS.

All Custom Payment Service (CPS)/POS transactions must have:

- A valid reimbursement attribute (RA) for the requested CPS/POS program.
- Valid combinations of these for CPS/POS programs:
 - Requested payment service (RPS) field and authorization characteristics indicator (ACI)
 - Requested payment service (RPS) field and cardholder ID method
 - Requested payment service (RPS) field, authorization characteristics indicator (ACI), POS entry mode code, and reimbursement attribute (RA)
 - Requested payment service (RPS), authorization characteristics indicator (ACI), and transaction code qualifier (TCQ)

CPS/POS Data Element Values

This table lists the valid data element combinations for the Brazil Custom Payment Service (CPS)/POS markets.

Requested Payment Service (RPS)		Authorization Characteristics Indicator (ACI)		Reimbursement Attribute	Cardholder I.D. Method
Code	Definition	Code	Definition		
Brazil					
A	CPS/Retail, including Petrol	A	Card present	A	1
		E	Card present with merchant name and location data	A	1
B	CPS/Retail—Restaurant	A	Card present	A	1
		E	Card present with merchant name and location data	A	1

CPS/POS Validation Code Protected Fields

For approved transactions, V.I.P. computes a validation code using key elements in the authorization message. The values of the protected data fields' key elements must be submitted unchanged from authorization through clearing. If any values are changed from authorization through clearing, the transaction is reclassified to a non-Custom Payment Service (CPS) transaction or is returned.

This table lists the protected fields.

V.I.P. System Field	BASE II Field (Definition)
2	Account Number
4	Authorized Amount
18	Merchant Category Code
22 (positions 1–2)	POS Entry Mode
38	Authorization Code
39	Authorization Response Code
49	Authorization Currency Code
61.1	Other Amount, Transaction, Cashback ¹
62.1	Authorization Characteristics Indicator
62.2	Transaction Identifier
62.4	Market-Specific Authorization Data Indicator
62.23	Product ID

¹ If the specified subfield is not present, V.I.P. substitutes the default value, which must be provided in the clearing transaction sent to BASE II.

CPS/POS Key Data Requirements

In addition to meeting the common Custom Payment Service (CPS)/POS data requirements, a transaction in the Brazil national market must also meet the key data requirements in this table to qualify for the CPS/Retail program and for its associated rate.

Data Element	BASE II Field	Value
Merchant Category Code (MCC)	TCR 0, positions 133–136	Must be same as in authorization.
POS Entry Mode	TCR 0, positions 162–163	90, 05
Authorization Characteristics Indicator (ACI)	TCR 0, position 151	A or E
Transaction Code Qualifier (TCQ)	All TCRs, position 3	0
Requested Payment Service (RPS)	TCR 0, position 145	A for Retail including Petrol B for Restaurant
Authorization Date	TCR 5, positions 5–19 ¹	In transaction identifier
Purchase Date	TCR 0, positions 58–61	Within 1 day of authorization date
Cardholder ID Method	TCR 0, position 160	1

Data Element	BASE II Field	Value
Central Processing Date (CPD) <i>only through 10/07</i>	TCR 0, positions 164–167	Within 2 days of purchase date
Reimbursement Attribute (RA) <i>only through 10/07</i>	TCR 0, position 168	A
Card Acceptor ID	TCR 1, positions 81–95	Must not be spaces or zeros.
Authorization Source Code	TCR 1, Position 128	Must not be a space.
Transaction Identifier	TCR 5, positions 5–19	Must not be zeros.
Authorized Amount	TCR 5, Positions 20–31	Must not be zeros
Authorization Currency Code	TCR 5, positions 32–34	Must be the same as Source Currency Code.
POS Terminal Capability	TCR 0, position 158	2–5, 8, 9
Validation Code	TCR 5, positions 37–40	Must not be spaces.

1 Embedded in the transaction identifier.

CPS/POS Documentation

This documentation is available:

- *BASE II Clearing Data Codes*
- *V.I.P. Authorization-Only Online Messages Processing Specifications*
- *VisaNet Authorization-Only Online Messages Technical Specifications*
- *PaymentService 2000 Technical Specifications—Brazil (DS-941075)*
- *U.S. Interchange Reimbursement Fee (IRF) Rate Qualification Guide*
- *Visa International VisaNet Merchant Data Standards Handbook*

Chapter 14

Chargeback Reduction Service

The VisaNet Chargeback Reduction Service (CRS) is a worldwide service designed to reduce client costs associated with dispute processing. As part of BASE II Clearing, this service accesses information readily available to Visa, yet time-consuming for clients to research.

CRS Overview

Chargebacks are transactions issuers return through interchange to acquirers. A chargeback results when the presentment of a sales draft, cash advance, debit, or credit transaction is disputed by the customer or examined by the issuer and found to be improper.

Chargeback Reduction Service (CRS) is a function of BASE II Clearing and is mandatory for all Visa clients.

CRS Description

Original purchase and cash transactions entered into interchange through the Edit Package at the acquirer's processing center are not affected by the service until received at the VisaNet

Interchange Center (VIC). Invalid drafts are returned to the acquirer's processing center, lessening chargebacks.

At the issuer's processing center, chargeback transactions pass through Chargeback Reduction Service (CRS) validation during Edit Package processing to ensure that they reflect the processing rules governing minimum amounts, client message text, chargeback reason codes, and number of presentments. Rejected transactions are prevented from entering into BASE II interchange at the VIC.

At the VIC, CRS evaluates sales drafts and chargebacks and inserts status indicators for the validation process. Availability of this information helps issuers and acquirers to substantially reduce research time and potential chargebacks.

CRS Status Indicators

These Chargeback Reduction Service (CRS) status indicators are added to the record under specific conditions.

- **Floor Limit Indicator**—The transaction was above or below the merchant's floor limit for the date of the purchase according to the *Visa Core Rules and Visa Product and Services Rules*. CRS validates these transaction elements to determine the floor limit:
 - Account type
 - Mail/telephone indicator
 - Merchant location
 - Merchant category code
 - Reimbursement fee category
 - Special condition indicator
 - CHIP/non CHIP transactions
 - Transaction date

When the transaction is validated at the VisaNet Interchange Center (VIC), the VIC inserts the appropriate code in TCR 0, field position 24, Floor Limit Indicator.

- **CRB/Exception File Indicator**—The account number on the card used in the transaction was listed in the Card Recovery Bulletin (for international transactions) or Account Screen Authorization File (ASAF) (for U.S. transactions) for the merchant's region on the date of the transaction. CRS evaluates these transaction elements to determine the account status on the Visa card pickup bulletin:
 - Account number
 - Mail/telephone indicator
 - Merchant location
 - Transaction date
 - Authorization code
 - Floor limit indicator

When the transaction is validated at the VIC, the VIC inserts the appropriate code in TCR 0, field position 25, CRB/Exception File Indicator.

- **PCAS Indicator**—The transaction’s authorization code was generated during Visa stand-in authorization (STIP—Stand-In Processor) using the Positive Cardholder Authorization Service (PCAS). CRS evaluates these elements to determine the authorization status:
 - Account number
 - Authorization code
 - Transaction date

When the transaction is validated at the VIC, the VIC inserts the appropriate code in TCR 0, field position 26, PCAS Indicator.

CRS validation does not take place if any data element needed to determine the status of the floor limit, Visa card pickup bulletin, or authorization code is missing from the transaction. BASE II Clearing applies the “validation not performed or insufficient information” indicator code to the transaction.

Note:

The floor limit, bulletin, and authorization indicators edits apply only to original sales drafts, not their second presentments.

CRS Sales Draft Validation

Original purchase and cash draft transactions entered into interchange at the acquirer’s processing center are validated by Chargeback Reduction Service (CRS) during BASE II processing.

Original purchase and cash draft transactions are returned to the acquirer when all three of these conditions are met.

- The account number is listed in the merchant’s Card Recovery Bulletin (CRB) or, for U.S. merchants, the Account Screen Authorization File (ASAF), as of 4:00 GMT on the date prior to the transaction.
- The transaction amount is equal to or below the merchant floor limit.
- No authorization code is present.

CRS Chargeback Validation

Chargeback Reduction Service (CRS) chargeback transactions are validated through outgoing Edit Package processing and BASE II processing at Visa.

Chargeback transactions are first validated at the issuer’s processing center. If a chargeback fails certain Edit Package edits, the chargeback is rejected; the issuer must resolve the problem before the transaction can be a part of outgoing interchange.

Chargebacks included in outgoing interchange are validated at Visa by CRS. CRS forwards valid chargebacks to acquirers.

CRS Edit Package Edits

Edit Package validation of Chargeback Reduction Service (CRS) chargebacks is based on these edits.

- **Client message text**—Transactions with certain T&E chargeback reason codes are rejected if there is nothing in the client message text field.
- **Allowable chargeback reason codes**—Chargebacks not having a reason code compatible with the transaction's merchant category code are rejected. This applies to Express Payment Service (EPS), Automatic Teller machine (ATM), and T&E transactions. For example, an international ATM transaction cannot be returned to the issuer with a chargeback reason code 75 (inadequate message content).
- **Number of presentments**—BASE II rejects certain chargebacks if the number of chargebacks allowed has been exceeded. Acquirers are limited to two presentments, including the original, and issuers are limited to one chargeback.

CRS VIC Edits

Chargeback transactions are analyzed again during BASE II processing at the VisaNet Interchange Center (VIC) for authorization-related data.

Chargeback Reduction Service (CRS) returns invalid chargebacks to issuers under these conditions:

- **Chargeback reason code 72**—The transaction was above the merchant floor limit, but the chargeback reason code indicated that the transaction was below the floor limit; or the reason code indicated that the transaction exceeds the floor limit when the transaction amount is actually under the merchant floor limit. (CRS allows a plus or minus 20% tolerance level, above or below the floor limit, before returning the transaction.)
- **Chargeback reason code 70**—The cardholder account number was not listed in the CRB or Account Screen Authorization File (ASAF) for the merchant's region on the date of the transaction, but the chargeback reason code indicates that it was listed. This is based on the CRB/Account Screen Authorization File (ASAF) history file that is maintained at each VIC.
- **Invalid reason code**—A chargeback of an ATM transaction contained an invalid reason code.

CRS Benefits

Chargeback Reduction Service (CRS) provides many benefits to acquirers and issuers in handling chargebacks.

Acquirer Benefits

With VisaNet Chargeback Reduction Service (CRS), acquirers receive additional information through BASE II to validate chargebacks and Request for Copy (RFC) service messages sent by issuers.

The service automatically returns some invalid, authorization-related chargebacks to issuers on behalf of acquirers, thus eliminating the acquirer's research and processing time for these chargebacks.

Issuer Benefits

With VisaNet Chargeback Reduction Service (CRS), issuers receive additional information through BASE II to assist in making chargeback decisions. With this data, issuers are able to prevent chargebacks, as well as make better-informed chargeback and representment decisions with lower research costs.

The service automatically returns some invalid, authorization-related sales drafts to acquirers on behalf of issuers, thus eliminating the issuer's research and processing time for these sales drafts.

Chapter 15

Fraud Reporting System

BASE II provides the Fraud Reporting System (FRS) to help clients report, track, and analyze fraudulent transactions.

FRS Overview

Fraud Reporting System (FRS) consolidates fraud information, helping clients detect fraud patterns and reduce losses. Participation in FRS is mandatory for issuers and for acquirers.

Fraud reporting by individual clients protects all clients from excessive losses due to fraudulent transactions. Visa consolidates worldwide fraud information to detect fraud patterns and develop new countermeasure programs. Visa works with clients, law enforcement agencies, and government agencies to investigate and curtail fraud sources.

Clients can use Visa Resolve Online (VROL) to report fraudulent transactions to FRS. See *Visa Resolve Online Member Implementation Guide*.

Visa offices use fraud information for:

- Fraud trend analysis.
- International Risk Management (IRM) risk program performance tracking.
- Management fraud activity and fraud summary reports.

- Client reporting compliance monitoring.
- IRM program information support.
- Client data and information queries.
- Region-specific reports distributed to clients.

Transaction data contained on the Visa fraud master file also supports essential Risk Management programs, including:

- Merchant Fraud Performance (MFP)
- Visa Issuer Fraud Detection (VIFD)
- High-Risk File Access/Common Data Infrastructure (CDI)

Clients ensure eligibility for maximum chargeback rights by correctly reporting fraud transactions. It is important that clients report fraud accurately and comprehensively to maximize client protection from fraudulent losses.

FRS Description

The Fraud Reporting System (FRS) offers clients a convenient, electronic way to report and monitor fraud activity.

Fraud occurs when an individual who is not the cardholder or designee uses a card or its account number to obtain goods or services without the cardholder's consent. Fraud also occurs when a card is obtained through misrepresentation of identification or financial status.

The Fraud Advice transaction (TC 40) transmits information about fraudulent transactions to and from clients.

The TC 40 record comprises four transaction component records (TCRs). Use of TCRs 0, 1, 2, and 3 depends on fraud reporting requirements based on specific programs, regional initiatives, or both.

Report New Fraud Transactions

Confirmed fraud transactions must be reported for on-us, domestic, and international transactions.

Fraud reporting is accomplished through these steps.

1. Gather the required fraud advice data as specified by the specific program or regional fraud reporting requirements.
 - Reporting of specific fraud advice information is required by some Visa programs, by regional directive, or both.
 - Some Visa regions require mandatory reporting of specific fraud advice fields.
2. Submit the completed fraud advice to Visa using the appropriate VisaNet system:
 - BASE II
 - V.I.P.
 - Visa Resolve Online (VROL)
3. Verify fraud reporting on the daily or weekly Fraud Activity Reports to determine if the reported transaction was:
 - Accepted.
 - Accepted but requires additional verification or correction and resubmission.
 - Rejected and requires correction and resubmission.

Clients must electronically report all fraud to Visa.

FRS Notification Code Assignment

The Notification Code (NC) field in the fraud advice record in the Fraud Reporting System (FRS) is used to define whether the transaction should be added (NC 1 and NC 2), changed (NC 3), deleted (NC 4), or reactivated (NC 5).

FRS Account Sequence Number Assignment

Fraud transaction Account Sequence Numbers can be assigned by Visa or by the client. A maximum of 1000 transactions can be reported on a given account through the Fraud Reporting System (FRS).

If the client assigns its Account Sequence Numbers, transactions should be assigned in consecutive order as received, beginning with 001. Assigning Account Sequence Numbers in a different (or random) order may make the level of fraud for a given merchant or cardholder appear much higher than it actually is on various Risk Management programs and reports. It may also be difficult to keep track of nonsequentially assigned transactions from both a case management and maintenance basis.

FRS Fraud Investigation Status

Some clients modify the merchant name field in the Fraud Reporting System (FRS) with special codes or symbols to flag fraud transactions for a particular status (for example, charged back, charged off, or investigative results). This action changes the merchant name and significantly interferes with the identification process of the various Visa Risk Management programs, which in turn may negatively affect client chargeback rights.

FRS Reporting Timeframes

Fraud advice records are added to the Visa Fraud Master File on a daily basis and assigned a posting date on the day they are processed by the Fraud Reporting System (FRS). The Posting Date is used to determine how many days from the Transaction Date the transaction is being reported and whether the fraud transaction meets specific reporting timeframes.

Transactions reported after 90 days (but less than 180 days) from the Transaction Date are accepted and added to the Visa Fraud Master File. The transaction is also flagged as late. On the Fraud Activity Report Warning Activity, a Warning Status Code of W20—Purchase Date appears with the record.

Note:

The reporting timeframe is 60 days in the AP Region.

Reporting a transaction late may affect the eligibility of the transaction for acceptance by various Visa programs, such as the Merchant Fraud Performance (MFP) program and chargeback rights.

If additions are attempted after 180 days, FRS rejects the transactions with a Reject Status Code R82—Late Transaction on the Fraud Activity Report Reject Activity.

FRS Transaction Edits

The Fraud Reporting System (FRS) performs all edits on submitted fraud transactions. When adding or modifying a fraud transaction, the transaction is rejected by Visa if one or more fields contain data that does not comply with stated fraud advice record data edit requirements or was submitted using incorrect procedures.

FRS maintains rejected fraud transactions in a separate file instead of applying them to the Fraud Master File. All rejected fraud transactions must be corrected and resubmitted to Visa by the client.

If reporting errors occur, clients are notified of the specific reporting problems in the daily or weekly Fraud Activity Warning and Reject Reports.

FRS Reports

When fraud transactions are submitted to the Fraud Reporting System (FRS), fraud reports are received that verify the status of the client's reporting. The Fraud Activity Reports and Fraud

Summary Reports list reported fraud transactions (additions, changes, deletions, and reactivations) that were correctly submitted and applied to the Visa Fraud Master File. The reports also list transactions with data errors, processing errors, or both.

Reports may be received in one of these frequencies and formats:

- Weekly print-image reports
- Daily print-image reports
- Daily transaction response records

All transactions submitted during a given period are listed on one of the three Fraud Activity Reports and are summarized on the Fraud Summary Reports. If previously active clients submit no transactions during a given reporting period, a “null” report is distributed indicating that no fraud advice activity was received. Clients do not receive any fraud reports during the period.

Reports are available on a daily basis as print-image or transaction response records. Weekly reports are available as print-image only.

Note:

Issuers will not receive any reports if they have not reported fraud in the last 90 days.

FRS Process Description

Clients are required to report specific information about the transaction in the Fraud Reporting System (FRS), such as transaction amount and date. Required information is obtained from several sources, which include the cardholder, the original transaction draft, and the authorization message.

FRS Message Flows

Clients report fraud transactions to V.I.P., which holds the messages until end-of-day processing and then forwards them to Fraud Reporting System (FRS). Responses to client-reported fraud transactions are received as incoming transactions and are returned as part of the next BASE II cycle. The Response Code is a fraud advice field that identifies the type of incoming fraud advice transactions received by the client.

The format and content of each incoming fraud advice and associated data are based on whether the Transaction Response Record is a detail or summary record. Detail records are Transaction Confirmed (TC), Transaction Reject (TR), and Transaction Warning (TW). The summary record is Transaction Summary (TS).

These types of daily Transaction Response Records (TCRs) for recipients are provided:

- Confirmations (detail records)
- Warnings (detail records)

- Rejections (detail records)
- Summary (one summary record)

BASE II issuers or acquirers may receive a combination of these options:

- Daily print-image reports (default Asia-Pacific)
- Daily transaction response records
- Weekly print-image reports (default U.S.)
- Monthly print-image reports (default)
- Quarterly print-image reports (default)

FRS Requirements

Participants must be connected to VisaNet to use Fraud Reporting System (FRS) and have the ability to create TC 40 fraud advices.

FRS Testing and Certification

Certification is required in some Visa Regions. Visa, however, encourages all participants to be certified before using the service. Clients can check with their Visa representative for certification requirements.

The VisaNet Certification Management Service (VCMS) provides testing and certification assistance for FRS participants. Clients can contact their Visa representative to make the arrangements.

FRS Service Monitoring

Service (compliance) monitoring applies to FRS; however, the process varies according to region. Service participants can contact their Visa representative for information about service monitoring.

FRS Planning and Implementation

Visa issuers and acquirers must comply with the fraud reporting rules as defined in the *Visa Core Rules and Visa Product and Services Rules*.

FRS Documentation

This documentation is available:

- *Fraud Reporting System (FRS) User's Guide*
- *Member Fraud Control Manual*
- *Visa Core Rules and Visa Product and Services Rules*

Chapter 16

Issuers' Clearinghouse Service

Visa Issuers' Clearinghouse Service (ICS) is a risk management tool that assists bankcard issuers in reducing application and identity fraud and credit losses. Visa ICS provides significant value over credit bureau data and other application risk-management tools.

ICS Overview

Visa Issuers' Clearinghouse Service (ICS) provides information that issuers use to identify fraudulent or questionable bankcard applications. The system also responds to queries on the creditworthiness of current cardholders and allows issuers to add reports of fraud and unauthorized use to the ICS database. Visa ICS reporting is required for all issuers.

Visa ICS is a component of Visa Advanced ID Solutions, a set of application risk tools that also includes:

- ID Analytics ID Score Plus
- Fraud Intelligence by eFunds
- Debit Report by eFunds

Visa ICS is used by issuer management, supervisory, and technical personnel in the U.S.

The service provides the issuer with incremental data to increase profits by preventing and reducing losses due to credit and fraud risks. ICS resources include:

- Information about application activity and fraudulent card use reported by all issuers.
- Alerts identifying invalid or questionable data.
- Bankruptcy alerts for new applicants.
- Retroactive alerts generated when data from a previously submitted transaction matches a new bankruptcy filing or meets the thresholds for excessive application, fraud, or unauthorized use activity.
- Statistics for developing fraud-loss control programs through the identification of fraudulent activity patterns.

Visa ICS is part of a broader set of fraud mitigation tools called **Visa Advanced ID Solutions** and the **Visa Advanced ID Suite (VAIS)** that enhances risk protection by providing broader risk coverage. Under VAIS, Visa ICS and eFunds Corporation to provide Fraud Intelligence™ by eFunds, a tool giving Visa clients access to Demand Deposit Account (DDA) fraud information, including deposit account fraud closures associated with social security numbers within the past five years, addresses within the past five years, and social security numbers within the past five years and/or associated with an address within the past 180 days.

ICS Description

Visa Issuers' Clearinghouse Service (ICS) is a nationwide, centralized, shared database containing information on Visa and MasterCard credit card applicants, incidents of fraud, unauthorized use, bankruptcy filings, and other questionable data. The database was jointly developed by Visa U.S.A. and MasterCard International, Inc. with guidance from U.S. issuers.

ICS Database

The Visa Issuers' Clearinghouse Service (ICS) database is a collection of databases containing information on Visa and MasterCard applicants nationwide.

A separate database stores information on each of the three data elements tracked by ICS: social security numbers (SSNs), addresses, and telephone numbers. The system also includes a history database that counts the number of times each of these data elements appears on Visa and MasterCard credit card applications. Databases that track documented fraud reports and bankruptcy petitions are also included in the history database.

The database is compiled from a number of nationwide sources daily. Specific components of the data collection and updating process involve issuers, third-party sources, and Bankruptcy Retrieval Services.

- **Visa ICS Issuer Data**—Current data in ICS on credit card applications comes from issuers required to report the name, SSN, address, and telephone number for all of their credit card applications, including declined applications and unauthorized use or fraud at the account level, with and without a loss.
- **Visa ICS Third-Party Sources**—The Social Security Administration supplies ICS with files on valid and invalid SSNs. These files contain information about SSNs of deceased persons and currently unassigned numbers. The U.S. Postal Service and other data sources provide information on valid and questionable addresses and telephone numbers.
- **Visa ICS Bankruptcy Retrieval Services**—The Bankruptcy Retrieval Services (BRS) sends ICS information about every bankruptcy petition filed in the United States. This information is collected daily from most U.S. courts.

ICS Database Access

Issuers access the Visa Issuers' Clearinghouse Service (ICS) database through a connection to the VisaNet BASE II System. Issuers have direct access as a BASE II endpoint or through a third-party processor acting as a BASE II endpoint.

Other options are available through a dedicated connection to the ICS processor. For other options, contact ICSDirect@visa.com.

ICS Transactions

The process of sending an application, inquiry, or fraud report to Visa Issuers' Clearinghouse Service (ICS) is generally referred to as an "add transaction."

This table shows the five basic types of add transactions.

Transaction Type	Purpose
Add Application	Notify ICS of approved or pending bankcard application.
Add Decline	Notify ICS of a declined bankcard application.
Add Inquiry	Inquire about a current cardholder or applicant for a debit card account.
Add Fraud	Report a fraudulent application or other incident of fraud or unauthorized use.
Debit Inquiry	Inquire about an applicant for a debit card account.

Transaction Type	Purpose
Non-Bankcard Inquiry	Inquiry about an applicant for nonbankcard applicants.
Add Delete	Delete a transaction previously reported in error.

Submit an add transaction to:

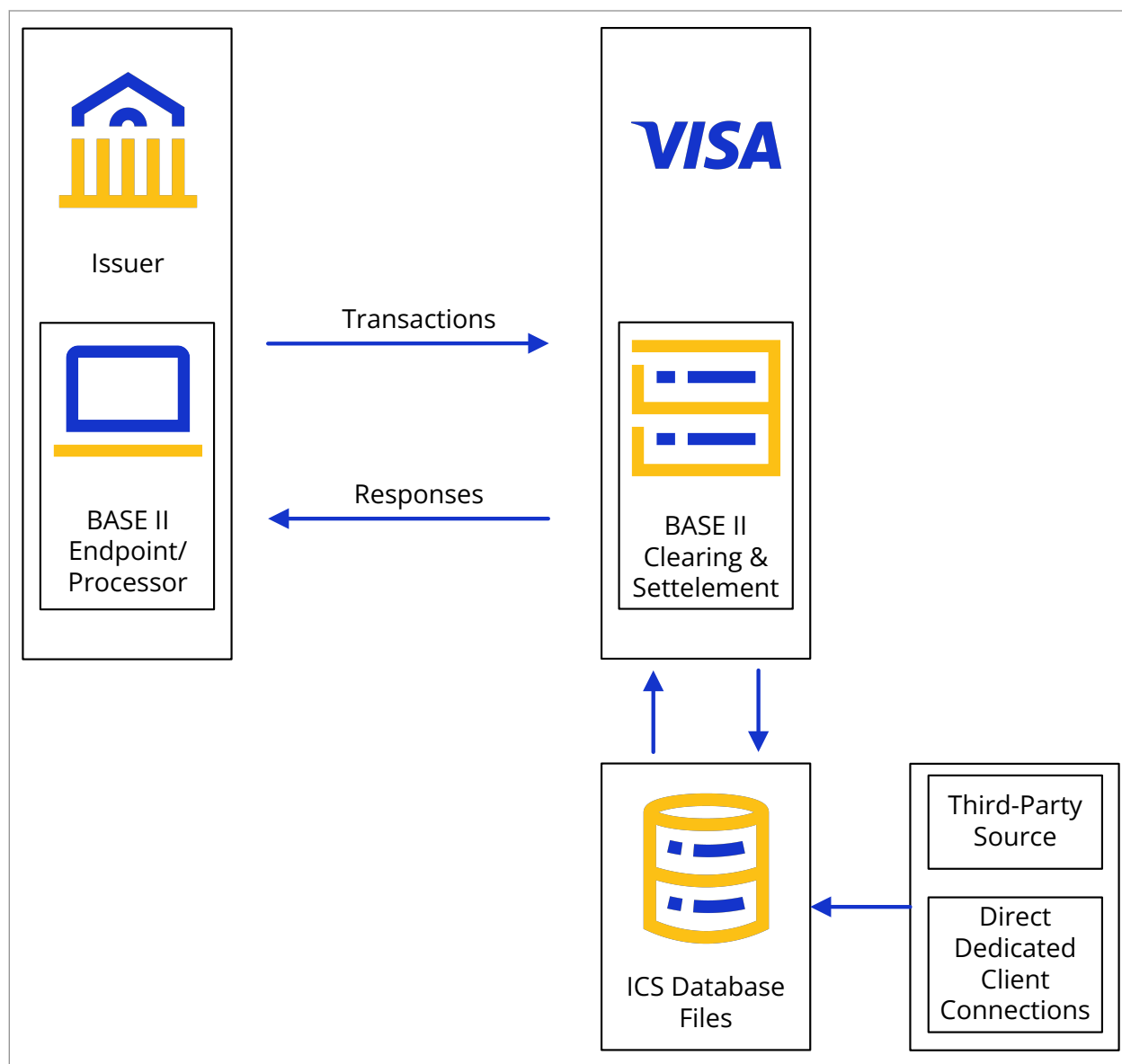
- Report a consumer application for a credit card, including declined applications.
- Report fraudulent or unauthorized use of a credit card.
- Report a suspicious application or a fraudulent application.
- Make a debit card or nonbankcard application inquiry.
- Investigate a retroactive alert or request information about a current cardholder or applicant. Inquiries of this kind are often made when evaluating a current cardholder's request for a credit limit increase or a replacement credit card.
- Delete an application or a fraud report incorrectly reported to ICS.

ICS Process Flow

Visa Issuers' Clearinghouse Service (ICS) processes issuer transaction files by verifying the transaction files and editing, comparing, and counting the data.

The flow of information from issuers and third-party sources through ICS and the BASE II System and back to issuers is illustrated in this figure.

ICS Process Flow



ICS Issuer Transaction File Verification

When a transaction is sent to Visa Issuers' Clearinghouse Service (ICS), the system first checks to make sure that all required data elements are present.

ICS verifies issuer transaction files by:

- Recording the date and time of receipt by ICS. This date does not necessarily match the entry date of the file.

The entry date is the date of input, in Julian format (YYDDD), that identifies the creation date of the transaction file record.

- Verifying transaction receipt from a validated issuer by checking the issuer Bank Identification Number (issuing BIN).

Issuers are identified by a unique six-digit issuing identifier assigned by Visa. When an issuer uses another party for settlement processing, or when an issuer uses a data processor, the issuing BIN in ICS transactions identifies the issuer, not the data processor.

- Verifying that the locator number is unique for each transaction by issuing BIN by day.

This issuer-assigned number contains up to 11 digits and must be unique for a given date and issuing BIN. Along with the issuing BIN and date, this number identifies ICS records, controls information retrieval and is required for processing add deletes. (Issuers may find it helpful to assign a locator number that corresponds to the application or account number of the original transaction submitted.)

The entry date, issuing BIN, and locator number together form a unique access reference number that identifies every record on file at ICS. Each issuer transaction has an access reference number used to retrieve it from the ICS/BNS database.

ICS Invalid Responses

Visa Issuers' Clearinghouse Service (ICS) recognizes two types of invalid response transactions: severe invalids and invalid responses.

Severe invalids are transactions that cannot be processed because the required record format or input is incorrect. The response is also invalid if it contains unmatched or missing data. This type of invalid response alerts the issuer to incorrect address, phone, zip code, or social security numbers.

Invalid responses indicate that Visa ICS cannot validate the data because of incorrect or missing data. They are flagged by ICS with invalid codes and returned to the issuer upon request. Invalids can be turned off in Member Options. Invalids are not processed or added to the database.

See *Issuers' Clearinghouse Service Quick Reference*.

ICS Edit Data

Visa Issuers' Clearinghouse Service (ICS) only requires one valid element of social security number, address, or phone to process any transaction. Issuers are encouraged to send all data.

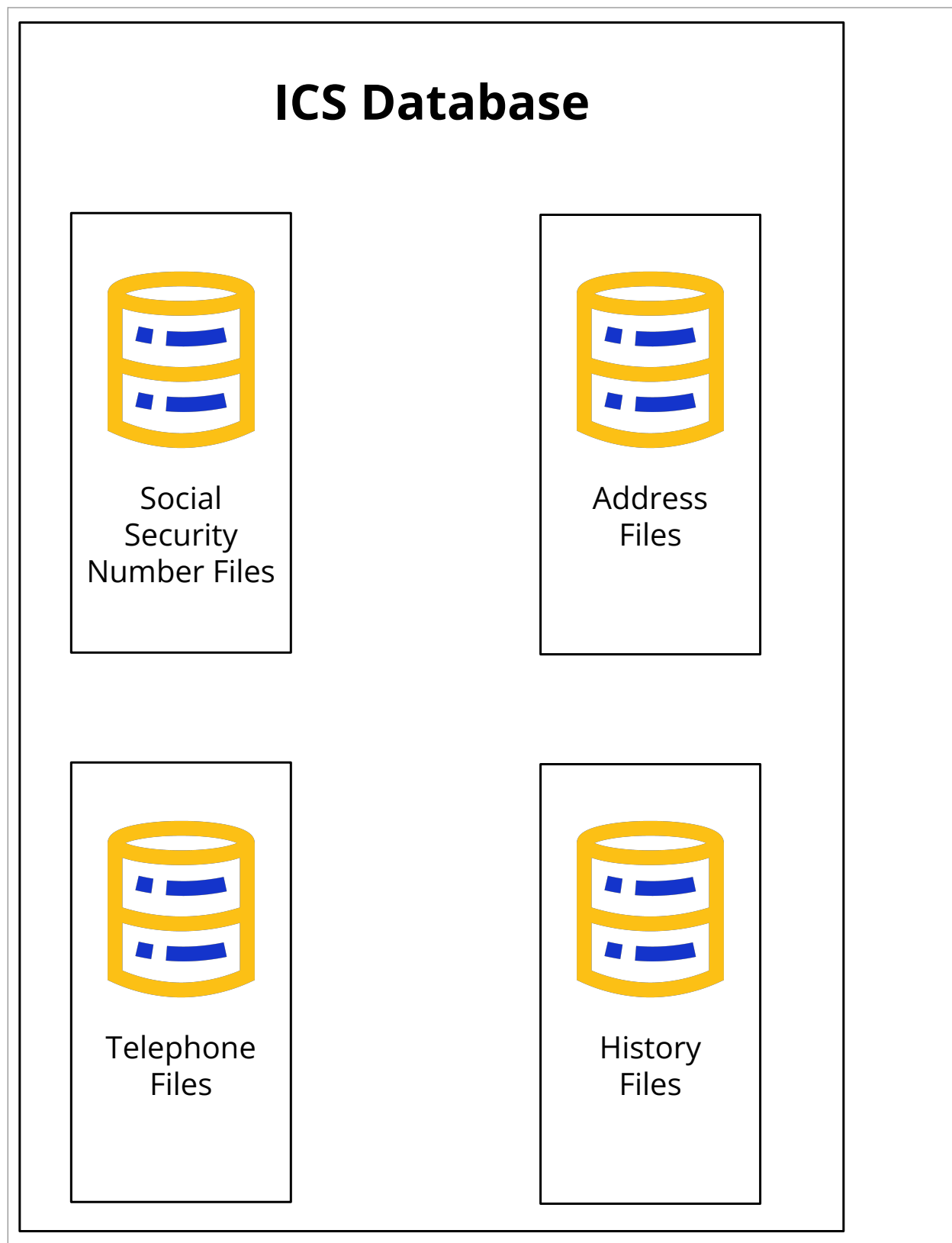
ICS processes all valid information. For example, if the SSN is invalid, the address and telephone number is processed normally and increment the database.

ICS Data Comparison

Visa Issuers' Clearinghouse Service (ICS) compares each SSN, address, and telephone number against the database and Member Options to determine if any information on the application is questionable or is associated with previous fraudulent activity or a bankruptcy petition.

Data elements on add applications and add declines are also counted and added to the ICS history file to help all issuers monitor excessive application activity. An overview of ICS database files is shown in the figure.

ICS Database Files



The Social Security Number File

The Social Security Number file contains the following information:

- Social security numbers entered by issuers.
- Valid SSN ranges and combinations from the Social Security Administration (SSA) and a list of SSNs for deceased persons along with supplemental information received from law enforcement.

Address Files

The Address files contain a Delivery Sequency File (DSF). The DSF is a computerized file that contains all delivery point addresses serviced by the USPS, with the exception of general delivery. DSF provides the following:

- Address validation
- Address standardization
- ZIP+4 validation
- Detection of potentially undeliverable addresses for reasons other than a change of address (for example vacant lots).
- Delivery-type code that indicates business or residential
- Seasonal delivery information

Home Telephone Number Files

The Home Telephone Number files contain the following information:

- Telephone numbers entered by issuers.
- Valid telephone area codes and valid telephone prefixes.
- Infobase questionable data files identify suspect telephone numbers. An example of questionable data would be when an applicant uses a telephone number of a penal institution, hospital, or hotel.

History Files

The SSN, address, and home telephone number are kept on file for each application and inquiry that ICS processes. Fraud History files are also maintained.

Fraud History Files

ICS Fraud History files contain claims of fraudulent use, Visa and MasterCard account numbers, arrest status information, and case numbers.

ICS Data Counts

Visa Issuers' Clearinghouse Service (ICS) uses counts to determine when issuers must be alerted to excessive activity.

As SSNs, addresses, and home telephone numbers are placed on file, Visa ICS counts the occurrences of each data element. For example, an SSN count of "1" means that the number was reported once to ICS. A count of "2" means that the number was reported twice, either by the same issuer or by two different issuers. Similar counts are kept for address and telephone number occurrences.

ICS monitors the number of thresholds for each data element to determine how to respond to a transaction. ICS is not a relational database. Each data element is matched individually. This allows issuers to see patterns that might not be visible at nationwide credit bureaus. For instance, an alert returned with a count of 1 tied to the applicant's social security number in the last 30 days and a count of 30 triggered by the address in the last 30 days may indicate a mail drop used by a fraud ring..

The counts and record data are retained by ICS from one or four years depending on type of activity:

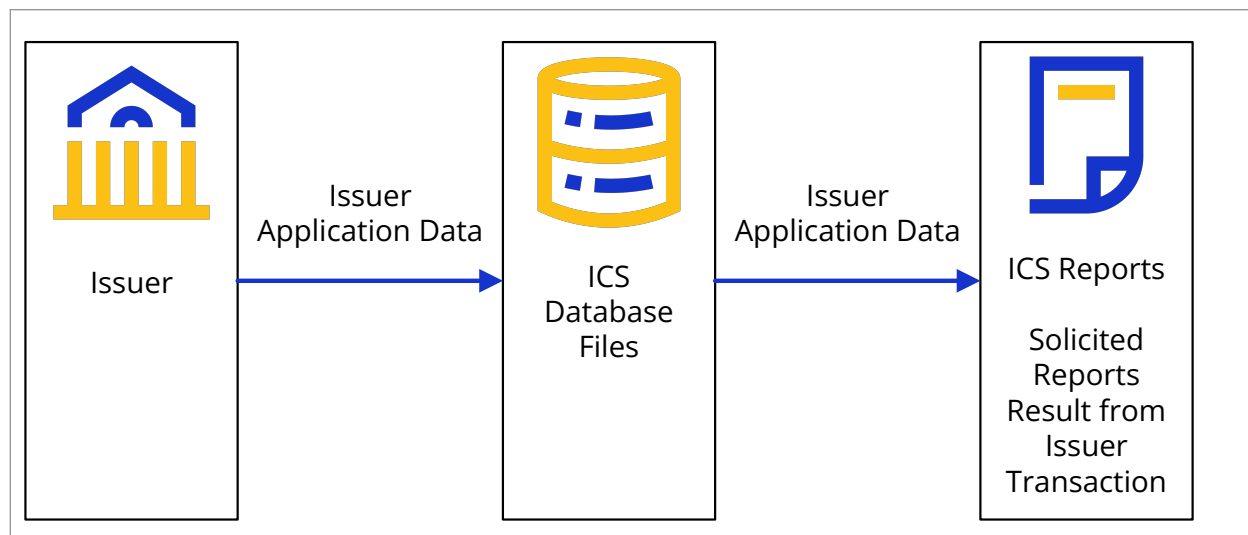
- Application data is retained for one year.
- Fraudulent applications and unauthorized use reports are retained for up to four years depending on the type of unauthorized use.

ICS Outputs

After processing issuer transactions, Visa Issuers' Clearinghouse Service (ICS) returns a response.

An overview of ICS reporting is shown in the ICS Reporting figure.

ICS Reporting



ICS Solicited Alerts

ICS sends these response transactions:

- **Bankruptcy Alert**—Informs issuers that the SSN submitted on an application or inquiry transaction was used in a bankruptcy filing..
- **Issuer Alert**—Advises issuers that application or unauthorized use activity has exceeded either issuer-specified Member Options, thresholds, or questionable data files.
- **Invalid Response**—Indicates that one or more issuer transaction data elements are invalid due to an issuer input error. ICS returns the transaction with related invalid codes.
- **Confirmation**—Indicates that all issuer transaction data elements are valid, none of the information triggered an alert, and the transaction was successfully processed. A confirmation is sent *only* if the issuer elects to receive Confirmation reports.

ICS Unsolicited Alerts

A Visa Issuers' Clearinghouse Service (ICS) Retroactive Alert indicates that activity or unauthorized use on one or more data elements from a previously submitted application equals or exceeds the issuer's activity on velocity, bankruptcy, or unauthorized use thresholds.

Note:

Alerts of fraudulent activity do not imply that the applicant or cardholder is responsible; in most cases, the applicant is the victim of such use. Issuers must carefully verify information related to certain types or patterns of use. For example, ICS flags addresses, telephone numbers, and SSNs used in bankcard applications where cards were previously reported as not received, or where an issuer-specified number of cards were reported lost or stolen, and subsequently used.

Using ICS Output

Issuers may use Visa Issuers' Clearinghouse Service (ICS) to evaluate account acquisitions, debit inquiries, collections, credit limit reviews, and fraud investigations.

Account Acquisition

ICS responses alert issuers to excessive credit card applications and instances of fraudulent use associated with an SSN, address, or home telephone number. Bankruptcy alerts advise issuers about consumer bankruptcy filings associated with an SSN. This information may be used to assess risks associated with new applications. ICS usually has bankruptcy filing data earlier than the nationwide credit reporting agencies.

Debit Inquiries and Non-Bankcard Inquiries

ICS responses alert issuers to excessive credit card applications and instances of fraudulent use associated with an SSN, address, or home telephone number.

Collection

Collection officers can use ICS for ad hoc inquiries. ICS also alerts issuers to accounts with problem histories. For example, an address associated with excessive applications may indicate a potential for credit abuse. Issuers may also be alerted to monitor excessive bankcard activity for a consumer who recently filed for bankruptcy. Issuers may use this information to assess risks and determine how aggressively to pursue collections.

Credit Limit Review

ICS responses help issuers determine whether to reissue cards or to increase credit limits. For example, ICS identifies excessive bankcard applications, which may indicate that a cardholder is overextended. ICS also provides information on fraudulent application activity.

Fraud Investigation

ICS responses help issuers identify possible patterns of fraudulent card use. For example, the issuer may want to question lost and stolen card claims if the cardholder has a history of reporting other issuers' cards as lost or stolen. This information may alert an issuer to investigate and prosecute cases of fraud. ICS may reduce losses by alerting issuers to potential problems.

ICS Requirements

Issuers can use the Options Selection Form to customize Visa Issuers' Clearinghouse Service (ICS) services, or accept ICS default services, thresholds, and time parameters. Options used to report to ICS may be set differently for each issuing BIN, market segment, or both.

An Options Selection Form is available from Visa Customer Service representatives. The form can be completed before implementing ICS to set options or after implementing to change options. Issuers that do not select options prior to implementation will receive the default values.

Note:

Issuers must check with their third-party processor's representative before selecting their ICS options to ensure that each option is supported by the processor. New issuers may need to allow three to four months to prepare for implementation of ICS.

ICS Legal Responsibilities

Under Fair Credit Reporting Act requirements, issuers may access Visa Issuers' Clearinghouse Service (ICS) for information or for making inquiries concerning only their own cardholders or applicants to evaluate a cardholder's continuing creditworthiness. Issuers must not base credit decisions solely on ICS responses. Issuers must decide whether and how to use the information to justify conducting an investigation.

Note:

Issuers may not use ICS information or responses to develop solicitation lists.

ICS Documentation

The following documentation is available:

- *Issuers' Clearinghouse Service User's Manual*
- *Issuers' Clearinghouse Service User's Training Guide*
- *Issuers' Clearinghouse Service Description*
- *Issuers' Clearinghouse Service Prescreen Handbook*
- *Issuers' Clearinghouse Service ICSDirect User's Guide*
- *Issuers' Clearinghouse Service ICSDirect Administrator's Guide*
- *Issuers' Clearinghouse Service Quick Reference Guide*

Chapter 17

Cross-Border Acquiring Service

The Cross-Border Acquiring Service allows acquirers to expand their markets outside their own countries while receiving the benefits of domestic interchange rates except where local laws or Visa regulations prohibit the practice.

BASE II Cross-Border Acquiring Service Overview

The Cross-Border Acquiring Service allows acquirers to expand their markets outside their own countries while receiving the benefits of domestic interchange rates except where local laws or Visa regulations prohibit the practice.

For example, a cardholder in country X buys a product or service from a merchant in country X, whose processor is located in country Y. The merchant's bank (or processor) participates in the Cross-Border Acquiring service and as a consequence abides by domestic processing rules and pays the domestic interchange rate in country Y instead of the international interchange rate.

Participants in the Cross-Border Acquiring Service include:

- Acquirers within the European Union (Europe region).
- Merchants located in the Canada region may have U.S. region acquirer/processor cross-border acquirer transactions if they are:
 - Mail/Phone Order Transactions (including computer) initiated by U.S. cardholders and advertised and cleared in U.S. dollars.
- Merchants located in the U.S. region may have Canada region acquirer/processor cross-border acquirer transactions if they are:
 - Mail/Phone Order Transactions (including computer) initiated by Canadian cardholders and advertised and cleared in Canadian dollars.
- International Airlines.
- Merchant outlet is a U.S. military base, embassy, or consulate on foreign territory.
- Europe region acquirers who cross-border acquire within the CEMEA region.
- CEMEA region acquirers who cross-border acquire within the Europe region.
- AP region acquirers who cross-border acquire within the AP region.

Chapter 18

Cross-Border Issuing Service

Cross-Border Issuing permits issuers to use a single new or existing Visa issuing BIN or an account range within an issuing BIN to issue cards in one or more countries or currencies. Clients have the option to license an issuing BIN to be used for Cross-Border Issuing or to use an existing issuing BIN. Account ranges from the Cross-Border Visa issuing BIN can be set up to issue cards in different countries.

BASE II Cross-Border Issuing Service Overview

Cross-Border Issuing permits dual-message issuers to use a single issuing BIN to issue cards in multiple countries and currencies. Dual message issuers may participate in the Cross-Border Issuing Service.

BASE II Cross-Border Issuing Service Description

Cross-Border Issuing supports cards in all regions. Clients license issuing BINs to issue cards in another country. A single issuing BIN can be used to issue cards in multiple countries.

Cross-Border Issuing cards maintain the features of existing card products and inherit the properties of those products. The major features of Cross-Border Issuing cards are:

- Issuance of the cards in multiple countries and currencies under a single issuing BIN.
- Identification of the country of issuance at the account range level instead of at the issuing BIN level.
- Application of fees, charges, and dispute rules according to the country in which the cards are issued.

Clients may participate in the Cross-Border Issuing Service by using one of these methods:

- Lead Bank and Partner Bank Cross-Border card issuance
- Direct Cross-Border card issuance

BASE II Cross-Border Card Issuing Bank Partnerships

A bank that does not have a subsidiary or an affiliate in a foreign country may choose to partner with a bank in that country to issue Cross-Border cards. The Lead Bank identifies the Partner Bank and initiates the partnership agreement. Visa International's Multinational Program Administrator office may assist in identifying a Partner Bank, providing sample contracts, and resolving issues related to multinational issuing.

Once a Cross-Border Issuing agreement is in place, the Lead Bank identifies an issuing BIN and an account range within that issuing BIN. The Lead Bank requests its Visa region to point that account range to the Partner Bank's issuing identifier for processing.

The Partner Bank is responsible for:

- Card issuance
- Authorization
- Cardholder billing
- Cardholder service
- Settlement

The primary financial liability resides with the Partner Bank. In accordance with *Visa Core Rules and Visa Product and Services Rules*, the issuing BIN licensee has the ultimate financial responsibility.

BASE II Direct Cross-Border Card Issuing

With Direct Cross-Border Issuing, a bank may choose to issue cards across borders in multiple countries and in multiple currencies, without a relationship with a Partner Bank.

These responsibilities remain with the issuer:

- Authorization
- Card issuance
- Cardholder billing
- Cardholder service

Clients can issue cards in other countries and currencies after obtaining a properly licensed issuing BIN for each issuing setup. With Cross-Border Issuing, clients can use a single issuing BIN and establish a specific account range to represent a country or currency.

To perform settlement, the issuing bank may work with either its own subsidiary or with a correspondent bank in the country of the settlement currency. The *Visa Core Rules and Visa Product and Services Rules* require that the settlement bank be located in the country of the designated settlement currency.

BASE II Cross-Border Issuing Service Requirements

Depending on the issuance scenario (Lead Bank/Partner Bank versus Direct Issuance), the client banks and Visa regions will have different responsibilities and tasks to perform for the Cross-Border Issuing Service.

The Cross-Border Issuing Service tasks include:

- The Lead Bank identifying a Partner Bank.
- The Lead Bank negotiating a contract between the Lead Bank and the Partner Bank.
- The Lead Bank and the Partner Bank defining liability and risk assessments within their contract.
- The Lead Bank identifying the account ranges for Cross-Border Issuing.
- The Partner Bank and Lead Bank defining their business requirements in the Member Information Questionnaire (MIQ).
- The Direct issuer identifying the Cross-Border Visa issuing BIN and account range.

- The Direct issuer maintaining financial responsibility for the Cross-Border Visa issuing BIN and account range.
- Directly issuing the Cross-Border cards and performing all the processing of the Cross-Border card transactions.
- The Direct issuer identifying a funds transfer for the Cross-Border account range.

The Visa regional implementation group will work with the client to establish parameters in the VisaNet systems for authorization, clearing, and settlement.

Project Initiation

Clients who want to issue cards under Cross-Border Issuing must participate in Visa International's Multinational Program Administration (MPA).

BASE II Cross-Border Issuing Service Documentation

The following documentation is available:

- *Visa Multinational Program Guide*
- *Cross-Border Issuing Member Implementation Guide*
- *Visa Core Rules and Visa Product and Service Rules, Volume 1, General Rules*

Chapter 19

BASE II Multiple Clearing

Multiple Clearing allows merchants to obtain submit multiple clearing records by select MCCs in connection with a single authorization request.

Multiple Clearing Split Shipment transactions for card absent transactions allows a merchant to obtain a single authorization and submit multiple clearing records for a single purchase. This type of transactions occurs when a merchant is shipping items from different locations or warehouses for a single purchase transaction and allows them to submit a clearing record as each item is shipped.

For merchants this capability provides;

- Greater Flexibility - Merchants can optionally split card absent transactions in a way that best fits their business structure.
- Increased Efficiencies - For split shipment transactions, merchants only submit a single authorization for multiple shipped items.
- Improved Customer Experience - Merchants can fulfill the order more quickly and efficiently.

Multiple Clearing Requirements

Merchants should employ the following to manage their processing of split-shipment card-absent transactions.

- Submit multiple clearing records with a multiple clearing sequence number in TCR 5 position 45-46 in ascending order. The first clearing record must have a multiple clearing sequence number set to 01.
- Include the total authorized amount, authorization code and transaction ID from the approved authorization message in each multiple clearing record submitted.
- At the time of check-out, disclose to the cardholder the possibility that multiple shipments may result in multiple postings to the cardholder's monthly statement.
- Communicate with the cardholder during the fulfillment process the details (e.g. date, amount, etc.) when a split-shipment is being sent for a transaction and the amount that will be posted on their statement.

For multiple clearing transactions other than the purchase of passenger transport tickets the Multiple Clearing Sequence Count (MCSC) TCR 5 position 47-48 with the first submitted clearing message must be a value greater than 1. If the total count of clearing messages to be submitted for the transaction is not known with the first clearing submission it is acceptable to use a best estimate or '99'. When using an estimated value the MCSC should be adjusted to reflect the actual final count of clearing message submissions once known.

- For multiple clearing transactions other than the purchase of passenger transport tickets sequential order of Multiple Clearing Sequence Number (MCSN) with each clearing message is preferred but not required.
- For multiple clearing transactions other than the purchase of passenger transport tickets, it is preferred but not required that with the last clearing message submission the Multiple Clearing Sequence Number (MCSN) and Multiple Clearing Sequence Count (MCSC) equal.

Chapter 20

VisaNet Certification Management Service

The VisaNet Certification Management Service (VCMS) for BASE II enables client and endpoint staff to test their BASE II software applications, transaction formats, and VisaNet data transmissions. The service helps identify and fix potential problems before activating new or enhanced services into production.

BASE II VCMS Overview

The VisaNet Certification Management Service (VCMS) provides a dedicated environment to clients for testing their systems with Visa. When using VCMS for BASE II to test, participants can verify that their transaction formats and offline clearing and settlement interfaces are in working order before connecting to VisaNet in a production environment.

VCMS for BASE II is for Visa processors, issuers and acquirers involved with testing their systems for BASE II data.

BASE II VCMS Description

The VisaNet Certification Management Service (VCMS) for BASE II supports testing by simulating VisaNet production systems, but in physically separate environments. VCMS ensures that messages pass format and edit checking correctly.

VCMS for BASE II provides these testing resources.

- Responders—Support lifecycle testing.
- Visa Test System-Clearing and Settlement (VTS-CS)—Allows clients to manage their BASE II Testing. It is a subscription-based tool available as a Visa Access service.
- Auto-Presentation—Allows issuers to process VCMS VisaNet Integrated Payment (V.I.P.) System online authorizations and receive settlement records for these items through VCMS for BASE II (for BASE II-connected issuers). Auto-Presentation is available in the VTS-CS tool.

VCMS testing for BASE II:

- Verifies that the components of participating systems that have been enhanced or changed can communicate accurately with VisaNet.
- Provides an environment to test internal systems, including testing for new clients, endpoints and new services.

Visa Regional Testing Representatives coordinate and validate client testing, including:

- Customized File Delivery Type (CFDT) is a setting in BASE II Config that allows endpoints to request various kinds of customized delivery files. This functionality can be used in VCMS to separate transactions originated by Responder identifiers from those originated by non-Responder identifiers.

To request a CFDT for transactions originated by Responder identifiers, contact a Visa Regional Testing Representative.

Note:

CFDT is available only for delivery files.

- Supporting creation of client test data.
- Reviewing BASE II and Edit Package reports to verify successful testing, depending on the region.

Participants can receive instructions for new services and business enhancements in the *VisaNet Business Enhancements Global Technical Letter and Implementation Guide* for each service. This guide lists the transaction type requirements that must be processed through VCMS BASE II to meet the specified success criteria.

BASE II VCMS and Production Differences

VisaNet Certification Management Service (VCMS) for BASE II offers the same core functionality as the corresponding Visa production systems excluding expedited delivery, except for SMS reports and raw data delivery.

Note: Not all reports are available in VCMS.

BASE II VCMS Testing Benefits

The VisaNet Certification Management Service (VCMS) is used to test various conditions in phases.

The VCMS testing environment provides:

- Testing for clearing and settlement interfaces in a dedicated VisaNet BASE II test environment.
- Access to production-ready code.
- Transaction lifecycle testing.
- Daily updating of the latest currency rates from production (other than pre-release).
- Testing for new client installations and existing client upgrades to new or enhanced VisaNet services.
- VisaNet Settlement Service (VSS) testing.
- A shared environment accessible worldwide.
- Improved client readiness for BASE II transaction processing in the production environment.
- Reduction of risk to production systems.

BASE II VCMS Documentation

The following documentation is available:

- *VisaNet Certification Management Service (VCMS) BASE II Testing Guide*
- *VisaNet Business Enhancements Global Technical Letter and Implementation Guide*

Chapter 21

BASE II Clearing System Restrictions

BASE II Clearing system uses embargo blocking to block U.S. client transactions in countries where U.S. government regulations prohibit U.S. transactions.

BASE II Clearing System Restrictions Overview

Visa blocks U.S. client transactions in countries where U.S. government regulations prohibit U.S. transactions (for example, no U.S. transactions are allowed in Iran). This functionality, known as **embargo blocking**, is performed by the BASE II Clearing system. Visa's policy is to promptly add or remove countries from the embargo blocking list as the U.S. sanctions change.

BASE II Clearing System Restrictions Description

Visa sets up embargo blocking for U.S. clients in V.I.P. and BASE II based on the U.S. government trade sanctions in force.

For U.S. clients, embargo blocking is automatically set in the VisaNet for countries that are subject to U.S. comprehensive sanctions. Visa's Sanctions Compliance team notifies BASE II administrators when a country needs to be added to or deleted from the system's Country Exclusion tables.

Once set, embargo blocking not only rejects embargoed country transactions on U.S. Visa cards but also rejects embargoed country traffic on Visa cards issued by foreign subsidiaries of U.S. banks

Embargo blocking can also be used as an optional feature by issuing countries who choose to block transactions with another country or countries (for example, the clients in country A are ordered by their government to set embargo blocking to reject all transactions from country B).

Embargo blocking of transactions originating in countries being embargoed by the U.S. government is mandatory for all U.S. clients.

BASE II applies blocking logic to prohibit U.S. clients from engaging in financial transactions involving specific countries.

Legal and Political Restrictions Acquirer Information

Acquirers should be aware that transactions, such as purchases and ATM withdrawals destined for an issuer in a country that has been listed for embargo blocking will be returned. Clients considering acquiring transactions from any country subject to an embargo should seek independent legal advice relating to this activity.

Legal and Political Restrictions Issuer Information

Issuers should be aware that transactions, such as purchases and ATM withdrawals originating in countries designated for embargo blocking will be returned to the acquirer.

BASE II Clearing System Restrictions Description

When a transaction is submitted to BASE II for clearing, the Legal and Political Process Restrictions system verifies the transaction information.

When a transaction is submitted to BASE II for clearing, the system:

- determines the issuer's region and zone (country) and merchant's region and zone
- edits the information against a Country Exclusion table containing the list of embargoed countries and the countries embargoing them

- determines if the card issuer is a subsidiary of a U.S. client
- declines and returns any transactions that meet the embargo criteria

If the transaction originated in an embargoed country, the acquiring bank does not get paid. For example, a U.S. cardholder visiting Iran makes a credit card purchase that is not authorized because the merchant is not linked to VisaNet. When the transaction is submitted by the merchant's acquirer, BASE II determines it occurred in a country on the embargoed list and was made with a U.S. client-issued card, and, if this is the case, returns the transaction to the acquirer.

BASE II Clearing System Restrictions Service and Support

For further information about embargo blocking, contact the Visa Sanctions Compliance team.